

Vision™ Air Handler

Sizes 003 thru 090

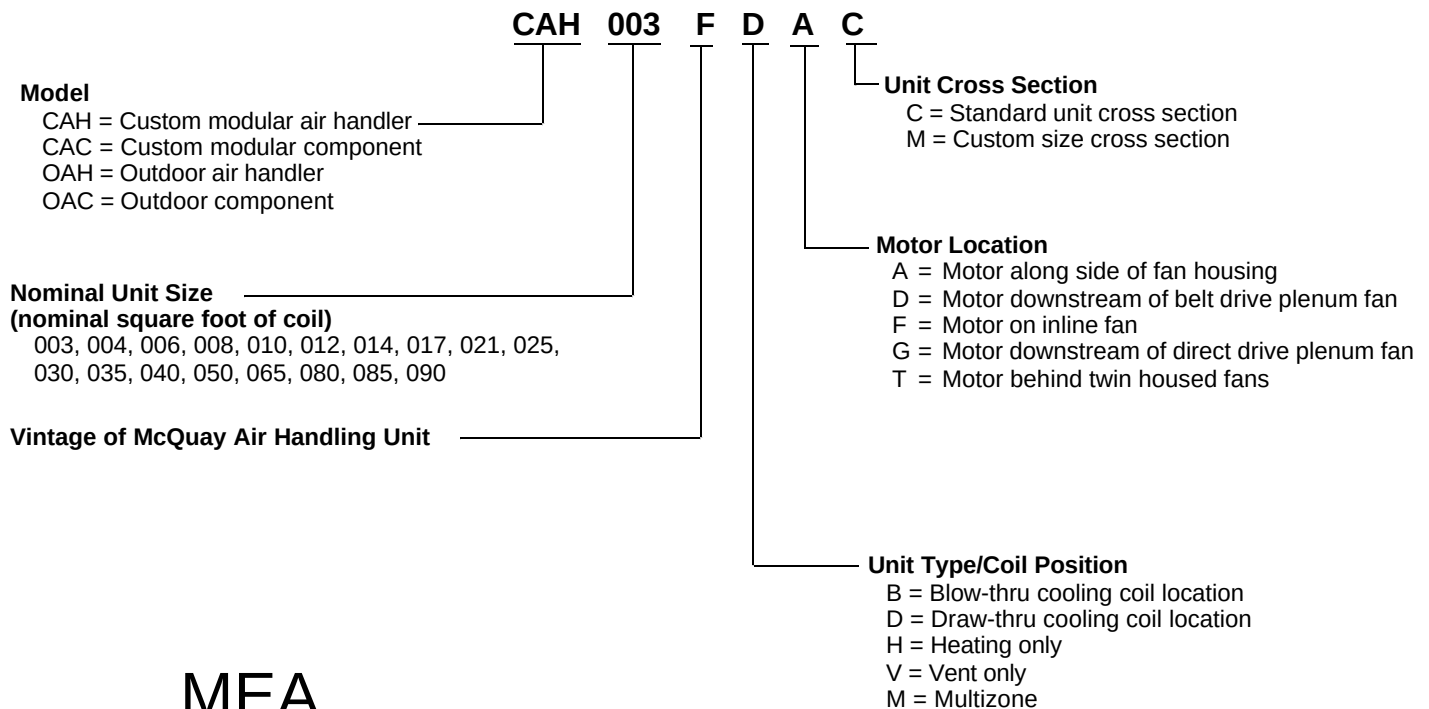


MEA

342-99-E

See page 1

Vision™ air handler model nomenclature



MEA

342-99-E

Agency listed



All standard units
All custom size units
All Canadian units

ARI certification



Standard & custom size units certified in accordance with the central station air handling units certification program, which is based on ARI Standard 430.*



Certified in accordance with the forced circulation air cooling and air heating coils certification program, which is based on ARI Standard 410.

*NOTE: Ratings at other than wide open position of the inlet vanes are outside the scope of ARI Standard 430.

Table of contents

Vision advantage at a glance	pages 2-3
Introduction	pages 4-5
Vision's unique standard features	pages 6-7
Vision's customized options.....	pages 8-9
Component types.....	pages 10-14
Air handler selection	page 15
Quick select table	pages 16-17
Application considerations	pages 18-20

Engineering and physical data

Component pressure drops	pages 22-28
Component and section weights	pages 29-32
Fan data	pages 33-36
Coil data	pages 37-39
Filter data.....	pages 40-41
Dimensional data.....	pages 42-71
Engineering guide specification	pages 72-76

"Vision" and "SelectTOOLS"™ are registered trademarks of McQuay International. All rights reserved.

© 2001 McQuay International. All rights reserved throughout the world.

Bulletin illustrations cover the general appearance of McQuay International products at the time of publication. We reserve the right to change design and construction specifications at any time without notice.

The Vision air handler frame channel design is covered under patent number 5,713,651.

The Vision advantage at a glance

Flexibility	
Vision's Unique Design	What It Does For You
Custom-modular platform	Allows "customizing" of the system with a wide selection of components and sizes.
Variable Dimensioning™ design	Allows cabinet to be sized in two-inch increments to fit almost any space.
Ships assembled or by section	Minimizes install time; reduces installation cost; allows passage through doors, tight spaces and in elevators (great for retrofit jobs).
Indoor or outdoor construction	Gives the option to apply an air handler where space permits.
Multiple coil, fan and filter selections	Allows optimum selections for cost, energy efficiency, performance, indoor air quality, and low noise.
Wide selection of base rail heights (4"-12")	Allows for adequate condensate trapping; eliminates need for costly house-keeping pad. Provides space required between condensate connection and traps on steam applications.
Painted cabinet	Enhances exterior appearance for units in visible locations.
Indoor air quality	
Vision's Unique Design	What It Does For You
Low-leakage cabinet construction (capable of less than 0.5 CFM/ft ² of cabinetry at 5.0" positive w.c.)	Minimizes air leakage, noise and unfiltered air.
Double sloped microbial-resistant coated galvanized (standard) or stainless steel drain pan	Inhibits bacterial growth; eliminates standing water that can support bacteria.
Smooth interior surfaces	Reduces potential for accumulating dirt and microbial growth; allows for easy cleaning.
Multiple filter types (flat, angular, bag and cartridge) with side-load and front-load capabilities	Gives full range of filter efficiencies, final filter arrangements, and filter section depth flexibility.
Double-wall construction	Prevents fibers from eroding into the air stream; prevents dirt and moisture from penetrating fiberglass insulation.
Patented splice collar	Prevents unfiltered, unconditioned air from entering the system. Allows for leak-resistant section to section joining in the field.
Easy-to-remove access panels	Allows for easy inspection and cleaning of drain pans; promotes regular inspections.
Hinged access doors with full-grip handles	Allows for proper inspection.

Operating efficiency	
Vision's Unique Design	What It Does For You
Low-leakage cabinet design	Increases operating efficiency; reduces energy loss and operating costs.
Patented gasketed frame channels	Minimizes direct exposure of metal to metal in cabinet framework to reduce cold bridging and condensate collection; lowers operating costs.
Energy recovery (plate heat exchanger or energy recovery wheel)	Reduces cost of heating or cooling outside air; recovers up to 50% of the energy normally exhausted from a building.
Fan selection options (housed forward curve or airfoil, belt drive or direct drive plenum, inline and twin fans)	Results in lowest possible BHP requirements.
Patented UltraSeal™ low-leak dampers	Maximizes operating efficiency; reduces operating cost.
Solid or perforated double wall liners	Acts as a barrier between internal and external panels to reduce condensation and leakage for energy savings.
Easy, low cost installation	
Vision's Unique Design	What It Does For You
Ships assembled or by section	Minimizes install time, reduces installation cost; allows passage through doors, tight spaces and in elevators (ideal for retrofit jobs).
Complete disassembly	Allows for complete knock down during installation if required for tight fit applications.
Patented section splicing	Saves installation time; creates an airtight environment.
Extended coil connections with gasketed airseal, external drains and vents	Eliminates need to remove panels for connection during installation; preserves air-tight environment; saves time and cost.
Bolted frame construction with removable panels	Provides complete accessibility to unit components.
Fan system factory tested and balanced	Saves time during installation and promotes proper operation.
Discharge or inlet plenum sections	Adapts easily to ductwork and provides a turning section for stacked components.
Easy maintenance and serviceability	
Vision's Unique Design	What It Does For You
Easy-to-remove access panels in a wide range of section depths	Makes it fast and easy to clean and inspect drain pan, clean or remove coil, clean interior.
Removable cross-member frame channels	Allows for top or side coil removal without disrupting unit integrity.
Extended fan bearing lube lines	Makes lubricating fan system easier.
Extended coil drain connections	Reduces coil venting time; helps coil condensate drain completely.
Hinged access doors with full-grip handles	Allows for proper inspection.

Introduction

Quality

For nearly 60 years, McQuay air handling equipment has been respected and regarded as a high quality product. McQuay has taken a major step in redefining the indoor air handler with the Vision air handler. Demands for improved indoor air quality, low sound, high operating efficiency and smaller mechanical rooms require a better product for today's air handler market. The McQuay Vision air handler has been designed to meet or exceed these demands.

The key to providing such a high quality product is in the basic design. The McQuay Vision air handler's patented construction has been designed to provide unequalled thermal efficiencies and to be airtight. In addition, Vision air handlers offer tremendous flexibility in sizing, component options, and unit arrangements to meet the indoor air quality, operating efficiency, sound and installation requirements for today's extensive commercial and custom markets.

Flexibility

By virtue of its unique frame design, the McQuay Vision air handler offers tremendous flexibility. This flexibility is reflected in the unique Variable Dimensioning™ design that allows Vision air handlers to be sized in two-inch increments to fit the available space. Numerous section and component options, and the ability to arrange components in whatever arrangement is required, allow Vision air handlers to be customized to the requirements of each job without expensive modifications in the field. Finally, Vision air handlers can be shipped as a completely assembled unit, in modules or by component sections for new or retrofit applications that require smaller sections for passage through the building.

Cabinet Construction

McQuay Vision air handler cabinetry consists of a box-type frame channel - the backbone of the unit - and easy-to-remove panels. The unique, patented frame channel design allows three identical pieces to be bolted together to form a corner of the unit. Channels are formed of G90 galvanized steel or G60 painted galvanized with factory applied neoprene gasketing on all flanges to minimize leakage. Gasketing is also factory applied at all contact surfaces between interior and exterior metal components to minimize thermal bridging. Panels are secured to the unit with fasteners that can be easily removed to access the unit interior. The frame channel and panels can be easily disassembled and reassembled, giving contractors tremendous flexibility in installing and servicing the units - even where space is very limited - and saving time and money. Patented splice joints help guide sections together for a tight fit, saving additional time and money on installation. Splice joints are also fully insulated at the factory.



Patented gasketed frame channel

This unique cabinet design results in air tight, thermally efficient units, which translates into energy savings over the life of the unit. Vision air handlers are capable of leakage rates less than 0.5 CFM per square foot of cabinetry at 5.0" positive static pressure. Units will not sweat when exposed to interior air temperatures down to 50°F, and exterior room conditions up to 70% relative humidity.

Vision air handlers are available with or without insulation, and with single or double wall construction. Double wall units are available with solid or perforated G90 galvanized and solid stainless steel liners. Fiberglass insulation is available in 1- or 2-inch thickness and $\frac{3}{4}$, 1-1/2, or 3 pound per cubic foot densities. The thickness of the insulation is never compromised for maximum thermal efficiency. A full thickness of uncompressed insulation is provided throughout the unit - including underneath the drain pan.

Base rails are provided as an option on unit sizes through 21 square feet and are standard on larger units. They are available to meet jobsite height requirements, ranging from 4" to 12" and can eliminate the need for a housekeeping pad - saving time and money on installation. A full perimeter base rail per shipping section is factory assembled.



Patented splice joint for section to section joining

Outdoor Construction

McQuay Vision air handlers can now be used anywhere an air handler is needed. New construction features provide weather protection to extend the range of uses for Vision air handlers to outdoor applications. Along with the frame channel and panel construction of indoor units, Vision outdoor units include a roofcap that covers all top surfaces and field assembled joints. The roofcap adds a complete layer of heavy galvanized or galvanized painted steel, forming a double shield to protect all critical surfaces and provide extra durability. Inlet hoods with screens help prevent water intake into the damper openings and are factory installed. Insulated, double-walled coil vestibules protect piping and control valves.

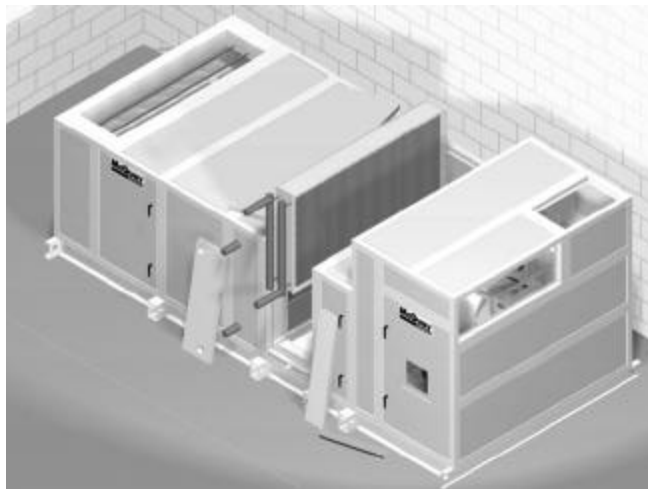
Most sizes and component selections that are available with indoor Vision air handlers are available in the outdoor construction. Refer to the Vision SelectTOOLS™ Software for the complete set of configurations available for the outdoor construction units.

Access and Serviceability

Equipment must be designed to perform efficiently and withstand the wear and tear of everyday use. It must also be designed to provide easy access to interior components for routine maintenance and service to maintain peak performance. The patented frame channels and easy-to-remove panels or hinged access doors of the Vision air handler cabinet provide complete access to the unit interior and components. Components, including the fan and coil assembly, can be removed through the side of the unit, top of the unit, or a combination of both, which reduces required service clearances. Coil removal on unstacked coils does not require access to the non-connection end of the coil. A unique coil tie down method simplifies coil removal and replacement.



Easy access for routine maintenance and service



Top and side panel, frame channel removal for component replacement

Vision SelectTOOLS™ Software Selection Program

Because the McQuay air handler is so flexible and has many different component types, there are virtually an infinite number of unit arrangements possible. To help the customer easily define their product requirements, McQuay provides a user friendly selection program, called the Vision SelectTOOLS™ Software Selection Program. This program configures and sizes both standard and custom units. Components can be selected in minutes. This Windows® based program will lead the user through the selection process by prompting for pertinent input data for all components required. Component sections are selected by placing them on a configuration screen. Once the unit layout is defined, the options and accessories are identified. The program gives immediate feedback regarding fan and coil selection, offering a choice of many different options based on the performance inputs. Once final component selections have been made, the program will provide all output needed for specification and submittal purposes, including fan curves, coil performance psychometric charts, weights, dimensional drawings, and a unit specification. A fully detailed .DXF file is also available. The Vision SelectTOOLS™ Software is a comprehensive, efficient and user friendly selection program.

"Windows" is a registered trademark of the Microsoft Corporation.

Vision's Unique Standard Features

- Custom-modular design**
- allows custom selection and configuration of components
 - allows design of system to meet space and performance requirements



- Patented UltraSeal™ low-leak dampers**
- maximizes operating efficiency
 - reduces operating cost



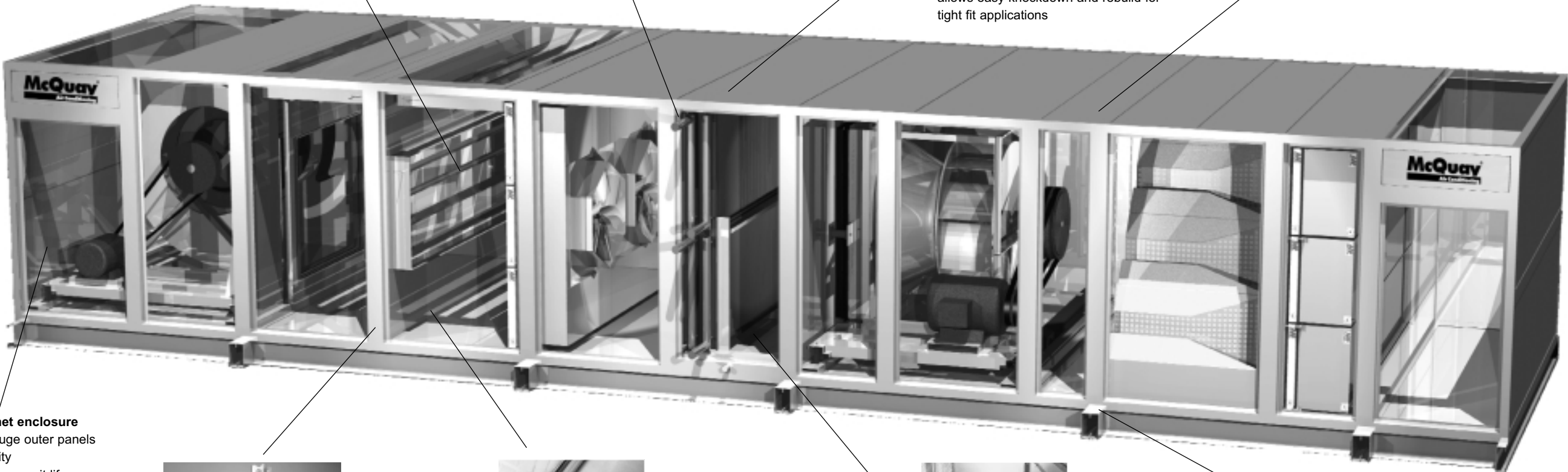
- Extended coil connections**
- saves installation costs
 - reduces maintenance time
 - aids proper drainage



- Frame channel disassembly**
- allows two-sided access
 - easier, faster service and maintenance
 - allows coils to be removed through side, top or both
 - allows easy knockdown and rebuild for tight fit applications



- Patented splice collar**
- reduces installation cost
 - creates an air tight environment



- Rugged cabinet enclosure**
- 16- or 18-gauge outer panels provide rigidity
 - promotes longer unit life
 - 1" or 2" insulation
 - 20-gauge galvanized, galvanized perforated or stainless steel liners



- Gasketed frame channel**
- eliminates metal-to-metal contact between paneling and framework
 - minimizes cabinet condensate and corrosion
 - facilitates top and side component removal
 - promotes long life



- Smooth interior surface**
- reduces dirt accumulation
 - facilitates cleaning
 - improves IAQ



- Visible double-sloped drain pan**
- makes inspection and cleaning easier
 - improves IAQ

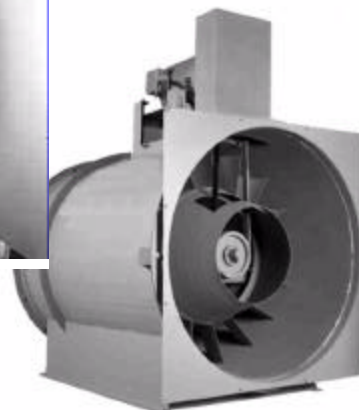


- Variable height base rails (4"-12")**
- eliminates costly housekeeping pad
 - allows for adequate condensate trapping

Vision's customized options

Use McQuay Vision as a product platform on which to build the ideal air handler for your specific application. Customized options include:

- Variable sizes
- Indoor and outdoor construction
- Painted cabinet
- Multiple section depths
- Variable base rail heights (4"-12")
- Various casing and drain pan materials
- Mixing boxes/economizers
- Sound attenuators
- Multiple coil section depths
- Energy recovery sections (heat wheels, fixed plate heat exchangers, and runaround coil loops)
- Multiple fan selections (forward curve, airfoil, inline, and belt or direct drive plenum fans)
- Twin fans
- Filters (flat, angular, bag and cartridge) available in side load and/or front-loading configuration
- Flush-mounted filter gauges
- Starters and inverters (VFDs)
- Electric heaters
- Inward opening doors
- Windows, doors, marine lights, and receptacles
- Manual selections to accommodate special components
- Flexibility in shipping arrangements



Fan Options



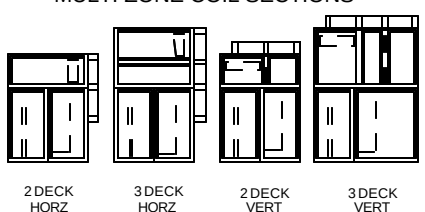
Multiple Filter Types



Blender / Air Mixer Options



Filter Gauges

RETURN / EXHAUST	ECONOMIZER	BLENDER	FILTER	ENERGY RECOVERY	F & BP DAMPER	COIL	ACCESS	HORIZONTAL COIL	SUPPLY FAN	DIFFUSER	COIL BLOW THRU	ACCESS	FILTER	ATTENUATOR	PLENUM
PLENUM FAN		2" & 4" ANGULAR	WHEEL	INTERNAL	HORIZ HEATING	LARGE	CLG. / HTG. COMBINATION	FC, AF FAN, & TWIN FAN			HORIZ		CARTRIDGE		DISCHARGE PLENUM
FC, AF FAN & TWIN FAN	INLET PLENUM	2" & 4" FLAT	FLAT PLATE CUBE	INTEGRAL	1 & 2 ROW STEAM AND 1 - 4 ROW WATER	MEDIUM	CLG. 10 ROW	PLENUM FAN			CLG. OR HTG.		BAG		
ESIFAN	MIXBOX WITH 4" FILTER	CARTRIDGE				SMALL	4 ROW	ESIFAN	MULTI-ZONE COIL SECTIONS  2 DECK HORZ 3 DECK HORZ 2 DECK VERT 3 DECK VERT						
MIXING BOX	MIXBOX WITH 2" FILTER	BAG		RIGHT ANGLE			SFA COIL MFA COIL F & BP	FAN AND VERTICAL COIL							
	MIXBOX WITH ANG FILTER														



Energy Recovery Equipment



Sound Attenuators



Marine Lights & Receptacles

Component types

Fans

Fan types available with the McQuay Vision air handling units are housed DWDI forward curved and airfoil fans, plenum fans, inline fans, and twin fans. Forward curved fans generally provide the lowest first cost option and are used for lower static pressure applications. McQuay housed forward curved fans will typically operate up to 6.0" of static pressure. Airfoil fans have a higher first cost, but are more efficient, quieter and can handle higher static pressures. McQuay housed airfoil fans will operate up to 9.0" of static pressure.

Plenum fans save space by eliminating turns in ductwork. They also provide a high degree of flexibility when locating the outlet ductwork. Plenum fans are also very good for blow-through applications as they generate a uniform outlet velocity profile. Both belt drive and direct drive plenum fans are available.

Inline fans are quieter and more efficient than most standard fans. Inline fans are most beneficial in VAV applications.

Twin fans (two housed forward curved fans mounted on one common shaft) are excellent in both retrofit units and stacked energy recovery units. Twin fans can handle high CFMs in very low profile units.

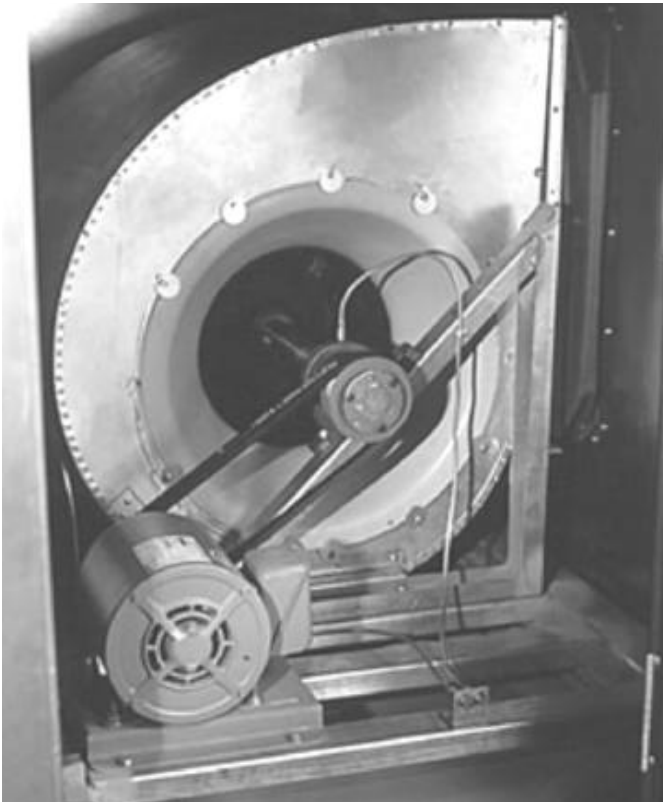
These different types of fans are available in numerous wheel diameters per unit size. The fan size selection is always optimized and identified by the Vision SelectTOOLS™ Software. The program offers a number of fan sizes to select from, considering performance, efficiency, sound generation and first cost. Fan and motor assemblies can be provided with 2" spring, rubber in shear, or rigid mount allowing the customer the option of isolating the fan and motor assembly internally or isolating the entire unit. All fans are dynamically balanced at the plant as an assembly including the motors and drives.

All fans rotate on a solid steel shaft of uniform diameter that has been ground, polished, and coated with a rust inhibitor. Fan shafts are selected to have a maximum operating speed well below the first critical speed.

Belt guards are available on plenum fans and inline fans. Seismic restraint is also available for applications requiring additional fan support.

The fan bearings provided will have a minimum L50 life of 200,000 hours, and are available as high as 1,000,000 hours. Bearings are selected for minimum noise levels and minimal service. Bearings are self aligning and prelubricated for immediate use. Bearings are mounted on rigid frames and positioned to provide proper balance. All bearings include copper lubrication lines that are extended to the access side of the fan cabinet with the grease fittings located near the access door opening.

All fan motors are internally mounted. The appropriate motor size will be selected by the Vision SelectTOOLS™ Software. Motors cannot be undersized, but may be oversized if desired. Motor options include open drip proof, totally enclosed fan cooled, 1 speed/1 winding, 2 speed/1 winding, and 2 speed/2 winding. Motor efficiencies available include standard efficiency, high efficiency, and premium efficiency depending on the motor type.



Housed airfoil fan



Spring isolation

Coils

The McQuay Vision air handler offers broad application flexibility in coil sections and coils. Coils can be arranged in draw through, blow through, or multizone configurations. Heating only, cooling only, or cooling and heating sections are available. All coils are installed with space between each coil to allow for access for cleaning and mounting of controls.

Cooling coil sections, and cooling and heating coil sections, are available in seven different section lengths to accommodate every application requirement. Drain pans extend the full length of the section. Removable access panels or doors may be provided in the deeper sections that will not interfere with piping connections extending through the unit side panels.

All cooling coils are mounted over a double sloped drain pan. The cooling coil rests on coil supports located over the drain pan. The drain pan extends beyond the leaving side of the coil to help recover condensate. The primary drain pan also extends under the coil headers and return bends to help remove condensate from the unit. A full thickness of insulation is always provided between the drain pan and the bottom outer panel. The drain pan is sloped in two planes to promote proper condensate removal. The galvanized drain pan is coated with an antimicrobial treatment as standard to further inhibit the growth of algae and fungi. A stainless steel drain pan is also available as an option.

Coil connections always extend through the unit cabinetry, allowing for the easy connection of valves and piping. Water coil vents and drains are located outside the cabinetry.

A range of different size face area coils is available, including small, medium, medium extended, large, extended and staggered. Generally small face area coils are used for heating applications, the large face area coils are used for cooling applications with bypass, and the extended face area coils maximize the unit cross section for restricted space applications. The extended coil is intended to be used to maintain coil face velocity limitations without going to a larger cabinet size. This ultimately saves valuable floor space and money.

McQuay unit coils can be selected from a full range of coil options. With McQuay being a major manufacturer of heat transfer coils, the coil options are virtually unlimited. The McQuay contractor coil line has many different coil types, ranging from hot and cold water, evaporator and steam coils. Standard McQuay coils are ARI certified. In addition to a broad range of circuitings, fin spacing and row depths, coils can be constructed of different material types for fins, tubes, connections, and casings. This provides the ability to specify the coil to exactly meet the application needs.

For more information on McQuay coils, consult the following catalogs:

- Cooling — water/evaporator Catalog 411
- Heating — water/booster Catalog 412
- Steam — standard/distributing Catalog 413



Coil connections, vents, and drains, along with drain pan connection extend through unit cabinetry

Filters

Good air quality is of utmost importance. A key ingredient in maintaining good air quality is air filtration. To provide a wide range of filtration options, McQuay International has enlisted the help of AAF International, a worldwide supplier of complete air filtration products. Their expertise has helped McQuay air conditioning equipment provide excellent filtration components.

The McQuay Vision air handler has been designed to house flat, angular, bag, or cartridge filters. These media types range in efficiencies up to 95% and MERV* 15. In addition to offering a full range of efficiencies, the filter media can be provided with an optional antimicrobial treatment. Antimicrobial treatments are highly recommended for a complete filtration system.

An optional filter pressure gauge may be ordered with each filter section to help promote regular servicing and prevent clogging.

Both bag and cartridge filters are provided with a pre-filter. Bag and cartridge filters can be either side load or front load, and access can be from either or both sides of the filter section. An access door is provided on either side or both sides of the unit. Filters can be positioned any place in the unit, and as many filter sections as required can be used. In many health and food industries, stringent filtration is required. Often times there is a need for a filter section to be the last component in the air stream. The McQuay air handler can meet these needs by providing a filter section located as the last component in the air stream and with full sheet metal liners.

Standard filter types available with the Vision Air Handler:

5700 - 2" depth. Flat panel filter designed for heavy dust loading conditions.

PerfectPleat - MERV 6 (30% efficiency), 2" or 4" depth. Made from pleated, self supporting DuraFlex™ by AAF.

AmAir 62plus - MERV 8 (70% efficiency), 2" or 4" depth. Standard AAF AmAir pleated filter with two layers of added polypropylene laminate to increase efficiency.

VariCel II MH - 95%, 85% and 65% efficiency, 4" depth, 1" track requirement. Mini pleat filters with metal cell sides and headers that provide superior moisture resistance.

VariCel SH - 95%, 85% and 65% efficiency, 12" depth, 1" track requirement. Steel interlocked header and cell sides hold the corrugated aluminum separated pleats allowing optimum airflow. Rated UL Class 1.

VariCel V - 95%, 85% and 65% efficiency, 12" depth, 1" track requirement. Mini pleats held in a V-bank configuration providing greater airflow capacity and longer service.

DriPak2000 bag - MERV 15, 14 and 11, (95%, 85% and 65% efficiency), 36", 30" and 22" depths, 1" track requirement. Extended surface pockets made from high-loft, layered synthetic media. Rated UL Class 1 MERV 8 efficiency. DriPak2000 filters also available in 19", 15" and 12" depths.

Visit AAF's web site www.americanairfilter.com for more technical filter information.

*What is the MERV Rating?

Minimum Efficiency Reporting Value (MERV) - ASHRAE Standard 52.2-1999 entitled "Method of Testing General Ventilation Air-Cleaning Devices for Removal by Particle Size" provides a methodology for determining filter efficiency at removing various sizes of particles as the filters become loaded. There are three ranges of particle sizes that define the MERValue:

Range 1 0.30 - 1.0 mm particle size.

Range 2 1.0 - 3.0 mm particle size.

Range 3 3.0 - 10.0 mm particle size.

The Table below shows a comparison of the MERV rating to the average arrestance percentage by the older ASHRAE Std. 52.1 method:

Std. 52.2(MERV)	Approx. Std 52.1		Particle Size Range
	Dust Spot Eff.	Arrestance	
15	>95%	n/a	1
14	90-95%	>98%	1
11	60-65%	>95%	2
8	30-35%	>90%	3
6	<20%	85-90%	3

Access

The access section can be selected to meet specific application criteria. Access sections are available to be placed anywhere in a unit in a variety of depths. Sections are available in depths of 16", 24", 30", 36", 42", 48", and 54". Typically access sections are used for field installed components, air monitoring devices or to provide ample space between components.

Mixing boxes and economizers

When outside and return air mixing is needed, either a mixing box or an economizer section can be selected. Either component will regulate the amount of outside and return air supplied to the conditioned space. The mixing box or economizer can make use of free cooling by opening outside air dampers when the ambient air will help to condition the supply air stream. Additionally, dampers may be individually sized to provide better mixing.

Both the mixing box and economizer are provided with the McQuay UltraSeal™ low leak damper. This damper has one of the lowest leakage rates in the industry, maximizing energy efficiency. At 4.0" static pressure and a face velocity of 1100 fpm through the dampers, the leakage rate is less than 0.2% (it is common to specify leakage rates at higher static pressures, but dampers should not exceed 2" of static pressure). The parallel airfoil blades are hollow core and fully gasketed. Continuous vinyl seals are provided between the damper blades. Stainless steel end seals and linkage built into high strength ABS plastic endcaps provide smooth quiet operation. When mixing boxes or economizers are used on outdoor units, optional rain hoods can be factory installed.

Face and bypass dampers

Face and bypass dampers can be provided for temperature modulation, by bypassing air around the coil. The opposed blades meter varying air volumes through the coil and bypass to attain the final air temperature demanded. McQuay Vision air handlers offer only low leak dampers in their face and bypass sections.

Three styles of face and bypass sections are available. Internal bypass is available for use with medium face area coils. External bypass and external right angle are used when larger face area coils are required. The damper blades are fabricated of continuous galvanized steel with the damper rods rotating in nylon bushings. Damper shaft extensions are supplied to facilitate damper motor location.

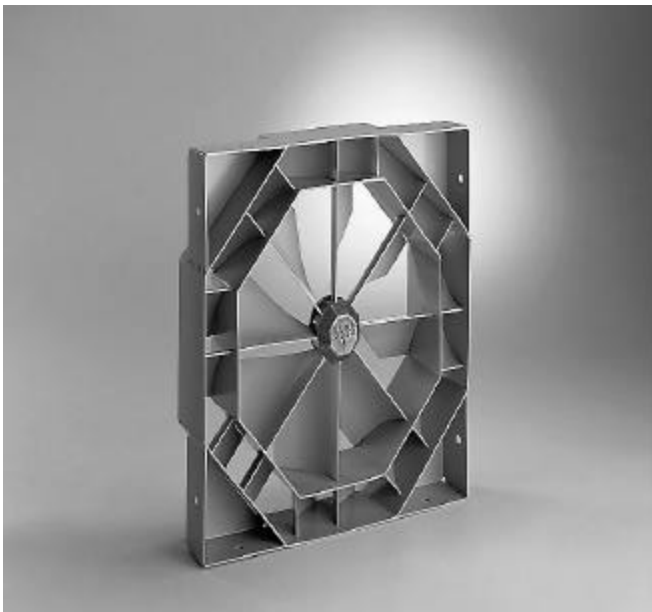


Internal face and bypass dampers

Blenders/Air Mixers

Stratification occurs from the mixing box when the airflow of two different temperature air streams do not mix completely. This incomplete mixing can continue through the air handler, and subject an unprotected coil (no glycol) to freezing temperatures, damaging the coil. With the increased minimum outdoor air requirements as identified by ASHRAE Standard 62, the likelihood for air stratification increases. An air handler must be able to handle the required amount of outdoor air, regardless of temperature, without risking damage to the coil.

Blenders/air mixers help to provide protection for coils from freeze-up due to stratification. Blenders/air mixers add additional turbulence to the passing air streams, boosting the air velocity for improved mixing. Proper distance is provided immediately downstream to give the air streams enough time to fully mix before reaching the next air handler component. Because blenders/air mixers are static devices, they require no maintenance. Different blender/air mixer lengths can be selected to satisfy either the acoustic, space, pressure drop, or initial cost requirements. The Vision SelectTOOLS™ Software will select the appropriate blender/air mixer for the application.



Blender

Attenuators

The issue of objectionable sound or 'noise' can no longer be treated lightly. Building occupants have become increasingly conscious of the quality of their environments with low sound levels being a key criteria. Studies have confirmed improved productivity when workers are performing in sound controlled environments. Consequently, building owners, engineers, and architects have become aware of the economic benefits associated with quiet environments.

McQuay's Vision air handler has been designed to provide the quietest sound levels possible. Factory installed attenuators are available for the discharge or return sections of the air handling unit to meet the most stringent sound attenuation requirements. Different attenuator lengths can be selected to satisfy either the acoustic, space, pressure drop, or initial cost requirements. Quality construction along with aerodynamic design, gives reliable performance, low pressure drop and low initial cost. The Vision SelectTOOLS™ Software will choose the correct attenuator for the application.



Attenuator section

Starters and VFDs

To provide safety, motor efficiency, and flexibility for the Vision air handler, a variety of factory-installed motor control options are available. These options include: Disconnect Switch, Starters, Combination Starter/Disconnect, Variable Frequency Drive (VFD) with Disconnect Switch, and VFD with Disconnect Switch & Bypass. Line reactors are available on both VFD with Disconnect Switch and VFD with Disconnect Switch & Bypass.

A unit mounted disconnect switch provides the means for disconnecting power in sight from the motor when required by the National Electric Code (NEC). The switch is a heavy-duty type with a non-fused disconnect switch in a NEMA 1 enclosure. A fused disconnect is also available.

Factory mounted starters offer remote start/stop operation and overload protection for an individual fan motor with minimum field cost to wire line and control circuit power to the unit. All starters provided are NEMA rated. Additional options include auxiliary contacts, control transformer, selector switch, push buttons, and pilot lights. A combination starter is a starter packaged with a safety switch. These motor controllers provide remote start/stop operation as well as disconnecting means in sight of the controller.

A VFD provides adjustable speed control of a single fan motor. Factory mounted outside the fan cabinet, the VFD provides not only a lower cost than field installation, but also an optimal installation location. VFDs are preprogrammed and include a keypad for local or remote control.

A unit installed VFD with disconnect provides local or remote speed control for a single fan motor with disconnecting means within sight when required by the NEC. A non-fused disconnect is offered packaged with the VFD in a separate NEMA 1 enclosure. Installation is limited to the connection of power and control wires at the device.

Packaged VFD with disconnect and bypass offers added motor control at the unit for critical fan applications. With bypass control, power to the VFD can be switched to allow full operation of the fan motor in cases of VFD failure. VFDs with disconnect switch, bypass and line reactors provide harmonic control when needed.

Energy Recovery

Maintaining acceptable indoor air quality is generally accomplished by introducing ventilation from outdoors. The air must be conditioned to match the indoor space requirements. Heat wheels and fixed plate heat exchangers are available as factory-installed options for Vision air handlers. These energy recovery components can recover 50% or more of the energy normally exhausted from a building. These devices capture heat from exhaust air as it passes through the air handler and transfer it to the supply air stream, reducing the cost of heating or cooling the outside air. Energy recovery components do this by transferring energy from a warm air stream to a colder air stream, heating cold outside air during the winter and cooling hot air during the summer.

The fixed plate creates a cross flow as it collects heat and conducts it to the other side of the plate. The fixed plate has no moving parts, is easy to clean, has a low pressure drop, can be sealed against cross contamination, and is offered in a wide range of configurations.

The heat wheel rotates at low speeds, capturing and transferring both sensible (heat) energy and latent (moisture) energy. The ability to transfer both sensible and latent energy gives the heat wheel several advantages. First, the required capacity of ventilation equipment is significantly reduced. Additionally, the heat wheel works at lower temperatures without frosting. The supply air from the heat wheel is not near saturation, and moisture in the ductwork is not an issue. Further, no condensate pan or drain is required. Finally, heat wheels provide humidification so that the humidifier can be downsized. It also helps to keep humidity in spaces where humidification is not applied, providing greater comfort to these zones.

Both types of energy recovery devices come with double wall construction with solid liners, and are configured on the inlets and outlets with splice collars to match the adjacent sections. The fixed plate requires two drip pans, which can be either microbial resistant galvanized steel or stainless steel.

Electric Heaters

Electric heaters extend the wide versatility of the Vision air handler. With negligible air pressure drop, accurate controllability, light weight, easy serviceability and inherent freeze protection, electric heaters are valuable alternatives to conventional steam and hot water heating coils.

ETL-approved electric heaters are available on all standard sizes (003-090) for horizontally-mounted draw through units in both left and right hand configurations. All units are open wire style construction, with automatic and manual backup limit controls, air switch, stainless steel terminals, power on pilot light, magnetic contactors and integral control boxes. Safety interlock switches, step controllers and vernier silicon controlled rectifiers (SCRs) are available options.

The heater requires its own electric service. Heaters are available in 208V, 240V, 480V and 600V (all 3 phase) and use an internal 24-Volt control circuit. Standard ranges of kW are available for each heater size and are designed to give an approximate temperature rise from 20°F to 60°F, depending on the airflow through the unit. Typical pressure drops range from .01-.04 inches of water, depending on the air velocity and number of rows of heating elements.

An integral control box with optional door handle power disconnect is included with the electric heater. Insulated panels are factory installed behind the control box. Any standard motor may be used, although the discharge air from the electric heater must not exceed 104°F since the rated ambient temperature of the motor will be exceeded. The heater is assembled into a separate section of the Vision air handler. Standard section widths include: 30", 34", 42" and 46" and vary depending on unit size, control type and kW used.

Air handler selection

Selection of the flexible Vision air handler depends on many different criteria. To simplify this process, McQuay International has designed an innovative computerized selection program. This program is called the Vision SelectTOOLS™ Software. It allows the user to develop an air handler from the ground up, and obtain all of the detail required for proper design and specification. The user can have a unit designed in a matter of minutes.

The program is completely integrated. All input data is carried through the selection process and considered as calculations are made. Because the program is integrated, the opportunity for errors is reduced. The software guides the user through the selection process. On-line editing helps select only viable options. The program is provided so that even the less experienced user can select air handlers accurately.

The design of an efficient air handler system is dependent on accurate system design and proper equipment selection. Factors which affect the unit selection include applicable codes, ventilation requirements, heating and cooling space loads, acceptable temperature differentials, thermal media and installation limitations. The selection of the unit can be broken down into four steps; unit type and size, coils, accessories, and fan and motor requirements.

The unit is generally selected based on the air volume required and the desired face velocity through the cooling coil. For cooling coils, 400-600 feet per minute is considered the optimum face velocity range for dehumidification and the prevention of any moisture carryover. The Vision SelectTOOLS™ Software will select the unit size based on air flow and face velocity requirements. Once the unit size is determined, the coils and all accessory components are selected. Once all components and coils are selected (identifying the total internal component pressure drop), the fan can be selected.

Selection of coils

The Vision air handler provides the ability to select from multiple face area coils per unit size. Once the coil size is selected, the row and fin requirements can then be determined based on performance criteria. McQuay International offers an extensive line of coil types and circuitings. This wide variety of circuiting, row, and fin spacing, in addition to different material types, provides a coil selection that will handle the load required.

Heating, cooling, and combination cooling and reheat sections are available. Coil sections come in many different depths to accommodate multiple rows of coils, and to provide access on the leaving air side or between coils for cleaning and inspection of the drain pan. Access in the coil section can be minimized or maximized depending on the space available and job needs. Coil sections can be placed as needed in the unit, and as many sections as required for conditioning of air can be provided.



WARNING

SMOKE CONTROL AND MANAGEMENT SYSTEMS

Improper smoke or fume air handling can result in severe personal injury or death. A registered Professional Engineer must design and approve the air conditioner and air handler application to make sure smoke and fume control meet local fire codes and NFPA requirements for the specific building application.

Selection of accessories

The Vision air handler product line offers a complete selection of component and section types, in a variety of unit arrangements and configurations. These component options are available to provide proper mixing, filtration, and temperature control of the air to meet the specific application requirements.

The outside and/or return air can be brought into the unit through a plenum, mixing box, or economizer. For mixing of the two air streams, dampers are required to modulate and direct outside and return air. This can be accomplished through the use of a mixing box or economizer. Both section types use the McQuay patented UltraSeal™ low leak dampers. Blenders/air mixers are also available to provide proper mixing of two air streams, to prevent stratification and avoid damage to equipment due to freezing temperatures.

To promote good air filtration, many different filter media types and arrangements are offered. Filters can be provided in angular or flat filter racks, a variety of media efficiencies, and with or without a prefilter. The filter section can be located anywhere in the unit, to filter air as it enters the unit or in a final filter arrangement before it leaves the unit. Bag or cartridge filters can be either front or side loading.

Also available are many different size access sections, for field installed components or to provide access between components. Other options include diffusers, plenums, face and bypass dampers, sound attenuators, and blenders/air mixers.

Standard access doors open outward for maximum accessibility to internal components. On pressurized sections, most doors will be provided as inward opening. Use of inward opening doors helps maintain the extremely low cabinet leakage rate of the Vision unit, and prevents doors from being inadvertently opened when the cabinet is under pressure.

Fan selection

Fan selection requires an accurate calculation of the resistance to the airflow through the entire system. This total resistance consists of the sum of two parts, the external and internal static pressure. The external static pressure is the static pressure found in the distribution system, external to the air handler. The internal unit resistance is the sum of the resistance of the coils and various other unit components and accessories. Component pressure drops are listed for specific air flows in the catalog. Use the Vision SelectTOOLS™ Software to determine internal component pressure drops of the system. Once the total static pressure is known, the software will identify the fans available to properly handle the air flow and static pressure for the system. The software will generate a full fan curve based on the fan selection and point of operation.

Quick select table

DESCRIPTION	UNIT SIZE								
	003	004	006	008	010	012	014	017	021
Air flow range (CFM)	900-2500	1200-3100	1700-4600	2200-6000	2900-7700	3600-9700	4200-11200	5000-13500	6000-16000
CFM @ 500 ft./min. through large face area coil	1550	1950	2850	3750	4800	6050	7000	8400	10050
Height x width (inches)	26 x 38	30 x 40	30 x 52	34 x 58	36 x 64	42 x 66	42 x 74	46 x 80	52 x 82
SECTION TYPE	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs
Fan Section									
Largest housed fan and motor available with top horizontal discharge	32/378	32/382	36/623	40/821	40/845	46/1060	46/1087	50/1424	52/1569
Largest inline fan and motor available	N/A	N/A	N/A	N/A	N/A	44/956	44/987	50/1314	54/1577
Largest belt drive plenum fan and motor	N/A	N/A	N/A	N/A	34/754	42/1075	42/1168	48/1584	52/1843
Largest direct drive plenum and motor	N/A	N/A	N/A	20/614	20/699	54/1286	54/1337	54/1406	54/1463
Mixing Box (includes damper weight)									
Mixing box only	20/177	20/171	20/201	20/235	22/289	24/305	24/329	26/371	30/427
Mixing box w/ 2" flat filter	24/208	24/225	24/268	24/312	26/380	28/411	28/438	30/499	34/568
Mixing box w/ 2" angular filter	42/311	42/345	42/403	42/457	44/535	46/569	46/613	48/723	52/800
Economizer (includes damper weight)									
	40/267	40/288	40/334	40/381	44/470	48/495	48/530	52/600	60/699
Blender (largest Kees and BPV)									
Kees	18/126	20/147	24/185	26/220	28/251	34/307	36/337	38/382	42/444
Blender Products IV	18/125	22/147	26/196	30/238	34/281	38/321	40/365	46/481	48/527
Side Load Filter Section									
Flat 2" & 4"	12/118	12/128	12/150	12/169	12/187	12/207	12/219	12/240	12/255
2" angular	32/238	30/230	30/267	30/293	30/320	30/340	30/366	30/419	30/437
Cartridge (12" deep w/2" pre-filter)	22/184	22/208	22/245	22/291	22/317	22/350	22/376	22/427	22/475
Bag (42" w/2" pre-filter)	42/265	42/290	42/338	42/385	42/422	42/452	42/484	42/526	42/567
Face & Bypass									
Internal	12/144	12/158	12/189	12/219	12/248	12/277	12/302	12/334	12/362
External	18/241	18/246	18/265	20/313	22/360	24/414	24/445	26/415	30/605
Coil Sections (includes coil weight)									
Heating only (2 row water)	12/201	12/179	12/210	12/242	12/276	12/306	12/331	12/366	12/412
Cooling only (4 row water)	18/216	18/255	18/307	18/357	18/406	18/463	18/582	18/595	18/745
Cooling only (6 row water)	24/324	24/333	24/408	24/450	24/518	24/592	24/680	24/765	24/858
Cooling & reheat (12 row cig. & 1 row htg.)	36/438	36/522	36/649	36/769	36/907	36/1067	36/1193	36/1357	36/1546
Vertical cooling only (6 row water)	30/322	30/378	30/451	30/522	36/630	36/729	36/813	36/906	42/1057
Vertical cooling & reheat (12 row clg. & 1 row htg.)	60/638	60/737	66/921	66/1080	66/1232	66/1433	72/1587	72/1799	72/2016
Multizone Coil Sections (includes coil weight)									
3 deck hor w/dampers (5 row clg. & 1 row htg.)	N/A	N/A	40/810	42/929	42/1028	52/1067	52/1360	54/1512	56/1905
Access Sections									
16" deep	16/177	16/108	16/145	16/161	16/174	16/186	16/199	16/215	16/223
24" deep	24/140	24/135	24/174	24/192	24/207	24/223	24/237	24/255	24/268
30" deep	30/159	30/170	30/196	30/216	30/232	30/250	30/266	30/286	30/301
36" deep	36/179	36/193	36/220	36/244	36/261	36/282	36/299	36/322	36/340
42" deep	42/198	42/213	42/242	42/267	42/286	42/309	42/237	42/353	42/374
48" deep	48/221	48/236	48/269	48/298	48/320	48/345	48/415	48/447	48/468
54" deep	54/240	54/253	54/287	54/334	54/358	54/384	54/408	54/439	54/466
Diffuser									
Diffuser with housed fan	10/86	10/92	10/110	12/133	12/143	16/175	16/185	16/201	16/216
Diffuser with inline fan	N/A	N/A	N/A	N/A	N/A	18/187	18/198	18/214	22/259
Attenuator									
Short	40/304	40/315	40/359	40/483	40/541	40/618	40/661	40/749	40/830
Medium	52/382	52/423	52/523	52/615	52/690	52/891	52/973	52/1105	52/1248
Long	64/444	64/513	64/670	64/789	64/884	64/1078	64/1178	64/1340	64/1551
Supply or Return Plenum									
Top, bottom or end opening	14/120	16/141	16/167	18/197	20/223	22/253	22/273	24/309	28/354

Quick select procedure

The quick select chart is provided to help quickly approximate a unit size based on air flow. This will establish the height, width, length, and weight for preliminary job requirements. The quick select chart does not represent the full range of section depths available for each section type, but will provide a typical section depth. For submittal purposes, the Vision Select-TOOLS™ Software must always be used to properly size coils,

fans, and other components.

To use the quick select chart, compare the application operating air flow requirement to the airflow ranges shown for each unit size. The air flow ranges shown on the quick select chart are based on coil and fan capabilities. For each unit size there are multiple coil sizes and numerous fan sizes available.

Quick select table - continued

DESCRIPTION	UNIT SIZE								
	025	030	035	040	050	065	080	085	090
Air flow range (CFM)	7300-19400	8500-22500	10000-26500	11500-31000	15000-40000	20000-54000	21500-57500	23100-61600	24600-65600
CFM @ 500 ft./min. through large face area coil	12150	14150	16700	19300	24500	33300	35900	38450	41000
Height x width (inches)	60 x 86	60 x 98	66 x 102	68 x 116	80 x 120	92 x 136	98/136	104/136	110/136
SECTION TYPE	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs	depth/wgt. inches/lbs
Fan Section									
Largest housed fan and motor available with top horizontal discharge	58/1911	58/2142	58/2198	70/3051	80/4115	92/5514	92/6343	92/6412	92/6472
Largest inline fan and motor available	64/2353	64/2444	70/2923	70/3176	90/5088	96/6048	96/6997	96/7229	96/6995
Largest belt drive plenum fan and motor	56/2431	58/2701	62/3104	62/3566	70/4723	82/6970	82/7315	82/7378	82/7427
Largest direct drive plenum and motor	66/2182	78/3289	84/3641	86/4298	86/4494	86/4867	N/A	N/A	N/A
Mixing Box (includes damper weight)									
Mixing box only	32/612	32/733	36/825	36/925	40/1078	46/1337	50/1816	54/1927	56/2013
Mixing box w/ 2" flat filter	36/780	36/923	40/1063	40/1235	44/1397	50/1765	54/2045	58/2162	60/2271
Mixing box w/ 2" angular filter	54/1044	54/1199	58/1354	58/1616	62/1806	68/2184	72/2585	76/2804	78/2900
Economizer (includes damper weight)									
	64/933	64/1073	72/1228	72/1340	80/1622	92/2057	100/3136	108/3349	112/3488
Blender (largest Kees and BPIV)									
Kees	46/707	48/847	58/1021	58/1122	68/1324	76/1727	80/2568	84/2719	84/2788
Blender Products IV	52/733	60/963	64/1067	70/1248	74/1496	88/1914	88/2716	92/2841	92/2914
Side Load Filter Section									
Flat (2" & 4")	12/392	12/500	12/530	12/592	12/631	12/737	12/809	12/822	12/854
2" angular	32/613	32/730	32/791	32/941	32/1003	32/1176	32/1303	32/1374	32/1390
Cartridge (12" deep w/2" pre-filter)	22/636	22/789	22/853	22/953	22/1025	22/1233	22/1351	22/1392	22/1500
Bag (30 or 36" deep w/2" pre-filter)	42/765	42/950	42/1006	42/1119	42/1216	42/1429	42/1587	42/1643	42/1669
Face & Bypass									
Internal	12/522	12/638	12/689	12/788	12/918	12/1086	12/1252	12/1278	12/1308
External	32/807	32/945	34/1052	38/1196	44/1370	50/1692	54/2175	56/2306	58/2366
Coil Sections (includes coil weight)									
Heating only (2 row water)	12/619	12/689	12/749	12/836	12/995	12/1198	12/1217	12/1441	12/1488
Cooling only (4 row water)	18/992	18/1214	18/1344	18/1508	18/1758	18/2229	18/2570	18/2734	18/2799
Cooling only (6 row water)	24/1206	24/1343	24/1485	24/1689	24/1968	24/2514	24/2538	24/3036	24/3168
Cooling & reheat (12 row clg. & 1 row htg.)	36/2075	36/2363	36/2654	36/3037	36/3586	36/4608	36/5287	36/5515	36/5558
Vertical cooling only (6 row water)	48/1511	48/1671	54/1878	N/A	N/A	N/A	N/A	N/A	N/A
Vertical cooling & reheat (12 row clg. & 1 row htg.)	72/2790	72/3108	72/3459	N/A	N/A	N/A	N/A	N/A	N/A
Multizone Coil Sections (includes coil weight)									
3 deck hor w/dampers (5 row clg. & 1 row htg.)	60/2110	60/2320	70/2685	70/3035	76/3601	88/5335	N/A	N/A	N/A
Access Sections									
16" deep	16/355	16/412	16/434	16/521	16/552	16/621	16/783	16/796	16/808
24" deep	24/409	24/469	24/495	24/585	24/621	24/696	24/922	24/938	24/953
30" deep	30/450	30/512	30/581	30/635	30/673	30/753	30/1025	30/1044	30/1063
36" deep	36/497	36/560	36/633	36/690	36/733	36/820	36/1142	36/1196	36/1185
42" deep	42/536	42/670	42/748	42/818	42/867	42/1003	42/1364	42/1389	42/1415
48" deep	48/641	48/716	48/757	48/871	48/952	48/1068	48/1486	48/1514	48/1543
54" deep	54/661	54/775	54/819	54/892	54/949	54/1063	54/1530	54/1559	54/1589
Diffuser									
Diffuser with housed fan	24/400	24/496	24/520	30/613	30/659	30/748	30/1041	30/1060	30/1080
Diffuser with inline fan	26/423	26/519	28/562	30/628	36/729	38/933	38/1289	38/1312	38/1336
Attenuator									
Short	40/1251	40/1434	40/1567	40/1746	40/1992	40/2681	40/3005	40/3148	40/3290
Medium	52/1697	52/1884	52/2174	52/2428	52/2845	52/3681	52/4026	52/4229	52/4431
Long	64/2077	64/2300	64/2605	64/2912	64/3426	64/4455	64/4872	64/5122	64/5373
Supply or Return Plenum									
Top, bottom or end opening	30/527	30/639	32/698	32/772	38/1018	42/1239	48/1957	52/2082	54/2172

Notes: Approximate shipping weights do not include base rails, except on sizes 025-065, which require a 6" base rail.

Unit sizes 080 and larger require an 8" base rail and double wall construction.

Coil weights based on aluminum fins and 8 fins per inch.

Except for unit sizes 080 - 090, approximate shipping weights do not include double wall construction. Weights shown on pages 10 and 11 do include double wall construction for all units.

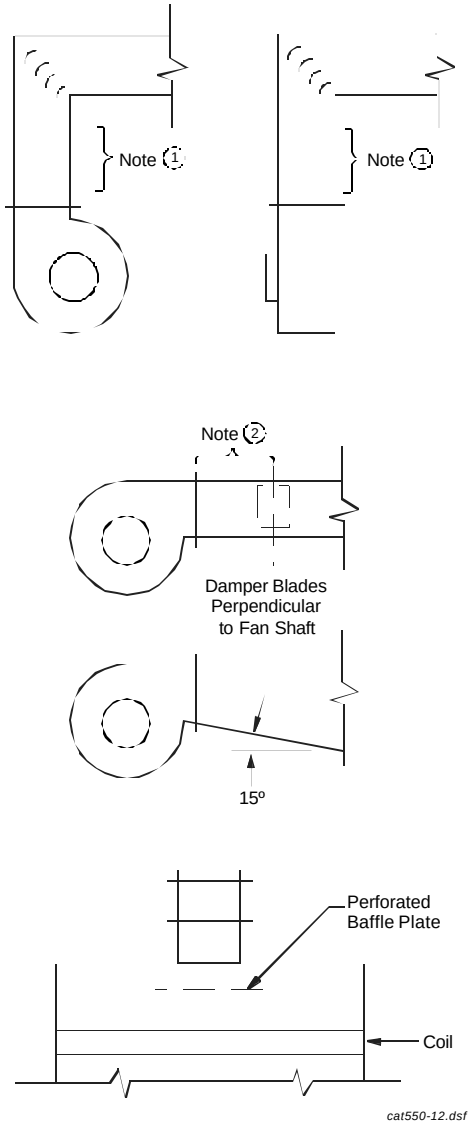
CFM ranges cover coil face velocities between 300-800 feet per minute, depending on the application of the unit. A CFM rating at 500 feet per minute through the large face area coil is rated as a reference point for cooling applications.

Once the unit size is determined, read the height and width from the chart. The approximate unit length and weight is determined by moving down the size column and summing the

depths and weights of each section that will make up the unit. This will provide an approximate size and weight for preliminary job requirements. It is important to remember that the Vision SelectTOOLS™ Software will optimize the selection criteria, a different sized unit may be selected once all the selection criteria is identified and calculated.

Application considerations

Figure 1. Discharge duct layout

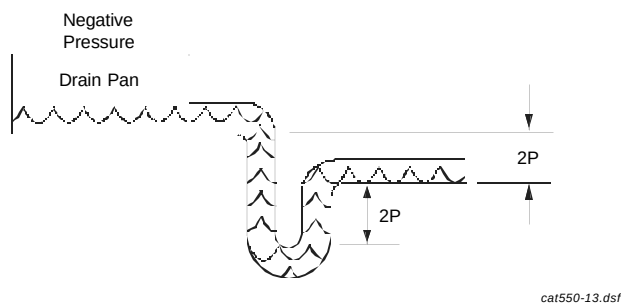


Notes:

- ① Elbows should not be closer than 1-1/2 to 2-1/2 times the largest dimension of fan discharge opening.
- ② Dampers should be placed at least 3 fan diameters downstream of the fan discharge.

Figures reprinted with permission from the
1979 ASHRAE Guide and Data Book (Equipment).

Figure 2. Drain pan taps



Installation flexibility

McQuay central station air handlers feature sectionalized design to provide maximum installation flexibility. Fan, coil, filter, mixing box, face and bypass, and access components allow the design flexibility of built-up systems with the cost advantage of factory fabricated units. Units can be shipped from the factory in as few or as many sections as required according to job site condition.

Mounting and access

Whether units are floor or ceiling mounted, care should be taken to keep the supporting structure level and rigid enough for satisfactory unit operation. Ideally, a heavy concrete slab should be used for floor mounted units, and main support beams for ceiling supported hung units. Ceiling suspended units must be trapeezed from the unit base rail, or field supplied materials. Long floor or ceiling spans should be avoided.

Units should be located so as to provide proper access for routine service. Clearance for filter removal on both sides of the filter section is usually necessary. Clearance should be provided as required for access panels. Room should be allowed for coil removal. Cooling units require clearance for a trap in the drain pan line.

Access to the interior of the Vision air handler is provided by hinged access doors or removable panels. For access between components, versatile access sections feature hinged access doors on either, or both sides.

Ductwork

Good ductwork layout will minimize system resistance and sound generation. Duct connections to and from units should allow straight, smooth airflow. Sharp turns in the fan discharge should be avoided, particularly turns opposed to wheel rotation. Turning vanes should be used. Discharge plenums or any abrupt change in duct size should be avoided. See Figure 1 for good fan outlet practices.

Piping and drain pan traps

Piping should be in accordance with accepted industry standards. Undue stress should not be applied at the connection to coil headers. Pipe work should be supported independently of the coils with adequate piping flexibility for thermal expansion. Drain lines and traps should be run full size from the drain pan connection. Drain pans should have traps to permit the condensate from the coils to drain freely. On a draw-through unit, the trap depth and the distance between the trap outlet and drain pan outlet should be twice the negative static pressure under normal unit operation.

Vibration isolation

To help keep noise and vibration compatible with the intended use of the conditioned air space, good acoustical and vibration engineering practices should be applied during the early stages of design.

Since most applications require vibration isolation, the Vision air handler is available with factory installed internal isolation. Internally isolated units feature spring or rubber in shear isolators sized specifically for each fan wheel and unit size. Because internal isolation minimizes vibration at the source (fan and motor), there is seldom a need for flexible connections on ductwork or coil piping. Internal isolation provides an opportunity for significantly reduced installation costs.

Multizone air handler applications

Blow-through air handlers are available in singlezone, two-deck, and three-deck configurations. The two- and three-deck units are offered with or without zone dampers. All unit configurations include a perforated plate fan discharge diffuser to provide even airflow downstream of the fan.

Multizone and dual duct air handlers typically provide comfort conditioning by distributing a constant air volume at variable temperature. In a typical system a portion of the air is heated by passing through the heating coil and the balance is cooled by the cooling coil. The heated and cooled airstreams are then mixed in the required proportion to provide the optimum temperature air to the conditioned space.

For dual duct applications, a pair of ducts bring heated and cooled air to the air mixing terminal boxes where the airstreams are mixed. By adding zone dampers to the dual duct unit, the air mixing takes place at the unit discharge and only one duct is required to distribute conditioned air to the building. The air mixing terminal boxes are also eliminated.

By adding a third bypass deck to the hot and cold decks, a triple deck multizone is created. The triple deck configuration offers significant energy conservation opportunities by allowing return or outside air to bypass both coils. The thermal inefficiency of mixing heated and cooled air is eliminated by the addition of the bypass deck. Bypass air is mixed with heated air for building zones that require heating. Bypass air is mixed with cooled air for building zones that require cooling.

Multizone air handling systems result in an absence of water, steam and condensate drain piping, wiring, electrical and mechanical equipment in the conditioned space for more usable commercial floor area and higher rental income.

Sound

The unit inlet, outlet, and radiated sound levels for each octave band based on your specific application, are calculated by the Vision SelectTOOLS™ Software. Sound performance data is derived from testing performed in accordance with the AMCA Standard 300. The effects of various components, casework, and unit configurations have been taken into account.

Air supply systems and fan laws

An air supply system consists of an air handler cabinet, heat exchanger, filters, ductwork, grilles and registers used to distribute air throughout the building. The system is independent of the fan used to supply the system.

The resistance of the system, referred to as static pressure (SP), is dependent upon the quantity of air (CFM) that is moved through it. The air quantity is determined by the cooling, heating and ventilating requirements.

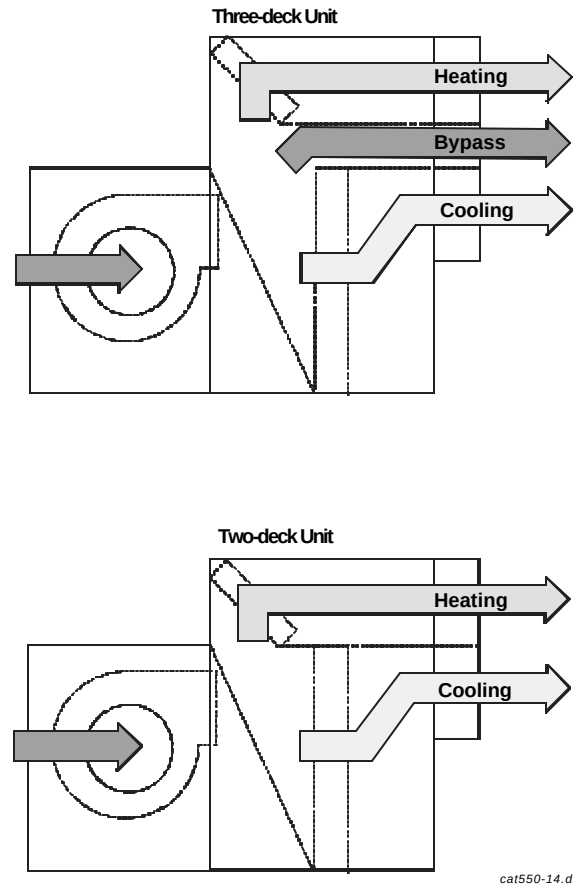
For any system, the static pressure will vary directly as the square of the air quantity. This relationship between CFM and SP establishes the system curve for that system and may be expressed as follows:

$$\left(\frac{CFM_1}{CFM_2}\right)^2 = \frac{SP_1}{SP_2} \quad \text{or} \quad SP_2 = SP_1 \left(\frac{CFM_2}{CFM_1}\right)^2$$

The system curve is unique for a particular system configuration. Any change to the system caused by dirty filters, damper changes, etc., will result in a new system curve.

For fans operating at low pressures (less than 10" W.G.), the effects of air compression are negligible. Disregarding air compression allows fan operation in a fixed system to be expressed by simple relationships. These relationships are known as fan laws and may be used to calculate the effects of fan speed and air density changes on this system.

Figure 3. Blow-through air handler configurations



1. The flow rate varies directly with the change in fan speed:

$$\frac{CFM_1}{CFM_2} = \frac{RPM_1}{RPM_2} \quad \text{or} \quad CFM_2 = CFM_1 \left(\frac{RPM_2}{RPM_1}\right)$$

A 10% increase in fan speed will give a 10% increase in air quantity.

2. The static pressure varies as the square of the change in fan speed:

$$\frac{SP_1}{SP_2} = \left(\frac{RPM_1}{RPM_2}\right)^2 \quad \text{or} \quad SP_2 = SP_1 \left(\frac{RPM_2}{RPM_1}\right)^2$$

A 10% increase in fan speed will give a 21% increase in static pressure.

3. The fan brake horsepower varies as the cube of the change in fan speed:

$$\frac{HP_1}{HP_2} = \left(\frac{RPM_1}{RPM_2}\right)^3 \quad \text{or} \quad HP_2 = HP_1 \left(\frac{RPM_2}{RPM_1}\right)^3$$

A 10% increase in fan speed will give a 33% increase in fan horsepower.

4. System static pressure and brake horsepower are directly proportional to the air density:

$$SP_2 = SP_1 \left(\frac{\text{density}_2}{\text{density}_1}\right) \left(\frac{RPM_2}{RPM_1}\right)^2$$

$$HP_2 = HP_1 \left(\frac{\text{density}_2}{\text{density}_1}\right) \left(\frac{RPM_2}{RPM_1}\right)^3$$

Consequently, the static pressure and brake horsepower decrease with an increase in air temperature or higher altitude, and increase with a decrease in air temperature or lower altitude.

To determine fan performance for temperatures and altitudes other than standard (70°F, 0 ft. altitude), the static pressure must be adjusted by the density ratio before the fan RPM and BHP requirements can be determined. Density ratios are expressed as temperature and altitude conversion factors in Table 1.

Table 1. Temperature and altitude conversion factors

AIR TEMP. (°F)	ALTITUDE (FEET)								
	0	1000	2000	3000	4000	5000	6000	7000	8000
-20	1.20	1.16	1.12	1.08	1.04	1.00	.97	.93	.89
0	1.15	1.10	1.08	1.02	.99	.95	.92	.88	.85
20	1.11	1.06	1.02	.98	.95	.92	.88	.85	.82
40	1.06	1.02	.98	.94	.91	.88	.84	.81	.78
60	1.02	.98	.94	.91	.88	.85	.81	.79	.76
70	1.00	.96	.93	.89	.86	.83	.80	.77	.74
80	.98	.94	.91	.88	.84	.81	.78	.75	.72
100	.94	.91	.88	.84	.81	.78	.75	.72	.70
120	.92	.88	.85	.81	.78	.76	.72	.70	.67
140	.89	.85	.82	.79	.76	.73	.70	.68	.65
160	.85	.82	.79	.76	.74	.70	.68	.65	.63
200	.80	.77	.75	.72	.69	.67	.64	.62	.60
250	.75	.72	.69	.67	.65	.62	.60	.58	.56

Fan and motor heat

Motor and drive heat - The total energy input to any fan motor is consumed in two ways - by heat dissipated through the motor frame and by work output. The amount of heat dissipated by the motor is a function of its operating efficiency:

$$\text{Motor heat} = \text{input} \times (1 - \text{motor efficiency})$$

A small amount of the motor work output is dissipated by the drive mechanism, which also results in a heat gain. Belt drive losses are a function of belt tension and number of belts as well as power transmitted. Typical belt drive losses range from 2% to 6% of BHP.

Whether motor and drive heat gain become part of an air handling system cooling load depends on the motor location relative to the conditioned space. For air handlers with internal motors, the motor and drive are within the conditioned space. Therefore, the motor and drive add heat to the system. This heat must be subtracted from the cooling capacity and added to the heating capacity of the unit.

Fan heat generation - All of the power input to a fan results in heat gain which must be considered as a cooling load. The amount of heat generated is directly proportional to the fan BHP:

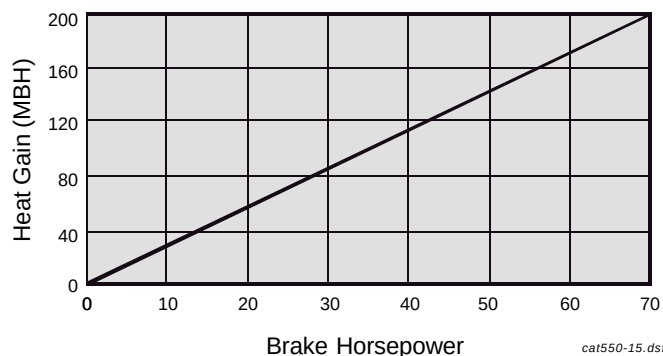
$$\text{Fan heat (Btuh)} = \text{BHP} \times 2545$$

Much of this heat generation occurs within the fan itself. Fans are not 100% efficient, and the energy losses which occur are converted directly into heat. The work done by the fan on the airstream increases the temperature, pressure and velocity of air. The heat of compression required to raise the airstream to this increased energy level is also a heat gain.

As the air travels throughout the building, its energy is deteriorated by friction, resulting in pressure drop. This is also heat gain, but it does not result in temperature rise because the air expands as the pressure is reduced. The expansion is a cooling process which offsets the heat generated by friction.

Typical fan and motor heat values are given in Figure 4.

Figure 4. Fan and motor heat gain



Air modulation

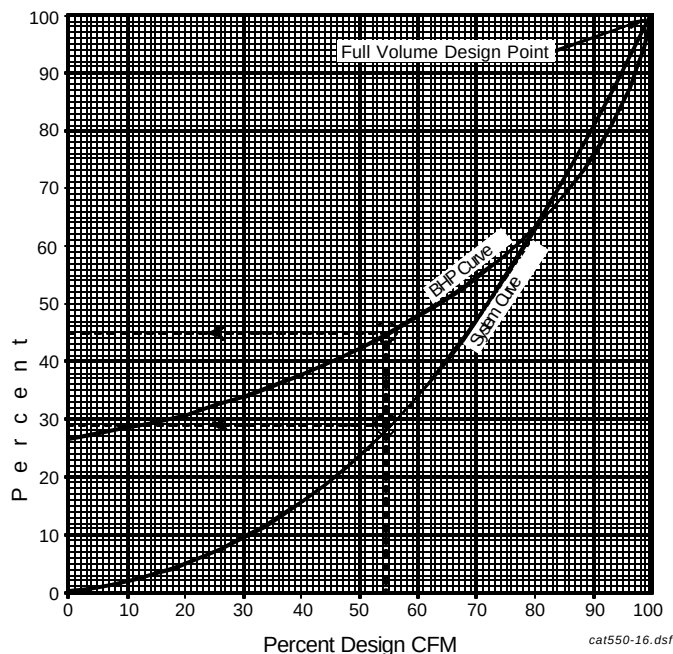
One of the most common methods of fan modulation is the use of inlet guide vanes. Fan volume reduction with inlet vanes is accomplished by pre-spinning the air in the direction of fan rotation. The effect of pre-spinning results in decreased air delivery, static pressure and brake horsepower. For each position of inlet guide vanes, a new fan curve is created. Brake horsepower reductions cannot be read directly off the fan curve on inlet vane applications because a new fan curve is generated as the inlet vane closes.

It is also difficult to estimate turn down capability on inlet vane applications. Any time a VAV system with terminal boxes is controlled by a static pressure sensor, a system resistance curve is developed which passes through the design operating point and a minimum static pressure control point. This system curve will affect where the fan will cross into the unstable operating range. An illustration of inlet vane turn down is presented in Figure 5.

Inlet vanes operate most efficiently when the fan is chosen at or near peak efficiency. Inlet guide vanes offer good power savings, wide modulation range, and low maintenance.

Note: Fan performance with inlet vane control at other than wide open position is not within the scope of our central station air handler certification program with ARI.

Figure 5. Inlet vane control reduction factors



Follow percent of design CFM up to system curve and left for percent of design static pressure. Follow percent of design CFM up to BHP curve and left for percent of rated horsepower.

Table 3. Filter media face areas and velocity through media

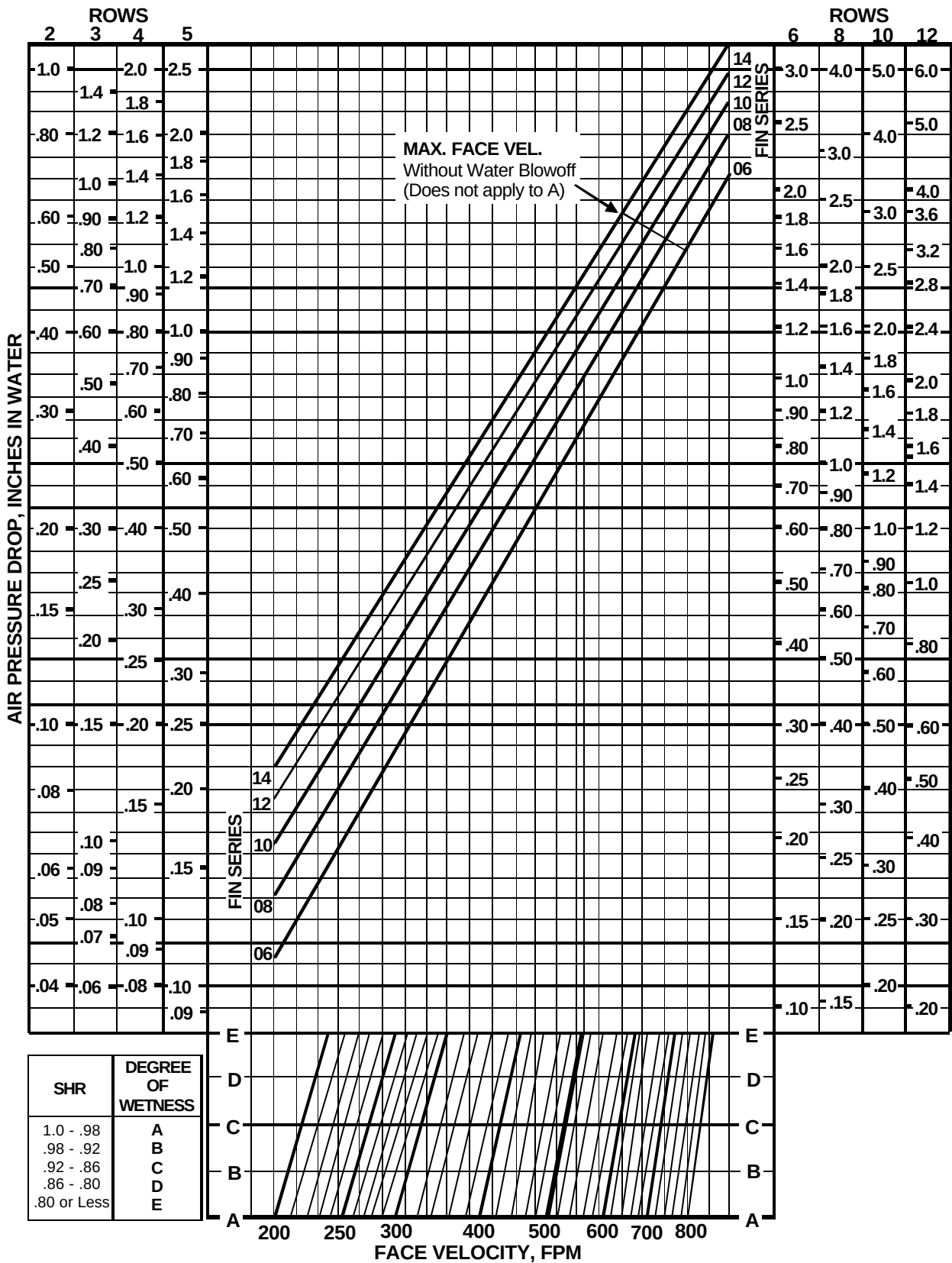
UNIT SIZE	CFM	FILTER FACE AREAS (SQ. FT.) & VELOCITIES (FPM)													
		Flat Panel		2" Angular		4" Angular		Varicel SH		Varicel II MH		Varicel V		DriPak 2000	
		Area	Vel	Area	Vel	Area	Vel	Area	Vel	Area	Vel	Area	Vel	Area	Vel
003	900	3.2	281	10.1	89	N/A	N/A	3.1	290	3.1	290	3.1	290	3.1	290
	1200	3.2	375	10.1	119	N/A	N/A	3.1	387	3.1	387	3.1	387	3.1	387
	1500	3.2	469	10.1	149	N/A	N/A	3.1	484	3.1	484	3.1	484	3.1	484
	1800	3.2	563	10.1	178	N/A	N/A	3.1	581	3.1	581	3.1	581	3.1	581
004	1200	5.7	211	11.4	105	N/A	N/A	5.6	214	5.6	214	5.6	214	5.6	214
	1600	5.7	281	11.4	140	N/A	N/A	5.6	286	5.6	286	5.6	286	5.6	286
	2000	5.7	351	11.4	175	N/A	N/A	5.6	357	5.6	357	5.6	357	5.6	357
	2400	5.7	421	11.4	211	N/A	N/A	5.6	429	5.6	429	5.6	429	5.6	429
006	1700	7.6	224	15.3	111	N/A	N/A	7.6	224	7.6	224	7.6	224	7.6	224
	2300	7.6	303	15.3	150	N/A	N/A	7.6	303	7.6	303	7.6	303	7.6	303
	2900	7.6	382	15.3	190	N/A	N/A	7.6	382	7.6	382	7.6	382	7.6	382
	3500	7.6	461	15.3	229	N/A	N/A	7.6	461	7.6	461	7.6	461	7.6	461
008	2200	8.8	250	17.7	124	17.7	124	8.1	272	8.1	272	8.1	272	8.1	272
	3000	8.8	341	17.7	169	17.7	169	8.1	370	8.1	370	8.1	370	8.1	370
	3800	8.8	432	17.7	215	17.7	215	8.1	469	8.1	469	8.1	469	8.1	469
	4600	8.8	523	17.7	260	17.7	260	8.1	568	8.1	568	8.1	568	8.1	568
010	2900	9.5	305	19.0	153	19.0	153	9.4	309	9.4	309	9.4	309	9.4	309
	3900	9.5	411	19.0	205	19.0	205	9.4	415	9.4	415	9.4	415	9.4	415
	4900	9.5	516	19.0	258	19.0	258	9.4	521	9.4	521	9.4	521	9.4	521
	5900	9.5	621	19.0	311	19.0	311	9.4	628	9.4	628	9.4	628	—	—
012	3600	13.2	273	20.3	177	20.3	177	13.1	275	13.1	275	13.1	275	13.1	275
	4800	13.2	364	20.3	236	20.3	236	13.1	366	13.1	366	13.1	366	13.1	366
	6000	13.2	455	20.3	296	20.3	296	13.1	458	13.1	458	13.1	458	13.1	458
	7200	13.2	545	20.3	355	20.3	355	13.1	550	13.1	550	13.1	550	13.1	550
014	4200	17.0	247	22.9	183	22.9	183	14.4	292	14.4	292	14.4	292	14.4	292
	5400	17.0	318	22.9	236	22.9	236	14.4	375	14.4	375	14.4	375	14.4	375
	6600	17.0	388	22.9	288	22.9	288	14.4	458	14.4	458	14.4	458	14.4	458
	8800	17.0	518	22.9	384	22.9	384	14.4	611	14.4	611	14.4	611	14.4	611
017	5000	19.0	263	35.7	140	36.0	139	18.9	265	18.9	265	18.9	265	18.9	265
	6700	19.0	353	36.0	186	36.0	186	18.9	354	18.9	354	18.9	354	18.9	354
	8400	19.0	442	36.0	233	36.0	233	18.9	444	18.9	444	18.9	444	18.9	444
	10100	19.0	532	36.0	281	36.0	281	18.9	534	18.9	534	18.9	534	18.9	534
021	6000	23.2	259	38.0	158	38.0	158	23.0	261	23.0	261	20.8	288	23.0	281
	8000	23.2	345	38.0	211	38.0	211	23.0	348	23.0	348	20.8	385	23.0	348
	10000	23.2	431	38.0	263	38.0	263	23.0	435	23.0	435	20.8	481	23.0	435
	12000	23.2	517	38.0	316	38.0	316	23.0	522	23.0	522	20.8	577	23.0	522
025	7300	26.6	274	39.9	183	39.9	183	26.5	275	26.5	275	26.5	275	26.5	275
	9800	26.6	368	39.9	246	39.9	246	26.5	370	26.5	370	26.5	370	26.5	370
	12200	26.6	459	39.9	306	39.9	306	26.5	460	26.5	460	26.5	460	26.5	460
	14600	26.6	549	39.9	366	39.9	366	26.5	551	26.5	551	26.5	551	26.5	551
030	8500	32.8	269	45.8	186	45.8	186	32.6	261	32.6	261	32.6	261	32.6	261
	11300	32.8	345	45.8	247	45.8	247	32.6	347	32.6	347	32.6	347	32.6	347
	14200	32.8	433	45.8	310	45.8	310	32.6	436	32.6	436	32.6	436	32.6	436
	17000	32.8	578	45.8	371	45.8	371	32.6	521	32.6	521	32.6	521	32.6	521
035	10000	39.4	254	63.3	158	63.3	158	39.1	256	39.1	256	37.7	265	39.1	256
	13400	39.4	340	63.3	212	63.3	212	39.1	343	39.1	343	37.7	355	39.1	343
	16700	39.4	424	63.3	264	63.3	264	39.1	427	39.1	427	37.7	443	39.1	427
	20000	39.4	508	63.3	316	63.3	316	39.1	512	39.1	512	37.7	531	39.1	512
040	12000	44.2	271	71.1	169	71.1	169	44.0	273	44.0	273	44.0	273	44.0	273
	16000	44.2	362	71.1	225	71.1	225	44.0	364	44.0	364	44.0	364	44.0	364
	20000	44.2	452	71.1	281	71.1	281	44.0	455	44.0	455	44.0	455	44.0	455
	24000	44.2	543	71.1	338	71.1	338	44.0	545	44.0	545	44.0	545	44.0	545
050	15000	55.3	271	73.7	2.4	73.7	204	55.0	273	55.0	273	55.0	273	55.0	273
	20000	55.3	362	73.7	271	73.7	271	55.0	364	55.0	364	55.0	364	55.0	364
	25000	55.3	452	73.7	339	73.7	339	55.0	455	55.0	455	55.0	455	55.0	455
	30000	55.3	542	73.7	407	73.7	407	55.0	545	55.0	545	55.0	545	55.0	545
065	19500	72.1	270	107.5	181	107.5	181	71.7	272	71.7	272	71.7	272	71.7	272
	26000	72.1	361	107.5	242	107.5	242	71.7	363	71.7	363	71.7	363	71.7	363
	32500	72.1	451	107.5	302	107.5	302	71.7	453	71.7	453	71.7	453	71.7	453
	39000	72.1	541	107.5	363	107.5	363	71.7	544	71.7	544	71.7	544	71.7	544
080	21500	80.2	268	107.5	200	107.5	200	78.2	275	79.7	270	78.2	275	79.7	270
	28700	80.2	358	107.5	267	107.5	267	78.2	367	79.7	360	78.2	367	79.7	360
	35900	80.2	448	107.5	334	107.5	334	78.2	459	79.7	450	78.2	459	79.7	450
	43100	80.2	538	107.5	401	107.5	401	78.2	551	79.7	541	78.2	551	79.7	541
085	23100	86	269	129	179	107.5	215	83.3	277	83.3	277	83.3	277	83.3	277
	30800	86	358	129	239	107.5	287	83.3	370	83.3	370	83.3	370	83.3	370
	38500	86	448	129	298	107.5	359	83.3	462	83.3	462	83.3	462	83.3	462
	46100	86	536	129	358	129	358	83.3	553	83.3	554	83.3	554	83.3	554
090	24600	90.4	272	129	191	129	191	86.3	285	89.8	274	87.4	281	89.8	274
	32800	90.4	363	129	254	129	254	86.3	380	89.8	365	87.4	375	89.8	365
	41000	90.4	454	129	318	129	318	86.3	475	89.8	456	87.4	469	89.8	456
	49200	90.4	544	129	381	129	381	86.3	570	89.8	548	87.4	563	89.8	548

Notes: 400 fpm maximum for angular filters.

Table 4. Filter media air pressure drop based on clean filters (continued)

UNIT SIZE	CFM	HIGH EFFICIENCY BAG											
		DriPak 2000											
		36"			30"			22"			19"	15"	12"
		65%	85%	95%	65%	85%	95%	65%	85%	95%	45%	45%	45%
003	900	0.15	0.22	0.29	0.15	0.21	0.32	0.17	0.28	0.42	0.11	0.10	0.13
	1200	0.20	0.29	0.39	0.19	0.29	0.43	0.23	0.37	0.56	0.18	0.14	0.17
	1500	0.24	0.36	0.49	0.24	0.36	0.53	0.28	0.46	0.70	0.25	0.19	0.21
	1800	0.29	0.43	0.59	0.29	0.43	0.64	0.33	0.56	0.83	0.33	0.24	0.25
004	1200	0.11	0.16	0.21	0.11	0.16	0.24	0.13	0.21	0.32	0.07	0.07	0.10
	1600	0.15	0.21	0.29	0.14	0.21	0.31	0.17	0.27	0.42	0.11	0.10	0.13
	2000	0.18	0.27	0.36	0.18	0.26	0.39	0.21	0.34	0.52	0.16	0.13	0.16
	2400	0.22	0.32	0.44	0.21	0.32	0.47	0.25	0.41	0.62	0.21	0.16	0.19
006	1700	0.12	0.17	0.22	0.11	0.17	0.25	0.14	0.21	0.33	0.08	0.07	0.11
	2300	0.15	0.23	0.31	0.15	0.22	0.33	0.18	0.29	0.44	0.12	0.11	0.14
	2900	0.19	0.28	0.39	0.19	0.28	0.42	0.22	0.37	0.55	0.17	0.14	0.17
	3500	0.23	0.34	0.47	0.23	0.34	0.51	0.27	0.44	0.66	0.23	0.18	0.20
008	2200	0.13	0.19	0.25	0.13	0.19	0.28	0.15	0.24	0.37	0.09	0.08	0.12
	3000	0.17	0.26	0.35	0.17	0.26	0.38	0.20	0.33	0.50	0.15	0.13	0.16
	3800	0.22	0.32	0.44	0.22	0.32	0.48	0.26	0.42	0.63	0.21	0.17	0.19
	4600	0.26	0.39	0.54	0.26	0.39	0.58	0.31	0.51	0.76	0.28	0.21	0.23
010	2900	0.16	0.23	0.31	0.15	0.23	0.34	0.18	0.30	0.45	0.13	0.11	0.14
	3900	0.21	0.31	0.42	0.21	0.31	0.46	0.24	0.40	0.60	0.20	0.16	0.19
	4900	0.26	0.39	0.53	0.26	0.39	0.57	0.30	0.50	0.75	0.28	0.21	0.23
	5900	—	—	—	—	—	—	—	—	—	—	—	—
012	3600	0.14	0.21	0.28	0.14	0.20	0.30	0.16	0.26	0.40	0.11	0.09	0.13
	4800	0.19	0.27	0.37	0.18	0.27	0.40	0.22	0.35	0.53	0.16	0.14	0.17
	6000	0.23	0.34	0.47	0.23	0.34	0.50	0.27	0.44	0.66	0.23	0.18	0.20
	7200	0.27	0.41	0.56	0.27	0.41	0.60	0.32	0.53	0.79	0.30	0.23	0.24
014	4200	0.14	0.21	0.28	0.14	0.21	0.31	0.17	0.27	0.41	0.11	0.10	0.13
	5400	0.18	0.27	0.36	0.18	0.27	0.40	0.21	0.35	0.52	0.16	0.13	0.16
	6600	0.22	0.33	0.45	0.22	0.33	0.48	0.26	0.42	0.64	0.21	0.17	0.20
	8800	0.29	0.43	0.60	0.29	0.43	0.65	0.34	0.56	0.84	0.33	0.24	0.25
017	5000	0.14	0.20	0.27	0.13	0.20	0.29	0.16	0.25	0.39	0.10	0.09	0.12
	6700	0.18	0.26	0.36	0.18	0.26	0.39	0.21	0.34	0.52	0.15	0.13	0.16
	8400	0.22	0.33	0.45	0.22	0.33	0.49	0.26	0.43	0.64	0.22	0.17	0.20
	10100	0.27	0.39	0.55	0.27	0.40	0.59	0.31	0.51	0.77	0.29	0.22	0.23
021	6000	0.13	0.20	0.26	0.13	0.19	0.29	0.16	0.25	0.38	0.10	0.09	0.12
	8000	0.18	0.26	0.35	0.17	0.26	0.38	0.21	0.33	0.51	0.15	0.13	0.16
	10000	0.22	0.32	0.44	0.22	0.32	0.48	0.25	0.42	0.63	0.21	0.17	0.19
	12000	0.26	0.39	0.53	0.26	0.39	0.57	0.30	0.50	0.75	0.28	0.21	0.23
025	7300	0.14	0.21	0.28	0.14	0.20	0.30	0.16	0.26	0.40	0.11	0.09	0.13
	9800	0.19	0.28	0.37	0.18	0.27	0.41	0.22	0.36	0.54	0.16	0.14	0.17
	12200	0.23	0.34	0.47	0.23	0.34	0.51	0.27	0.44	0.66	0.23	0.18	0.20
	14600	0.27	0.41	0.56	0.28	0.41	0.61	0.32	0.53	0.79	0.30	0.23	0.24
030	8500	0.13	0.20	0.26	0.13	0.19	0.29	0.16	0.25	0.38	0.10	0.09	0.12
	11300	0.18	0.26	0.35	0.17	0.26	0.38	0.21	0.33	0.51	0.15	0.13	0.16
	14200	0.22	0.32	0.44	0.22	0.32	0.48	0.26	0.42	0.63	0.21	0.17	0.19
	17000	0.26	0.39	0.53	0.26	0.39	0.58	0.30	0.50	0.75	0.28	0.21	0.23
035	10000	0.13	0.19	0.26	0.13	0.19	0.28	0.15	0.25	0.38	0.09	0.09	0.12
	13400	0.17	0.26	0.35	0.17	0.25	0.38	0.20	0.33	0.50	0.15	0.12	0.16
	16700	0.21	0.32	0.43	0.21	0.32	0.47	0.25	0.41	0.62	0.20	0.16	0.19
	20000	0.26	0.38	0.52	0.26	0.38	0.56	0.30	0.49	0.74	0.27	0.21	0.22
040	12000	0.14	0.20	0.28	0.14	0.20	0.30	0.16	0.26	0.40	0.10	0.09	0.13
	16000	0.18	0.27	0.37	0.18	0.27	0.40	0.21	0.35	0.53	0.16	0.13	0.17
	20000	0.23	0.34	0.46	0.23	0.34	0.50	0.27	0.44	0.66	0.23	0.18	0.20
	24000	0.27	0.40	0.56	0.27	0.40	0.60	0.32	0.52	0.79	0.30	0.22	0.24
050	15000	0.14	0.20	0.27	0.13	0.20	0.29	0.16	0.26	0.39	0.10	0.09	0.13
	20000	0.18	0.27	0.36	0.18	0.26	0.39	0.21	0.34	0.52	0.16	0.13	0.16
	25000	0.22	0.33	0.45	0.22	0.33	0.49	0.26	0.43	0.65	0.22	0.17	0.20
	30000	0.27	0.40	0.55	0.27	0.40	0.59	0.31	0.51	0.77	0.29	0.22	0.23
065	19500	0.14	0.21	0.28	0.14	0.21	0.31	0.17	0.27	0.41	0.11	0.10	0.13
	26000	0.19	0.28	0.38	0.19	0.28	0.41	0.22	0.36	0.55	0.17	0.14	0.17
	32500	0.24	0.35	0.48	0.24	0.35	0.52	0.27	0.45	0.68	0.24	0.19	0.21
	39000	0.28	0.42	0.58	0.28	0.42	0.62	0.33	0.54	0.81	0.31	0.23	0.25
080	21500	0.12	0.16	0.23	0.15	0.19	0.27	0.17	0.26	0.38	0.07	0.09	0.13
	28700	0.15	0.21	0.31	0.19	0.26	0.36	0.22	0.35	0.51	0.11	0.13	0.16
	35900	0.19	0.26	0.39	0.24	0.32	0.45	0.27	0.43	0.63	0.16	0.18	0.20
	43100	0.23	0.31	0.47	0.28	0.39	0.54	0.32	0.52	0.76	0.21	0.22	0.24
085	23100	0.12	0.16	0.23	0.15	0.20	0.28	0.17	0.27	0.40	0.08	0.10	0.13
	30800	0.16	0.22	0.32	0.20	0.27	0.37	0.23	0.36	0.52	0.12	0.14	0.17
	38500	0.19	0.27	0.40	0.24	0.33	0.46	0.28	0.44	0.65	0.16	0.18	0.21
	46100	0.23	0.32	0.48	0.29	0.40	0.55	0.33	0.53	0.77	0.22	0.23	0.24
090	24600	0.12	0.16	0.23	0.15	0.20	0.28	0.17	0.26	0.39	0.07	0.09	0.13
	32800	0.16	0.21	0.31	0.20	0.26	0.37	0.22	0.35	0.52	0.12	0.13	0.17
	41000	0.19	0.27	0.39	0.24	0.33	0.46	0.28	0.44	0.64	0.16	0.18	0.20
	49200	0.23	0.32	0.47	0.28	0.39	0.55	0.33	0.53	0.76	0.21	0.22	0.24

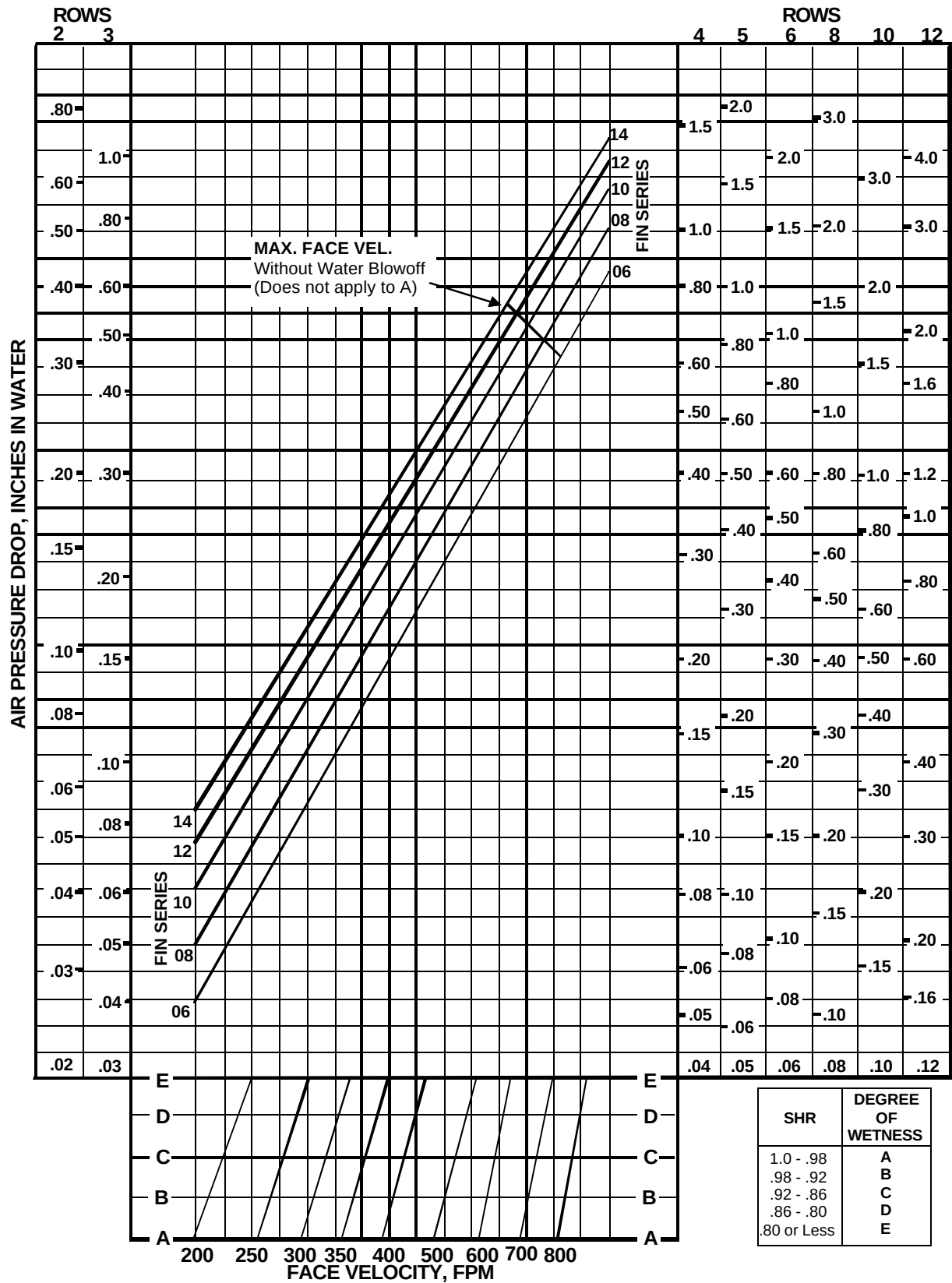
Figure 6. Coil air pressure drop — HI-F5 fin type (2 thru 12 rows)



Note: The letters A, B, C, D or E following the face velocity indicate the degree of wetness at which the coil would be operating. Dry coils are shown by the letter A, wet coils by the letter E. Intermediate conditions are shown by the letters B, C and D. Air pressure drop for odd fin spacings may be found by interpolation.

cat550-17.dsf

Figure 7. Coil air pressure drop — E-F5 fin type (2 thru 12 rows)



cat550-18.dsf

Note: The letters A,B,C,D or E following the face velocity indicate the degree of wetness at which the coil would be operating. Dry coils are shown by the letter A, wet coils by the letter E. Intermediate conditions are shown by the letters B, C and D. Air pressure drop for odd fin spacings may be found by interpolation.

Figure 8. Coil air pressure drop — HI-F5 fin type

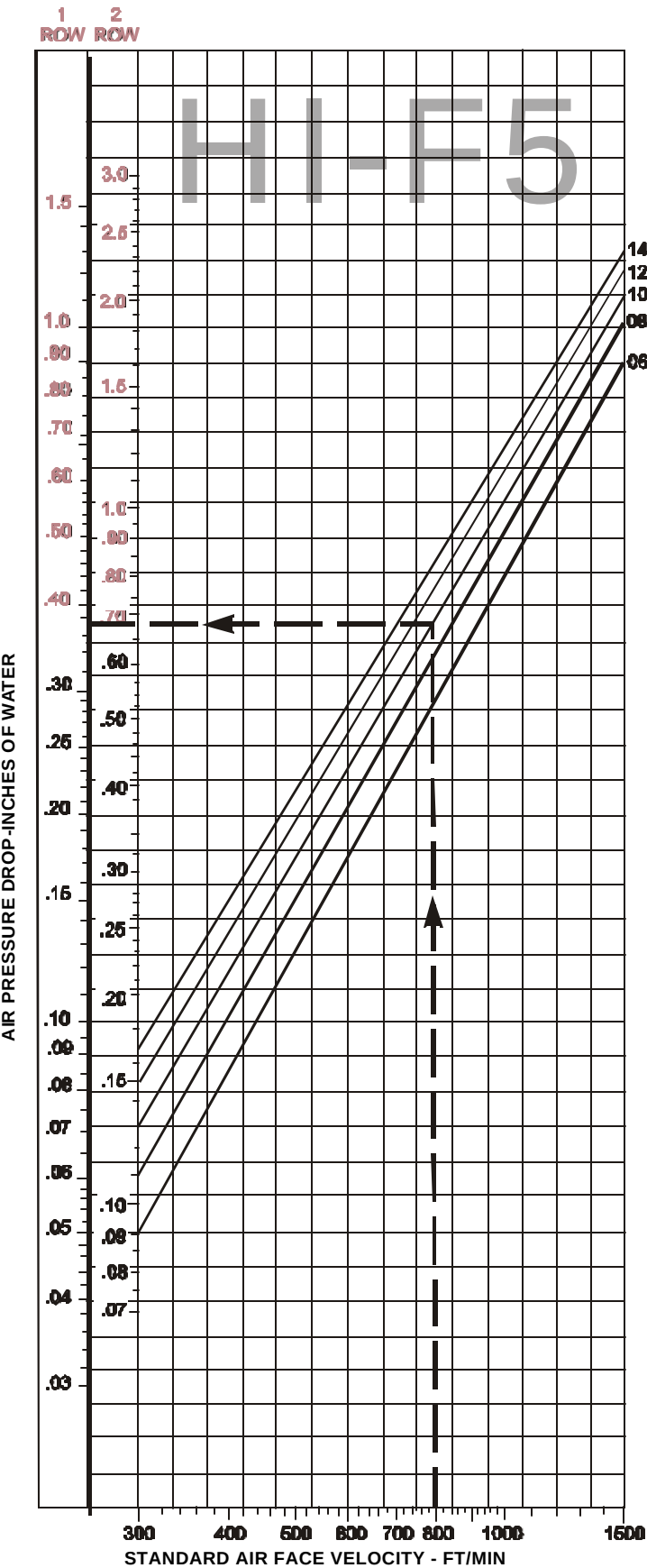
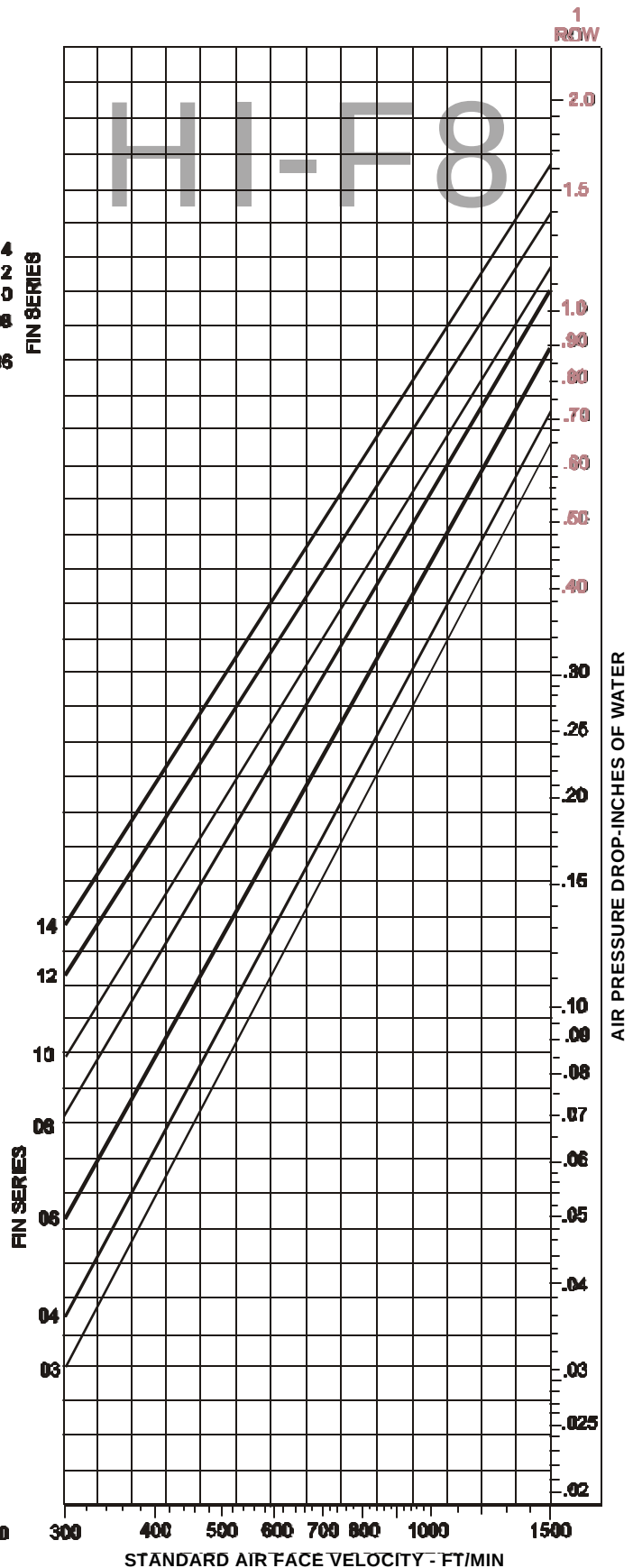


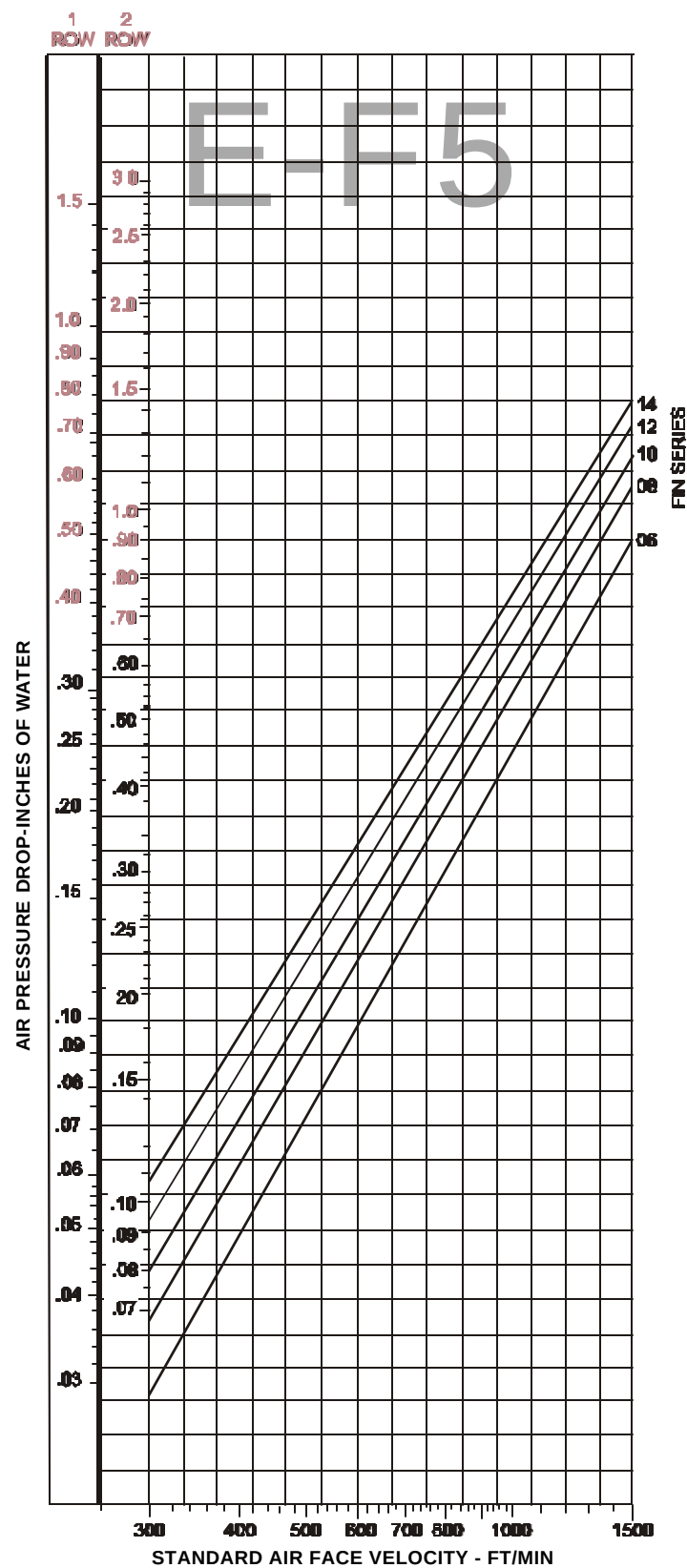
Figure 9. 1" Steam coil air pressure drop



NOTE: Air pressure drop for odd fin spacings may be found by interpolation.

cat550-19.cdr

Figure 10. Coil air pressure drop — E-F5 fin type



cat550-20.cdr

NOTE: Air pressure drop for odd fin spacings may be found by interpolation.

Component and section weights

Table 6. Approximate shipping weights

DESCRIPTION	UNIT SIZE																	
	003		004		006		008		010		012		014		017		021	
	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg
Mixing Box with Dampers																		
Mixing box	242	110	262	119	307	139	350	159	419	190	440	200	475	215	532	242	608	276
Mixing box w/ flat filter	281	127	301	137	354	161	406	184	483	219	515	234	552	250	620	281	700	318
Mixing box w/ angular filter	394	179	429	195	505	229	576	261	668	303	715	325	773	351	882	400	1032	468
Plenum																		
Plenum top, bottom or end	182	82	208	94	245	111	289	131	329	149	371	169	401	182	472	214	541	245
Economizer with Damper																		
	388	176	416	189	480	218	528	289	329	290	669	304	714	324	802	364	927	421
Blender																		
	227	103	262	119	353	160	421	191	472	214	529	240	605	274	689	313	756	343
Filter - Side Load																		
2" angular	296	134	320	145	374	170	407	185	444	201	495	224	534	242	603	273	627	284
2" flat or 4" flat	182	82	192	87	225	102	254	115	277	126	305	138	325	148	354	161	374	170
4" cartridge w/ prefilter	159	72	171	78	204	93	235	107	263	119	284	129	307	139	332	151	360	163
12" cartridge w/ prefilter	250	113	275	125	321	146	380	172	411	186	446	202	485	220	535	243	583	265
12" or 15 bag w/ prefilter	237	107	253	115	296	134	340	154	370	168	393	178	424	193	459	208	491	223
19" or 22" bag w/ prefilter	267	121	288	130	335	152	381	173	417	189	446	202	480	218	513	233	553	251
30" or 36" bag w/ prefilter	320	145	343	155	397	180	449	203	489	222	520	236	558	253	599	272	643	292
F & BP Damper																		
External	237	108	255	116	301	137	351	159	426	193	516	234	557	253	621	282	743	337
Internal	164	74	182	83	215	98	246	112	285	129	317	144	343	156	376	171	408	185
Coil — Heating, Cooling, & Combination Section (coil weight not included)																		
Heating 1,2 row (12" deep)	171	78	184	83	214	97	237	107	256	116	271	123	292	133	315	143	330	150
Heating 3,4 row (14" deep)	204	92	219	99	256	116	284	129	308	140	327	148	352	160	381	173	399	181
Cooling 4 row (18" deep) — horizontal only	232	105	251	114	293	133	324	147	350	159	370	168	401	182	434	197	454	206
Cooling/combination option (24" deep) ②	267	121	289	131	336	152	371	169	399	181	423	192	460	209	496	225	519	235
Cooling/combination option (30" deep) ②	302	137	327	148	380	172	419	190	450	204	476	216	518	235	557	253	583	265
Cooling/combination option (36" deep) ②	337	153	365	165	422	191	466	211	499	226	529	240	575	261	619	281	649	294
Cooling/combination option (42" deep) ②	373	169	403	183	466	211	513	233	550	249	582	264	634	287	681	309	713	323
Cooling/combination option (48" deep) ②	407	185	440	200	508	231	559	254	600	272	635	288	738	335	794	360	829	376
Cooling/combination option (54" deep) ②	424	193	459	208	529	240	618	280	662	301	699	317	762	346	819	371	856	388
Multizone (coil weight not included)																		
2 deck horizontal	—	—	—	—	852	387	965	438	1041	472	1256	570	1356	615	1492	677	1610	730
3 deck horizontal	—	—	—	—	948	430	1068	485	1150	522	1380	626	1486	674	1667	756	1802	818
2 deck vertical	—	—	—	—	832	377	942	427	1016	461	1206	547	1299	589	1456	661	1576	715
3 deck vertical	—	—	—	—	978	444	1140	517	1225	556	1404	637	1508	684	1748	793	1909	866
Access																		
16" access	183	83	196	89	225	102	248	113	268	122	284	129	305	138	328	149	345	157
24" access	221	100	236	107	271	123	299	136	322	146	342	155	366	166	393	178	414	188
30" access	250	113	267	121	306	139	337	153	362	164	385	175	412	187	443	201	466	211
36" access	281	127	300	136	344	156	378	172	407	185	432	196	462	210	496	225	522	237
42" access	309	140	331	150	378	172	416	189	447	203	476	216	507	230	545	247	574	260
48" access	347	158	373	169	424	193	467	212	500	227	535	243	615	279	661	300	697	316
54" access	358	162	383	174	436	198	515	234	553	251	589	267	628	285	675	306	710	322
DWDI Fans — Top, Bottom or Upblast Discharge																		
Largest fan available ③	431	196	438	199	639	290	762	359	848	385	1018	462	1126	511	1465	665	1490	676
Fan section drain pan	20	9	20	9	24	11	32	15	36	16	43	19	48	22	58	26	61	28
DWDI Fans — Downblast Discharge																		
Largest fan available ③	449	203	458	208	675	306	817	370	967	439	1089	494	1264	573	1429	649	1547	702
Optional Fans (largest fan available)																		
Plenum	—	—	—	—	—	—	922	419	1055	479	1240	563	1347	612	1773	805	2077	943
Inline	—	—	—	—	—	—	—	—	—	—	1195	543	1246	566	1469	667	1962	891
Diffuser																		
Diffuser for DWDI and SWSI fans	174	79	187	85	216	98	242	110	260	118	308	140	329	149	357	162	377	171
Diffuser for inline fans	—	—	—	—	—	—	—	—	—	—	344	156	366	166	396	180	497	226
Attenuator																		
Short	368	167	409	186	467	212	599	272	665	302	746	339	800	363	896	406	983	446
Medium	473	214	535	243	650	295	752	341	836	379	1040	472	1132	513	1274	578	1423	645
Long	565	256	641	291	813	369	942	427	1049	476	1249	567	1359	617	1534	696	1750	794

Notes:

① All weights shown include double wall construction and a 6" base rail for units up to size 050. An 8" base rail is mandatory on units sizes 080 and larger.

② To determine the weight of a vertical coil section, combine the weight of the coil section desired and a plenum section.

③ Fan weights are based on the largest airfoil (sizes 3,4 & 6 forward curved) fan available without inlet vanes.

Table 6. Approximate shipping weights (continued)

DESCRIPTION	UNIT SIZE																	
	025		030		035		040		050		065		080		085		090	
	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg
Mixing Box with Dampers																		
Mixing box	688	312	789	358	887	402	1045	474	1236	561	1515	687	1873	850	1987	902	2075	942
Mixing box w/ flat filter	798	362	918	416	1048	475	1244	564	1408	639	1783	809	2102	954	2222	1009	2333	1059
Mixing box w/ angular filter	1158	525	1306	593	1486	674	1533	696	1723	782	2070	939	2644	1200	2864	1300	2961	1344
Plenum																		
Plenum top, bottom or end	619	281	700	318	795	361	812	368	1048	475	1130	591	2015	915	2141	972	2234	1014
Economizer with Damper																		
	1040	472	1163	527	1320	599	1398	634	1691	767	1218	989	3136	1424	3349	1520	3488	1584
Blender																		
	856	388	1064	483	1195	542	1233	559	1407	638	1812	822	2774	1259	2900	1317	2975	1351
Filter - Side Load																		
2" angular	716	325	811	368	922	418	844	383	905	411	1049	476	1362	618	1434	651	1451	659
2" flat or 4" flat	407	185	484	220	509	231	576	261	615	279	731	332	867	394	882	400	917	416
4" cartridge w/ prefilter	489	222	553	251	590	268	683	310	750	340	887	402	1140	518	1173	533	1221	554
12" cartridge w/ prefilter	635	288	748	339	807	366	944	428	1034	469	1273	578	1410	640	1451	659	1561	709
12" or 15 bag w/ prefilter	545	247	615	279	649	294	751	341	809	367	951	431	1127	512	1158	526	1194	542
19" or 22" bag w/ prefilter	612	278	689	313	698	317	821	372	909	412	1072	486	1264	574	1320	599	1328	603
30" or 36" bag w/ prefilter	748	339	896	406	937	425	1085	492	1181	536	1407	638	1587	720	1643	746	1669	758
F & BP Damper																		
External	903	410	998	453	1097	498	1247	566	1415	642	1814	823	2175	987	2306	1047	2366	1074
Internal	535	243	614	279	662	300	769	349	961	436	1139	517	1252	568	1278	580	1308	594
Coil — Heating, Cooling, & Combination Section (coil weight not included)																		
Heating 1,2 row (12" deep)	354	161	385	175	406	184	580	263	604	274	680	309	827	375	956	434	909	413
Heating 3,4 row (14" deep)	373	169	406	184	428	194	611	277	633	287	717	325	900	409	917	416	991	450
Cooling 4 row (18" deep) — horizontal only	515	234	561	255	588	267	695	315	741	336	844	383	1022	464	1041	473	1057	480
Cooling/combination option (24" deep)②	589	267	639	290	669	304	819	372	878	398	1002	455	1151	523	1173	533	1191	541
Cooling/combination option (30" deep)②	661	300	718	326	751	341	913	414	978	444	1118	507	1281	582	1305	592	1326	602
Cooling/combination option (36" deep)②	735	333	796	361	834	378	1009	458	1101	500	1259	571	1411	641	1450	658	1461	663
Cooling/combination option (42" deep)②	807	366	935	424	980	445	1178	534	1252	568	1446	656	1654	751	1684	765	1711	777
Cooling/combination option (48" deep)②	935	424	1014	460	1061	482	1272	577	1374	623	1571	713	1783	809	1816	824	1847	839
Cooling/combination option (54" deep)②	1045	474	1129	512	1185	537	1394	632	1474	669	1686	765	1855	842	1887	857	1918	871
Multizone (coil weight not included)																		
2 deck horizontal	1946	883	2117	960	2485	1127	2983	1353	3394	1540	4501	2042	—	—	—	—	—	—
3 deck horizontal	2162	981	2348	1065	2720	1234	3266	1482	—	—	—	—	—	—	—	—	—	—
2 deck vertical	2019	916	2219	1007	2523	1145	2987	1355	4123	1871	—	—	—	—	—	—	—	—
3 deck vertical	2433	1104	2694	1222	3415	1549	3916	1777	—	—	—	—	—	—	—	—	—	—
Access																		
16" access	373	169	428	194	451	205	508	230	544	247	614	279	783	355	796	361	808	367
24" access	447	203	508	231	536	243	607	275	651	295	734	333	922	419	938	426	953	433
30" access	461	209	523	237	550	249	682	309	730	331	823	373	1026	466	1044	474	1063	483
36" access	566	257	635	288	668	303	763	346	818	371	922	418	1142	518	1196	543	1185	538
42" access	578	262	710	322	746	339	909	412	974	442	1128	512	1367	621	1389	631	1415	642
48" access	744	338	828	376	873	396	997	452	1095	497	1224	555	1486	675	1515	688	1543	701
54" access	776	352	861	391	907	412	1091	495	1173	532	1324	601	1530	695	1559	708	1589	721
DWDI Fans — Top, Bottom or Upblast Discharge																		
Largest fan available ③	1733	786	1977	897	2075	941	3294	1494	4453	2020	5841	2650	6345	2881	6404	2907	6473	2939
Fan section drain pan	68	31	81	37	86	39	122	55	119	54	189	86	189	86	209	95	209	95
DWDI Fans — Downblast Discharge																		
Largest fan available ③	1741	790	1988	902	2088	948	3068	1392	4726	2144	6156	2793	6460	2933	6538	2968	6659	3023
Optional Fans (largest fan available)																		
Plenum	2525	1146	2865	1301	3263	1481	3767	1710	4963	2253	7254	3293	7395	3357	7444	3380	7510	3410
Inline	2834	1287	2947	1338	3447	1565	3742	1699	5650	2565	6002	2725	7064	3207	6822	3097	6893	3129
Diffuser																		
Diffuser for DWDI and SWSI fans	489	222	550	249	580	263	717	325	776	352	882	400	1098	498	1112	505	1133	514
Diffuser for inline fans	585	266	693	315	760	345	1402	637	1724	783	2198	998	1289	585	1312	596	1336	607
Attenuator																		
Short	1280	581	1503	682	1692	768	1868	848	2182	990	2363	1072	3181	1444	3334	1514	3488	1584
Medium	1693	768	1900	862	2190	993	2343	1063	2759	1252	3652	1657	4026	1828	4229	1920	4431	2012
Long	2090	948	2336	1060	2645	1200	2882	1308	3398	1542	4425	2008	4873	2212	5122	2325	5373	2439

Notes:

- ① All weights shown include double wall construction and a 6" base rail for units up to size 050. An 8" base rail is mandatory on units sizes 080 and larger.
- ② To determine the weight of a vertical coil section, combine the weight of the coil section desired and a plenum section.
- ③ Fan weights are based on the largest airfoil (sizes 3,4 & 6 forward curved) fan available without inlet vanes.

Table 7. Unit coil weights

UNIT SIZE	003		004		006		008		010		012		014		017		021	
ROWS	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg
1	18	8	23	10	29	13	36	16	43	20	53	24	60	27	67	30	78	35
2	26	12	33	15	43	20	54	24	66	30	83	38	92	42	107	49	125	57
3	37	17	47	21	63	29	81	37	99	45	122	55	139	63	161	73	207	94
4	46	21	58	26	78	36	101	46	123	56	151	69	172	78	200	91	257	117
5	55	25	70	32	94	43	120	55	147	67	181	82	206	93	239	108	307	139
6	64	29	81	37	109	49	140	64	171	78	210	95	239	108	278	126	357	162
8	82	37	104	47	140	63	179	81	219	99	269	122	306	139	356	161	457	207
10	100	45	126	57	170	77	218	99	267	121	328	149	373	169	434	197	557	253
12	118	53	149	68	201	91	258	117	315	143	386	175	440	200	512	232	657	298

UNIT SIZE	025		030		035		040		050		065		080		085		090	
ROWS	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg
1	108	49	123	56	143	65	156	71	212	96	267	121	434	195	449	202	466	210
2	166	75	189	86	222	101	258	117	345	157	446	202	663	298	678	305	711	320
3	247	112	291	132	339	154	403	183	559	254	731	332	1239*	558	1297*	584	1324*	596
4	306	139	361	164	420	191	500	227	694	315	907	412	1711**	770	1802**	811	1892**	851
5	366	166	432	196	502	228	598	271	829	376	1084	492	1959**	882	2070**	932	2180**	981
6	425	193	502	228	584	265	695	315	964	437	1260	572	2252**	1013	2383**	1072	2512**	1130
8	544	247	643	292	748	339	890	404	1234	560	1613	732	2828**	1273	3009**	1354	3182**	1432
10	663	301	783	355	911	413	1084	492	1504	682	1966	892	NA	NA	NA	NA	NA	NA
12	782	355	924	419	1075	488	1279	580	1774	805	2318	1052	NA	NA	NA	NA	NA	NA

NOTE:

*3-row coils (unit sizes 080-090) based on staggered medium face area 5EJ with 6 FPI and standard fin, tube and casing materials.
 **4-8 row coils (unit sizes 080-090) based on staggered medium face area with 12 FPI and standard fin, tube and casing materials. 1 & 2 row (unit sizes 003-065) based on large face area 5WQ & 5WH coils with 8 FPI and standard fin, tube and casing materials.
 3-row and greater (unit sizes 003-065) based on large face area 5W* or 5E* coils with 8 FPI and standard fin, tube and casing materials.
 Staggered coils are not available in 10 and 12-row coils.

Table 8. Weights for single speed and dual speed motors: 900 RPM, 1200 RPM, 1800 RPM, 3600 RPM, 1800/900 RPM and 1800/1200 RPM

900 rpm

Motor HP	1	1-1/2	2	3	5	7 1/2	10	15	20	25	30	40	50	60	75
NEMA frame	182T	184T	213T	215T	254T	256T	284T	286T	324T	326T	364T	365T	404T	405T	444T
Motor weight (lbs.)	56	64	94	111	150	207	300	300	385	415	580	580	750	800	1100
Motor weight (kg)	25	29	43	50	68	94	136	136	175	189	349	264	461	364	500

1200 rpm

Motor HP	1	1-1/2	2	3	5	7-1/2	10	15	20	25	30	40	50	60	75
NEMA frame	145T	182T	184T	213T	215T	256T	256T	284T	286T	324T	326T	365T	365T	404T	405T
Motor weight (lbs.)	44	66	85	114	145	224	248	330	377	450	487	703	720	1153	1200
Motor weight (kg)	20	30	39	52	66	102	113	150	153	205	221	320	327	524	545

1800 rpm

Motor HP	1/4	1/3	1/3	3/4	1	1½	2	3	5	7 1/2	10	15	20	25	30	40	50	60	75
NEMA frame	48	48	56	56	143T	145T	145T	182T	184T	213T	215T	254T	256T	284T	286T	324T	326T	364T	365T
Motor weight (lbs.)	19	24	35	35	44	54	54	84	100	139	165	242	273	351	432	531	592	714	895
Motor weight (kg)	9	11	16	16	20	24	24	38	45	63	75	110	124	159	196	241	269	324	406

3600 rpm

Motor HP	1	1-1/2	2	3	5	7-1/2	10	15	20	25	30	40	50	60	75
NEMA frame	56T	143T	145T	145T	182T	184T	213T	215T	254T	256T	284TS	286TS	324TS	326TS	405TS
Motor weight (lbs.)	32	40	40	50	76	91	110	147	182	223	359	369	447	558	713
Motor weight (kg)	15	18	18	23	35	41	50	67	83	101	163	168	203	254	324

1800/900 rpm*

Motor HP	1	1-1/2	2	3	5	7-1/2	10	15	20	25	30	40	50
NEMA frame	143	145	145	182	184	213	215	256	256	286	286	324	326
Motor weight (lbs.)	25	28	31	63	72	104	130	244	232	250	275	350	390
Motor weight (kg)	11	13	14	29	33	47	59	111	105	114	125	159	177

1800/1200 rpm

Motor HP	1	1-1/2	2	3	5	7-1/2	10	15	20	25	30
NEMA frame	145	182	182	184	215	256	256	284	286	286	286
Motor weight (lbs.)	30	56	65	73	113	211	237	318	348	349	295
Motor weight (kg)	14	25	30	33	51	96	108	145	158	159	134

*Data included reflects the largest NEMA frame and/or the heaviest motor weight per motor HP.

Table 9. Base rail weights

6" BASE RAIL WEIGHTS	STANDARD UNIT NOMINAL SIZE																	
	003		004		006		008		010		012		014		017		021	
	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg
End cross members (pair)	29	13	30	14	37	17	41	19	45	20	46	21	51	23	55	25	56	25
Add'l. cross member supports	10	5	10	5	14	6	15	7	17	8	17	8	19	9	21	10	21	10

6" BASE RAIL WEIGHTS (continued)	STANDARD UNIT NOMINAL SIZE																	
	025		030		035		040		050		065		080		085		090	
	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg
End cross members (pair)	57	26	63	29	65	29	68	31	71	32	75	34	75	34	75	34	75	34
Add'l. cross member supports	22	10	26	12	27	12	30	14	32	15	35	16	35	16	35	16	35	16

Weight factor for 6" high rail = .52 lb./inch (.00929 kg/mm).

10" BASE RAIL WEIGHTS	STANDARD UNIT NOMINAL SIZE																	
	003		004		006		008		010		012		014		017		021	
	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg
End cross members (pair)	39	18	40	18	50	23	55	25	60	27	62	28	69	31	74	33	75	34
Add'l. cross member supports	14	6	15	7	19	9	21	10	24	11	24	11	27	12	30	14	30	14

10" BASE RAIL WEIGHTS (continued)	STANDARD UNIT NOMINAL SIZE																	
	025		030		035		040		050		065		080		085		090	
	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg	lbs.	kg
End cross members (pair)	78	35	86	39	89	40	92	42	95	43	107	49	107	49	107	49	107	49
Add'l. cross member supports	32	15	36	16	38	17	43	20	47	21	50	23	50	23	50	23	50	23

Weight factor for 10" high rail = .74 lb./inch (.0132 kg/mm).

The base rail weight is determined by adding together the weight of the rail supports that run the length and width of the unit. The lengthwise support depends on the overall unit length. The cross member weight is dependent on the number of cross channels needed to support the unit. Each shipping section has a cross member located on each end. When a unit ships in one piece, the rail will have 2 end channel supports. When the unit ships in numerous shipping sections, each shipping section will have 2 end pieces. Fan, coil, and attenuator sections need additional cross member support on the entering and leaving air side of the section. If the coil section is adjacent to a fan, only one cross member will be provided between them. A heating only coil section needs an additional cross member support on the entering air side. Any shipping section over 8' long will require an additional cross member.

Base rails are required on sizes 025 and above.

Example:

Determine the weight of a 6" high base rail for a size 8 unit that is 108" long, consisting of a fan, access, coil, and angular filter in a single shipping section.

Weight factor for unit length for a 6" curb is .52 lb./in.

$$108" \times .52 \text{ lb./in.} = 56 \text{ lb.} \dots\dots\dots 56 \text{ lb.}$$

Weight of end cross members for each shipping section is 41 lb. each pair.

$$1 \times 41 \text{ lb.} = 41 \text{ lb.} \dots\dots\dots 41 \text{ lb.}$$

Weight of additional cross members to support fan and coil sections. In this example the fan and coil are separated by an access section, therefore 4 extra cross member pieces will be required for the fan and coil.

$$4 \times 15 = 60 \text{ lb.} \dots\dots\dots 60 \text{ lb.}$$

$$\text{Total base rail weight} \dots\dots\dots 157 \text{ lb.}$$

Fan data

Table 10. Physical data - forward curved & airfoil housed fans - Unit sizes 003 - 035

Forward Curved - English Units										
Diameter Class I (inches)	9 x 4	9 x 7	9 x 9	10.62	12.62	15	18	20	22.25	24.5
Maximum rpm	N/A	2189	2223	1934	1614	1328	1155	1050	944	858
Shaft & bearing diameter (inches)	N/A	1	1	1	1	1.188	1.188	1.938	2.1875	2.1875
Outlet area (square feet)	N/A	0.65	0.84	1.04	1.46	2.05	2.87	5.18	6.27	7.63
Diameter Class II (inches)	9 x 4	9 x 7	9 x 9	10.62	12.62	15	18	20	22.25	24.5
Maximum rpm	2244	2854	2896	2518	2091	1725	1450	1432	1288	1169
Shaft & bearing diameter (inches)	1	1	1	1	1.188	1.188	1.938	2.188	2.4375	2.4375
Outlet area (square feet)	0.48	0.65	0.84	1.04	1.46	2.05	2.87	5.18	6.27	7.63
Forward Curved - SI Units										
Diameter Class I (millimeters)	229 x 102	229 x178	229 x 229	270	321	381	457	508	565	622
Maximum rpm	N/A	2189	2223	1934	1614	1328	1155	1050	944	858
Shaft & bearing diameter (millimeters)	N/A	25	25	25	25	30	30	49	56	56
Outlet area (square meters)	N/A	0.060	0.078	0.097	0.136	0.190	0.267	0.481	0.58	0.71
Diameter Class II (millimeters)	229 x 102	229 x178	229 x 229	270	321	381	457	508	565	622
Maximum rpm	2244	2854	2896	2518	2091	1725	1450	1432	944	858
Shaft & bearing diameter (millimeters)	25	25	25	25	30	30	49	56	62	62
Outlet area (square meters)	0.045	0.060	0.078	0.097	0.136	0.190	0.267	0.481	0.58	0.71
Airfoil - English Units										
Diameter Class I (inches)	13.22	14.56	16.19	19.69	21.56	24				
Maximum rpm	3000	3000	2300	2000	1700	1500				
Shaft & bearing diameter (inches)	1.188	1.438	1.438	1.938	1.938	2.1875				
Outlet area (square feet)	2.11	2.85	3.52	4.68	5.82	7.01				
Diameter Class II (inches)	13.22	14.56	16.19	19.69	21.56	24				
Maximum rpm	4335	3918	3457	2858	2547	2255				
Shaft & bearing diameter (inches)	1.438	1.688	1.688	2.188	2.438	2.4375				
Outlet area (square feet)	2.11	2.85	3.52	4.68	5.82	7.01				
Airfoil - SI Units										
Diameter Class I (millimeters)	336	370	411	500	548	610				
Maximum rpm	3000	3000	2300	2000	1700	1500				
Shaft & bearing diameter (millimeters)	30	37	37	49	49	56				
Outlet area (square meters)	0.196	0.265	0.327	0.435	0.541	0.65				
Diameter Class II (millimeters)	336	370	411	500	548	610				
Maximum rpm	4335	3918	3457	2858	2547	2547				
Shaft & bearing diameter (millimeters)	37	43	43	56	62	62				
Outlet area (square meters)	0.196	0.265	0.327	0.435	0.541	0.65				

Figure 11. Torque for FC variable inlet vanes (in.-lb.)
Unit sizes 003-035

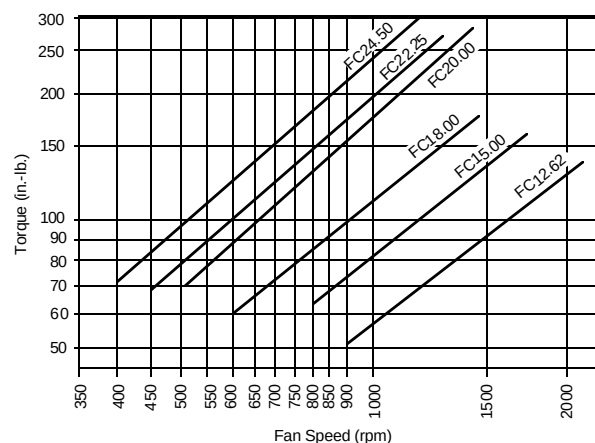
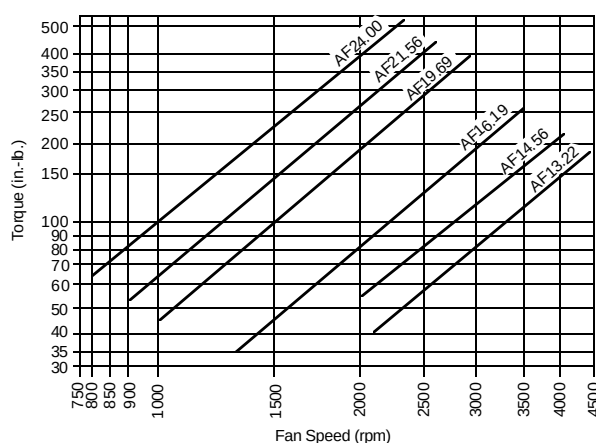


Figure 12. Torque for AF variable inlet vanes (in.-lb.)
Unit sizes 003-035



NOTE: Inlet vanes on fans 30HP or more are not acceptable per ASHRAE 90.1-1999.

cat550-22.dsf

Table 11. Physical data - forward curved & airfoil fans - Unit sizes 040 - 090

Forward Curved - English Units								
Diameter Class I (inches)	20	22.38	25	27.62	30.25	33	36	40.25
Maximum rpm	1010	930	790	690	650	600	560	420
Shaft & bearing diameter (inches)	1.438	1.438	1.688	1.688	1.688	2.188	2.188	2.938
Outlet area (square feet)	4.2	5.1	6.71	8.07	9.3	12.05	12.8	16.77
Diameter Class II (inches)	20	22.38	25	27.62	30.25	33	36	40.25
Maximum rpm	1281	1178	1011	910	835	763	715	550
Shaft & bearing diameter (inches)	1.688	2.188	2.438	2.438	2.688	2.688	2.688	3.438
Outlet area (square feet)	4.2	5.1	6.71	8.07	9.3	12.05	12.8	16.77
Forward Curved - SI Units								
Diameter Class I (millimeters)	508	568	635	702	768	838	914	1022
Maximum rpm	1010	930	790	690	650	600	560	420
Shaft & bearing diameter (millimeters)	37	37	43	43	43	56	56	75
Outlet area (square meters)	0.390	0.474	0.623	0.750	0.864	1.119	1.189	1.558
Diameter Class II (millimeters)	508	568	635	702	768	838	914	1022
Maximum rpm	1281	1178	1011	910	835	763	715	550
Shaft & bearing diameter (millimeters)	43	56	62	62	68	68	68	87
Outlet area (square meters)	0.390	0.474	0.623	0.750	0.864	1.119	1.189	1.558
Airfoil - English Units								
Diameter Class I (inches)	20	22.25	24.5	27	30	33	36.5	40.25
Maximum rpm	2077	1875	1691	1479	1328	1209	1073	972
Shaft & bearing diameter (inches)	1.688	1.938	2.188	2.188	2.438	2.438	2.438	2.938
Outlet area (square feet)	4.14	5.12	6.21	7.54	9.31	11.27	13.79	16.77
Diameter Class II (inches)	20	22.25	24.5	27	30	33	36.5	40.25
Maximum rpm	2703	2413	2199	1928	1730	1579	1401	1264
Shaft & bearing diameter (inches)	2.188	2.188	2.438	2.438	2.688	2.938	2.938	3.438
Outlet area (square feet)	4.14	5.12	6.21	7.54	9.31	11.27	13.79	16.77
Airfoil - SI Units								
Diameter Class I (millimeters)	508	565	622	686	762	838	927	1022
Maximum rpm	2077	1875	1691	1479	1328	1209	1073	972
Shaft & bearing diameter (millimeters)	43	49	56	56	62	62	62	75
Outlet area (square meters)	0.385	0.476	0.577	0.700	0.865	1.047	1.281	1.558
Diameter Class II (millimeters)	508	565	622	686	762	838	927	1022
Maximum rpm	2703	2413	2199	1928	1730	1579	1401	1264
Shaft & bearing diameter (millimeters)	56	56	62	62	68	75	75	87
Outlet area (square meters)	0.385	0.476	0.577	0.700	0.865	1.047	1.281	1.558

Figure 13. Torque for FC variable inlet vanes (in.-lb.)
Unit sizes 040-090

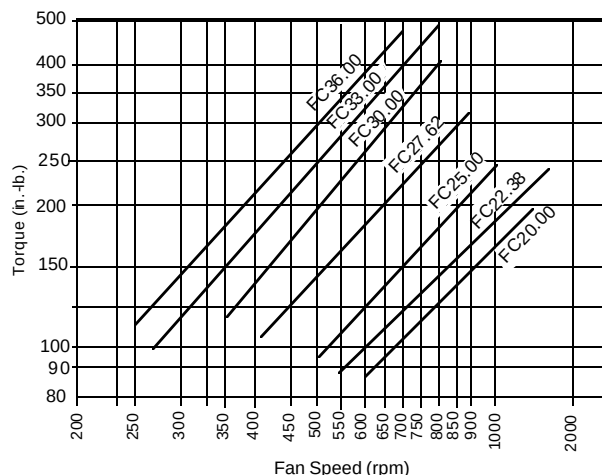
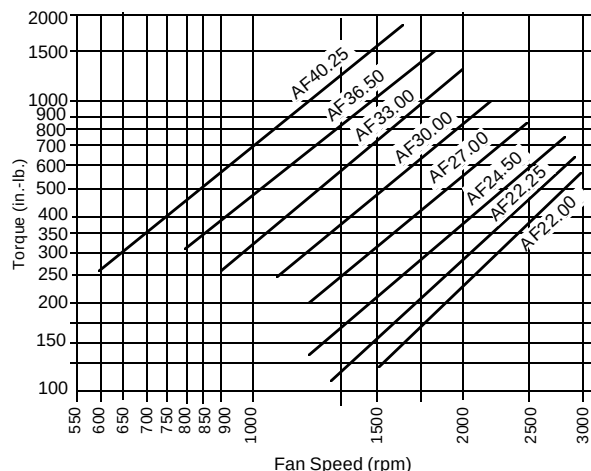


Figure 14. Torque for AF variable inlet vanes (in.-lb.)
Unit sizes 040-090



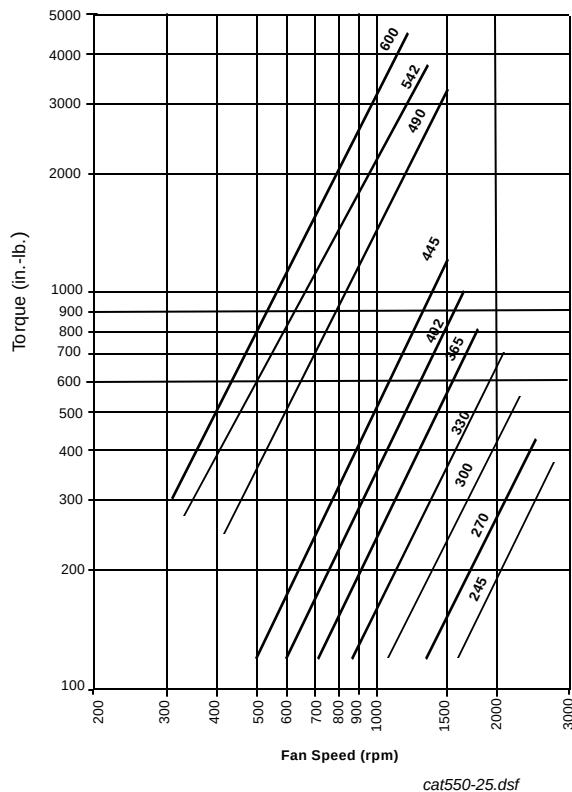
NOTE: Inlet vanes on fans 30HP or more are not acceptable per ASHRAE 90.1-1999.

cat550-24.dsf

Table 12. Physical data - belt drive & direct drive plenum fans

Plenum — English Units																
Diameter Class I (inches)	13.5	15	16.5	18.25	20	22.25	24.5	27	30	33	36.5	40.25	44.5	49	54.25	60
Maximum rpm	2895	2589	2376	2256	2077	1875	1691	1479	1328	1209	1073	972	882	799	725	651
Shaft & bearing diameter (inches)	1	1	1	1.188	1.188	1.188	1.438	1.688	1.688	1.688	1.688	1.938	1.938	2.188	2.438	2.438
Outlet area (square feet)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Diameter Class II (inches)	13.5	15	16.5	18.25	20	22.25	24.5	27	30	33	36.5	40.25	44.5	49	54.25	60
Maximum rpm	3786	3384	3100	2959	2703	2413	2199	1928	1730	1579	1401	1264	1150	1043	938	847
Shaft & bearing diameter (inches)	1.188	1.188	1.438	1.688	1.688	1.688	1.688	1.688	1.938	1.938	1.938	2.188	2.188	2.438	2.688	2.938
Outlet area (square feet)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Plenum — SI Units																
Diameter Class I (millimeters)	343	381	419	464	508	565	622	686	762	838	927	1022	1130	1245	1378	1524
Maximum rpm	2895	2589	2376	2256	2077	1875	1691	1479	1328	1209	1073	972	882	799	725	651
Shaft & bearing diameter (millimeters)	25.4	25.4	25.4	30.2	30.2	30.2	36.5	42.9	42.9	42.9	42.9	49.2	49.2	55.6	61.9	61.9
Outlet area (square meters)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Diameter Class II (millimeters)	343	381	419	464	508	565	622	686	762	838	927	1022	1130	1245	1378	1524
Maximum rpm	3786	3384	3100	2959	2703	2413	2199	1928	1730	1579	1401	1264	1150	1043	938	847
Shaft & bearing diameter (millimeters)	30	30	37	42	42	42	42	42	49	49	49	56	56	62	68	75
Outlet area (square meters)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Figure 15. Torque for Inlet Vane at 100% WOV (lb.-in.)
for SWSI plenum fans with nested inlet vanes



NOTE: Inlet vanes on fans 30HP or more are not acceptable per ASHRAE 90.1-1999.

Table 13. Physical data - inline fans

Inline Fans — English Units												
Unit Size, Class I (inches)	150	165	182	200	222	245	270	300	330	365	402	445
Maximum rpm	2727	2488	2236	2041	1835	1665	1476	1330	1208	1072	973	880
Shaft & bearing diameter (inches)	1.438	1.438	1.438	1.438	1.438	1.438	1.688	1.938	2.188	2.188	2.188	2.438
Outlet area (square feet)	2.24	2.72	3.32	4.00	4.93	5.98	7.27	8.97	10.8	13.3	16.1	19.7
Unit Size, Class II (inches)	150	165	182	200	222	245	270	300	330	365	36.5	40.25
Maximum rpm	3409	3111	2796	2551	2294	2082	1846	1662	1510	1341	1216	1100
Shaft & bearing diameter (inches)	1.438	1.438	1.438	1.688	1.688	1.688	1.938	2.188	2.188	2.438	2.688	2.688
Outlet area (square feet)	2.24	2.72	3.32	4.00	4.93	5.98	7.27	8.97	10.8	13.3	16.1	19.7
Inline Fans — SI Units												
Unit Size, Class I (millimeters)	150	165	182	200	222	245	270	300	330	365	402	445
Maximum rpm	2727	2488	2236	2041	1835	1665	1476	1330	1208	1072	973	880
Shaft & bearing diameter (millimeters)	36.5	36.5	36.5	36.5	36.5	36.5	42.9	49.2	55.6	55.6	55.6	61.9
Outlet area (square meters)	0.21	0.25	0.31	0.37	0.46	0.56	0.68	0.83	1.00	1.24	1.50	1.83
Diameter Class II (millimeters)	150	165	182	200	222	245	270	300	330	365	36.5	40.25
Maximum rpm	3409	3111	2796	2551	2294	2082	1846	1662	1510	1341	1216	1100
Shaft & bearing diameter (millimeters)	36.5	36.5	36.5	42.9	42.9	42.9	49.2	55.6	55.6	61.9	68.3	68.3
Outlet area (square meters)	0.21	0.25	0.31	0.37	0.46	0.56	0.68	0.83	1.00	1.24	1.50	1.83

Table 14. Physical data - twin fans

Twin Fans — English Units						
Diameter Class I (inches)	9.5	10.62	12.62	15	18.12	20
Maximum rpm	2400	2000	1600	1400	1200	1000
Shaft & bearing diameter (inches)	1.438	1.438	1.688	2.188	2.438	2.688
Outlet area (square feet)	2.57	3.13	4.36	6.15	8.55	10.91
Twin Fans — SI Units						
Diameter Class I (millimeters)	241	270	321	381	460	508
Maximum rpm	2400	2000	1600	1400	1200	1000
Shaft & bearing diameter (millimeters)	36.5	36.5	42.9	55.6	61.9	68.3
Outlet area (square meters)	0.23	0.29	0.40	0.56	0.77	1.00

Coil data

Table 15. Unit coil dimensional data — English units of measure

STANDARD CHILLED WATER, DX, or 3 & 4 ROW HEATING COILS (English Units — Inches and square feet)							CLEANABLE COILS OR 1 & 2 ROW HEATING COILS (English Units — inches and square feet)					
Unit Size		Ext/Stagg	Large	Ext/Stagg	Medium	Small		Ext/Stagg	Large	Ext/Stagg	Medium	Small
		Large		Medium				Large		Medium		
003	FH x FL	18 x 3 1	18 x 25	12 x 31	12 x 25	—	FH x FL	18 x 28	18 x 22	12 x 28	12x22	—
	Face Area	3.9	3.1	2.6	2.1	—	Face Area	3.5	2.8	2.3	1.8	—
004	FH x FL	21 x 33	21 x 27	15 x 33	15 x 27	12 x 27	FH x FL	21 x 30	21 x 24	15 x 30	15x24	12 x 24
	Face Area	4.8	3.9	3.4	2.8	2.3	Face Area	4.4	3.5	3.1	2.5	2
006	FH x FL	21 x 45	21 x 39	15 x 45	15 x 39	12 x 39	FH x FL	21 x 42	21 x 36	15 x 42	15x36	12 x 36
	Face Area	6.6	5.7	4.7	4.1	3.3	Face Area	6.1	5.3	4.4	3.8	3
008	FH x FL	24 x 51	24 x 45	18 x 51	18 x 45	15 x 45	FH x FL	24 x 48	24 x 42	18 x 48	18x42	15 x 42
	Face Area	8.5	7.5	6.4	5.6	4.7	Face Area	8	7	6	5.3	4.4
010	FH x FL	27 x 57	27 x 51	21 x 57	21 x 51	18 x 51	FH x FL	27 x 54	27 x 48	21 x 54	21x48	18 x 48
	Face Area	10.7	9.6	8.3	7.4	6.4	Face Area	10.1	3350	7.9	7	6
012	FH x FL	33 x 59	33 x 53	24 x 59	24 x 53	21 x 53	FH x FL	33 x 56	9	24 x 56	24 x 50	21 x 50
	Face Area	13.5	12.1	9.8	8.8	7.7	Face Area	12.8	11.5	9.3	8.3	7.3
014	FH x FL	33 x 67	33 x 61	24 x 67	24 x 61	21 x 61	FH x FL	33 x 64	33 x 58	24 x 64	24 x 58	21 x 58
	Face Area	15.4	14	11.2	10.2	8.9	Face Area	14.7	13.3	10.7	9.7	8.5
017	FH x FL	36 x 73	36 x 67	27 x 73	27 x 67	24 x 67	FH x FL	36 x 70	36 x 64	27 x 70	27x64	24 x 64
	Face Area	18.3	16.8	13.7	12.6	11.2	Face Area	17.5	16	13.1	12	10.7
021	FH x FL	42 x 75	42 x 69	33 x 75	33 x 69	30 x 69	FH x FL	42 x 72	42 x 66	33 x 72	33 x 66	30 x 66
	Face Area	21.9	20.1	17.2	15.8	14.4	Face Area	21	19.3	16.5	15.1	13.8
025	FH x FL	48 x 79	48 x 73	—	36 x 73	33 x 73	FH x FL	—	—	42 x 70	36 x 70	33 x 70
	Face Area	26.3	24.3	—	18.2	16.7	Face Area	—	—	20.4	17.5	16
	FH x FL	(2) 27 x 73	(2) 24 x 73	(2)21 x 73	—	—	FH x FL	(2) 27 x 70	(2) 24 x 70	—	—	—
	Face Area	27.4	24.3	21.3	—	—	Face Area	26.2	23.3	—	—	—
030	FH x FL	48 x 91	48 x 85	—	36 x 85	33 x 85	FH x FL	—	—	42 x 82	36 x 82	33 x 82
	Face Area	30.3	28.3	—	21.2	19.5	Face Area	—	—	23.9	20.5	18.8
	FH x FL	(2) 27 x 85	(2) 24 x 85	(2)21 x 85	—	—	FH x FL	(2) 27 x 82	(2) 24 x 82	—	—	—
	Face Area	31.9	28.3	24.8	—	—	Face Area	30.8	27.3	—	—	—
035	FH x FL	54 x 95	54 x 89	—	42 x 89	39 x 89	FH x FL	—	—	—	42 x 86	39 x 86
	Face Area	35.6	33.4	—	24.1	22.3	Face Area	—	—	—	23.3	21.5
	FH x FL	(2) 30 x 89	(2) 27 x 89	(2)24 x 89	—	—	FH x FL	(2) 30 x 86	(2) 27 x 86	(2)24 x 86	—	—
	Face Area	37.1	33.4	29.7	—	—	Face Area	35.8	32.3	28.7	—	—
040	FH x FL	54 x 109	54 x 103	—	42 x 103	39 x 103	FH x FL	—	—	—	42x 100	39 x 100
	Face Area	40.9	38.6	—	30	27.9	Face Area	—	—	—	29.2	27.1
	FH x FL	(2) 30 x 103	(2) 27 x 103	(2) 24 x 103	—	—	FH x FL	(2) 30 x 100	(2) 27x 100	(2) 24 x 100	—	—
	Face Area	42.9	38.6	34.3	—	—	Face Area	41.7	37.5	33.3	—	—
050	FH x FL	—	—	—	48 x 107	42 x 107	FH x FL	—	—	—	—	42 x 104
	Face Area	—	—	—	35.7	31.2	Face Area	—	—	—	—	30.3
	FH x FL	(2) 39x107	(2) 33 x 107	(2) 27 x 107	(2) 24 x 107	—	FH x FL	(2) 39x104	(2) 33x104	(2) 27x104	(2) 24 x 104	—
	Face Area	58	49	40.1	35.7	—	Face Area	56.3	47.7	39	34.7	—
065	FH x FL	—	—	—	—	54 x 123	FH x FL	—	—	—	—	—
	Face Area	—	—	—	—	46.1	Face Area	—	—	—	—	—
	FH x FL	(2) 45 x 123	(2) 39 x 123	(2) 36 x 123	(2) 30 x 123	(2)27 x 123	FH x FL	—	(2) 39 x 120	(2) 36 x 120	(2) 30 x 120	(2) 27 x 120
	Face Area	76.9	66.6	61.5	51.2	46.1	Face Area	—	65	60	50	45
080	FH x FL	—	—	—	—	54 x 123	FH x FL	—	—	—	—	—
	Face Area	—	—	—	—	46.1	Face Area	—	—	—	—	—
	FH x FL	(2) 48 x 123	(2) 42 x 123	(2) 36 x 123	(2) 30 x 123	(2) 27 x 123	FH x FL	—	(2) 42 x 120	(2) 36 x 120	(2) 30 x 120	(2) 27 x 120
	Face Area	82	71.8	61.5	51.2	46.1	Face Area	—	70	60	50	45
085	FH x FL	—	—	—	—	—	FH x FL	—	—	—	—	—
	Face Area	—	—	—	—	—	Face Area	—	—	—	—	—
	FH x FL	(2) 51 x 123	(2) 45 x 123	(2) 39 x 123	(2) 33 x 123	(2) 30 x123	FH x FL	—	—	(2) 39 x 120	(2) 33 x 120	(2) 30 x 120
	Face Area	87.1	76.9	66.6	56.4	51.2	Face Area	—	—	65	55	50
090	FH x FL	—	—	—	—	—	FH x FL	—	—	—	—	—
	Face Area	—	—	—	—	—	Face Area	—	—	—	—	—
	FH x FL	(2) 54 x 123	(2) 48 x 123	(2) 42 x 123	(2) 36 x 123	(2) 33 x 123	FH x FL	—	—	(2) 42 x 120	(2) 36 x 120	(2) 33 x 120
	Face Area	92.2	82	71.8	61.5	56.4	Face Area	—	—	70	60	55

Coil section depth limitation by number of rows in coil for single banks of coils (excludes staggered coils)

Section depth	Maximum number of rows
12"	2 row water or steam
14"	4 row water (except 5WM & 5WD)
16"	4 row 5WM & 5WD water

Cooling only section (no moisture eliminator)

Section depth	Maximum number of rows
18"	4 row DX or water (except 5WM & 5WD)
24"	8 row DX or water
30"	10 row DX or 12 water

Combination cooling and 1 or 2 row reheat (no moisture eliminator)

Section depth	Maximum number of rows
24"	6 row water or DX
30"	10 row water or DX
36"	12 row water

Notes: The spacing between coil casings is a minimum of 4".
If more access between coils is required, increase the section depth.
Cooling coil sections are available in section depths of 18", 24", 30", 36", 42", 48", 54".
If a moisture eliminator is required, use the next larger section size.

Table 15. Unit coil dimensional data — SI units of measure

STANDARD CHILLED WATER, DX, or 3 & 4 ROW HEATING COILS (SI Units — millimeters and square meters)							CLEANABLE COILS OR 1 & 2 ROW HEATING COILS (SI Units — millimeters and square meters)					
Unit Size		Ext/Stagg	Large	Ext/Stagg	Medium	Small		Ext/Stagg	Large	Ext/Stagg	Medium	Small
		Large		Medium				Large		Medium		
003	FH x FL	457 x 787	457 x 635	305 x 787	305 x 635	—	FH x FL	457 x 711	457 x 559	305 x 711	305 x 559	—
	Face Area	0.36	0.29	0.24	0.20	—	Face Area	0.33	0.26	0.21	0.17	—
004	FH x FL	533 x 838	533 x 686	381 x 838	381 x 686	305 x 686	FH x FL	533 x 762	533 x 610	381 x 762	381 x 610	305 x 610
	Face Area	0.45	0.36	0.32	0.26	0.21	Face Area	0.41	0.33	0.29	0.23	0.19
006	FH x FL	533 x 1143	533 x 991	381 x 1143	381 x 991	305 x 991	FH x FL	533 x 1067	533 x 914	381 x 1067	381 x 914	305 x 914
	Face Area	0.61	0.53	0.44	0.38	0.31	Face Area	0.57	0.49	0.41	0.35	0.28
008	FH x FL	610 x 1295	610 x 1143	457 x 1295	457 x 1143	381 x 1143	FH x FL	610 x 1219	610 x 1067	457 x 1219	457 x 1067	381 x 1067
	Face Area	0.79	0.70	0.59	0.52	0.44	Face Area	0.74	0.65	0.56	0.49	0.41
010	FH x FL	686 x 1448	686 x 1295	533 x 1448	533 x 1295	457 x 1295	FH x FL	686 x 1372	686 x 1219	533 x 1372	533 x 1219	457 x 1219
	Face Area	0.99	0.89	0.77	0.69	0.59	Face Area	0.94	0.84	0.73	0.65	0.56
012	FH x FL	838 x 1499	838 x 1346	610 x 1499	610 x 1346	533 x 1346	FH x FL	838 x 1422	838 x 1270	610 x 1422	610 x 1270	533 x 1270
	Face Area	1.25	1.12	0.91	0.82	0.72	Face Area	1.19	1.07	0.86	0.77	0.68
014	FH x FL	838 x 1702	838 x 1549	610 x 1702	610 x 1549	533 x 1549	FH x FL	838 x 1626	838 x 1473	610 x 1626	610 x 1473	533 x 1473
	Face Area	1.43	1.30	1.04	0.95	0.83	Face Area	1.37	1.24	0.99	0.90	0.79
017	FH x FL	914 x 1854	914 x 1702	686 x 1854	686 x 1702	610 x 1702	FH x FL	914 x 1778	914 x 1626	686 x 1854	686 x 1626	610 x 1676
	Face Area	1.70	1.56	1.27	1.17	1.04	Face Area	1.63	1.49	1.22	1.11	0.99
021	FH x FL	1067x1905	1067x1753	838x1905	838x1753	762x1753	FH x FL	1067x 1829	1067x 1676	838 x 1829	838 x 1676	762 x 1676
	Face Area	2.03	1.87	1.60	1.47	1.34	Face Area	1.95	1.79	1.53	1.40	1.28
025	FH x FL	1219x2007	1219x1854	—	914 x 1854	838 x 1854	FH x FL	—	—	1067x 1778	914 x 1778	838 x 1778
	Face Area	2.44	2.26	—	1.89	1.55	Face Area	—	—	1.89	1.63	1.49
	FH x FL	(2)686x1854	(2)610x1854	(2)533x1854	—	—	FH x FL	(2)686x1778	(2)610x1778	—	—	—
	Face Area	2.55	2.60	1.98	—	—	Face Area	2.43	2.16	—	—	—
030	FH x FL	1219 x 2312	1219 x 2159	—	914 x 2159	838 x 2159	FH x FL	—	—	1067x 2083	914 x 2083	838 x 2083
	Face Area	2.81	2.63	—	1.97	1.81	Face Area	—	—	2.22	1.90	1.75
	FH x FL	(2)686x2159	(2)610x2159	(2)533x2159	—	—	FH x FL	(2)686x2083	(2)610x2083	—	—	—
	Face Area	3.45	2.63	2.30	—	—	Face Area	2.86	2.54	—	—	—
035	FH x FL	1372 x 2413	1372 x 2261	—	991 x 2261	914 x 2261	FH x FL	—	—	—	1067 x 2184	991 x 2184
	Face Area	3.31	3.10	—	2.24	2.07	Face Area	—	—	—	2.33	2.16
	FH x FL	(2) 762x2261	(2)686x2261	(2)607x2261	—	—	FH x FL	(2)762x2184	(2)686x2184	(2)610x2184	—	—
	Face Area	3.46	3.10	2.76	—	—	Face Area	3.33	3.00	2.66	—	—
040	FH x FL	1372 x 2769	1372 x 2616	—	1067 x 2616	991 x 2616	FH x FL	—	—	—	1067 x 2540	991 x 2540
	Face Area	3.80	3.59	—	2.79	2.59	Face Area	—	—	—	2.71	2.52
	FH x FL	(2)762x2616	(2)686x2616	(2)610x2616	—	—	FH x FL	(2)762x2540	(2)686x2540	(2)610x2540	—	—
	Face Area	3.99	3.59	3.18	—	—	Face Area	3.87	3.48	3.09	—	—
050	FH x FL	—	—	—	1219 X 2718	1067 X 2718	FH x FL	—	—	—	—	1067x2642
	Face Area	—	—	—	3.31	2.90	Face Area	—	—	—	—	2.81
	FH x FL	(2)991x2718	(2)838x2718	(2)686x2718	(2)610x2718	—	FH x FL	(2)991x2642	(2)838x2642	(2)686x2642	(2)610x2642	—
	Face Area	5.39	4.55	3.72	3.32	—	Face Area	5.23	4.43	3.62	3.22	—
065	FH x FL	—	—	—	—	1372 x 3124	FH x FL	—	—	—	—	—
	Face Area	—	—	—	—	4.28	Face Area	—	—	—	—	—
	FH x FL	(2)1143x3124	(2)991x3124	(2)915x3124	(2)762x3124	(2)686x3124	FH x FL	—	(2)991x3048	(2)915x3048	(2)762x3048	(2)686x3048
	Face Area	7.14	6.19	5.71	4.76	4.28	Face Area	—	6.04	5.57	4.65	4.18
080	FH x FL	—	—	—	—	1372 x 3124	FH x FL	—	—	—	—	—
	Face Area	—	—	—	—	4.28	Face Area	—	—	—	—	—
	FH x FL	(2)1219x3124	(2)1067x3124	(2)915x3124	(2)762x3124	(2)686x3124	FH x FL	—	(2)1067x3048	(2)915x3048	(2)762x3048	(2)686x3048
	Face Area	7.61	6.67	5.71	4.75	4.28	Face Area	—	6.50	5.57	4.64	4.18
085	FH x FL	—	—	—	—	—	FH x FL	—	—	—	—	—
	Face Area	—	—	—	—	—	Face Area	—	—	—	—	—
	FH x FL	(2)1296x3124	(2)1143x3124	(2)991x3124	(2)838x3124	(2)762x3124	FH x FL	—	—	(2)991x3048	(2)838x3048	(2)762x3048
090	Face Area	8.09	7.14	6.18	5.24	4.75	Face Area	—	—	6.04	5.11	4.64
	FH x FL	—	—	—	—	—	FH x FL	—	—	—	—	—
	Face Area	—	—	—	—	—	Face Area	—	—	—	—	—
090	FH x FL	(2)1372x3124	(2)1219x3124	(2)1067x3124	(2)915x3124	(2)838x3124	FH x FL	—	—	(2)1067x3048	(2)915x3048	(2)838x3048
	Face Area	8.56	7.61	6.67	5.71	5.24	Face Area	—	—	6.50	5.57	5.11

Coil section depth limitation by number of rows in coil for single banks of coils (excludes staggered coils)

Section depth	Maximum number of rows
305 mm	2 row water or steam
356 mm	4 row water (except 5WM & 5WD)
406 mm	4 row 5WM & 5WD water

Cooling only section (no moisture eliminator)

Section depth	Maximum number of rows
457 mm	4 row DX or water (except 5WM & 5WD)
610 mm	8 row DX or water
762 mm	10 row DX or 12 water

Combination cooling and 1 or 2 row reheat (no moisture eliminator)

Section depth	Maximum number of rows
610 mm	6 row water or DX
762 mm	10 row water or DX
914 mm	12 row water

Notes: The spacing between coil casings is a minimum of 102 mm. If more access between coils is required, increase the section depth. Cooling coil sections are available in section depths of 457 mm, 610 mm, 762 mm, 914 mm, 1067 mm, 1219 mm, 1372 mm. If a moisture eliminator is required, use the next larger section size.

Multizone coils

The face area of the multizone cold deck coil is the same as other cooling coils shown in Table 15. Multizone hot deck coils have a face area that is different from other heating coils. The face area for the multizone hot deck is generally smaller.

The minimum section depth for multizone sections is shown on pages 55-58. Additional access is available downstream from the cooling coil. Refer to Vision SelectTOOLS™ Software for more detailed information.

MULTIZONE HOT DECK COILS (English Units — inches and square feet)			
Unit Size		1 & 2 Row	3 & 4 Row
006	FH x FL	12 x 36	12 x 39
	Face Area	3	3.3
008	FH x FL	15 x 42	15 x 45
	Face Area	4.4	4.7
010	FH x FL	15 x 48	15 x 51
	Face Area	5	5.3
012	FH x FL	18 x 50	18 x 53
	Face Area	6.3	6.6
014	FH x FL	18 x 58	18 x 61
	Face Area	7.3	7.6
017	FH x FL	21 x 64	21 x 67
	Face Area	9.3	9.8
021	FH x FL	24 x 66	24 x 69
	Face Area	11	11.5
025	FH x FL	27 x 70	27 x 73
	Face Area	13.1	13.7
030	FH x FL	27 x 82	27 x 85
	Face Area	15.4	15.9
035	FH x FL	30 x 86	30 x 89
	Face Area	17.9	18.5
040	FH x FL	30 x 100	30 x 103
	Face Area	20.8	21.5
050	FH x FL	36 x 104	36 x 107
	Face Area	26	26.8
065	FH x FL	42 x 120	42 x 123
	Face Area	35	35.9
080	FH x FL	N/A	N/A
	Face Area	N/A	N/A
085	FH x FL	N/A	N/A
	Face Area	N/A	N/A
090	FH x FL	N/A	N/A
	Face Area	N/A	N/A

MULTIZONE HOT DECK COILS (SI units - millimeters and square meters)			
Unit Size		1 & 2 Row	3 & 4 Row
006	FH x FL	305 x 914	305 x 991
	Face Area	0.28	0.31
008	FH x FL	381 x 1067	381 x 1143
	Face Area	0.41	0.44
010	FH x FL	381 x 1219	381 x 1295
	Face Area	0.46	0.49
012	FH x FL	457 x 1270	457 x 1346
	Face Area	0.59	0.61
014	FH x FL	457 x 1473	457 x 1549
	Face Area	0.68	0.71
017	FH x FL	533 x 1626	533 x 1702
	Face Area	0.86	0.91
021	FH x FL	610 x 1676	610 x 1753
	Face Area	1.02	1.07
025	FH x FL	686 x 1778	686 x 1854
	Face Area	1.22	1.27
030	FH x FL	686 x 2083	686 x 2159
	Face Area	1.43	1.48
035	FH x FL	762 x 2184	762 x 2261
	Face Area	1.66	1.72
040	FH x FL	762 x 2540	762 x 2616
	Face Area	1.93	2
050	FH x FL	914 x 2642	914 x 2718
	Face Area	2.42	2.49
065	FH x FL	1067 x 3048	1067 x 3124
	Face Area	3.25	3.34
080	FH x FL	N/A	N/A
	Face Area	N/A	N/A
085	FH x FL	N/A	N/A
	Face Area	N/A	N/A
090	FH x FL	N/A	N/A
	Face Area	N/A	N/A

Side load filter data

Table 16. Size and quantity of filters used per unit size

UNIT SIZE	FILTER SIZE (Inches)	NUMBER OF FILTERS USED PER FILTER SECTION						
		Flat Panel	2" Angular	4" Angular	Varicel SH	Varicel II MH	Varicel V	DriPak 2000
003	12 X 24		2					
	20 X 20							
	20 X 24		2					
	24 X 12			N/A				
	24 X 20	1			1	1	1	1
004	24 X 24							
	12 X 24	1	2		1	1	1	1
	20 X 20							
	20 X 24							
	24 X 12			N/A				
006	24 X 20							
	24 X 24	1	2		1	1	1	1
	12 X 24							
	20 X 20							
	20 X 24							
008	24 X 12			N/A				
	24 X 20							
	24 X 24	2	4		2	2	2	2
	12 X 24	1	2	2	1	1	1	1
	20 X 20							
010	20 X 24	1	2	2	2	2	2	2
	24 X 12							
	24 X 20							
	24 X 24	1	2	2				
	12 X 24				1			
012	20 X 20							
	20 X 24		4	4				
	24 X 12	2			2	2	2	2
	24 X 20							
	24 X 24	2	2	2	2	2	2	2
014	12 X 24	1			1	1	1	1
	20 X 20							
	20 X 24				1	1	1	1
	24 X 12	3			2	2	2	2
	24 X 20							
017	24 X 24	3	6	6	2	2	2	2
	12 X 24		3	3				
	20 X 20							
	20 X 24		6	6				
	24 X 12							
021	24 X 20	6			6	6	6	6
	24 X 24		3	3				
	12 X 24							
	20 X 20	4			4	4		4
	20 X 24	4	12	12	4	4		4
025	24 X 12							
	24 X 20						3	
	24 X 24						3	
	12 X 24				2	2	2	2
	20 X 20							
030	20 X 24	6	9	9				
	24 X 12							
	24 X 20							
	24 X 24	2	3	3	6	6	6	6
	12 X 24							
035	20 X 20							
	20 X 24							
	24 X 12							
	24 X 20						12	
	24 X 24							

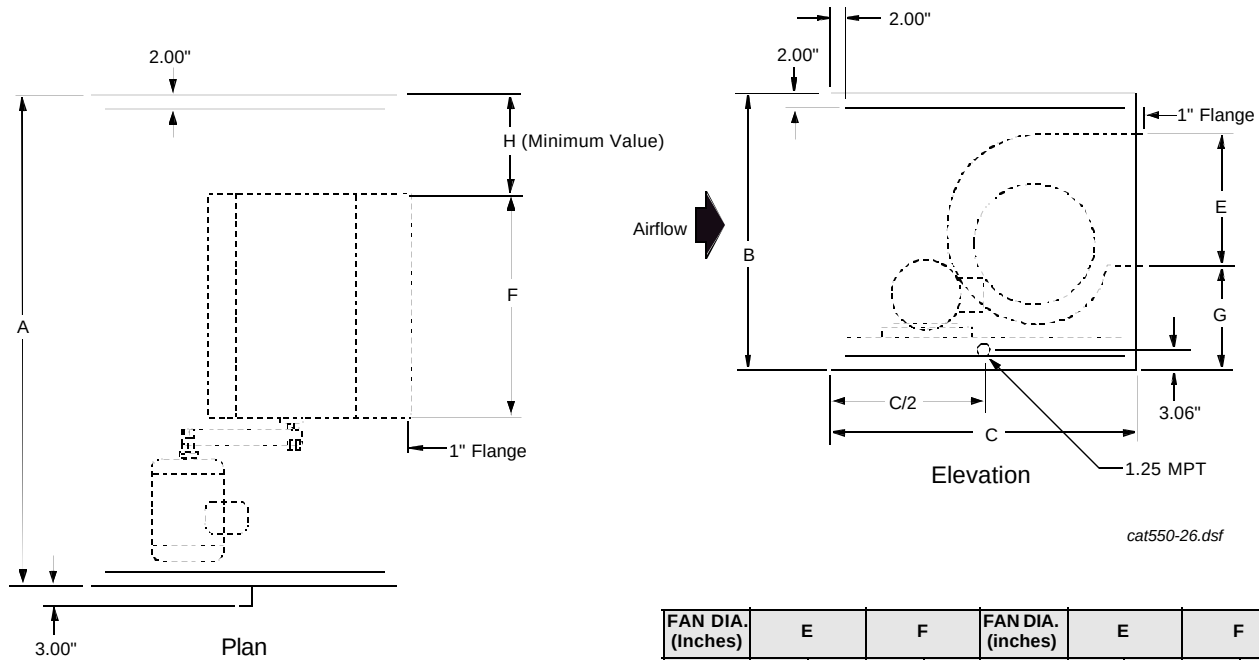
Table 16. Size and quantity of filters used per unit size (continued)

UNIT SIZE	FILTER SIZE (Inches)	NUMBER OF FILTERS USED PER FILTER SECTION						
		Flat Panel	2" Angular	4" Angular	Varicel SH	Varicel II MH	Varicel V	DriPak 2000
040	12 X 24		4	4			2	
	20 X 20	6			6	6		6
	20 X 24		16	16				
	24 X 12						4	
	24 X 20	9			9	9		9
	24 X 24		4	4			8	
050	12 X 24							
	20 X 20							
	20 X 24	3	24	24	3	3	3	3
	24 X 12							
	24 X 20							
	24 X 24	12			12	12	12	12
065	12 X 24				3		3	
	20 X 20	9				9		9
	20 X 24	3	10	10		3		3
	24 X 12				5		5	
	24 X 20	9				9		9
	24 X 24	3	20	20	15	3	15	3
080	12 X 24				3		3	
	20 X 20	3				3		3
	20 X 24	9	10	10		9		9
	24 X 12							
	24 X 20	3			5	3	5	3
	24 X 24	9	20	20	15	9	15	9
085	12 X 24	4			4	4	4	4
	20 X 20							
	20 X 24	20	12	12				
	24 X 12							
	24 X 20							
	24 X 24	4	24	24	20	20	20	20
090	12 X 24						3	
	20 X 20	12			15	12		12
	20 X 24	3	12	12		3		3
	24 X 12						5	
	24 X 20	12			15	12	5	12
	24 X 24	3	24	24		3	15	3

Notes: Number of filters used per flat panel, 2" angular, and 4" angular filter section is based on Pleated (30%).

Dimensional data

Top horizontal housed fan discharge



Note:

The H dimension shown is the minimum value.
This dimension will vary with the motor size used.

FAN DIA. (Inches)	E		F		FAN DIA. (Inches)	E		F	
	in.	mm	in.	mm		in.	mm	in.	mm
FC					Airfoil				
9x4	11.05	281	7.61	193	13.22	16.68	424	19.94	506
9x7	11.05	281	9.99	254	14.56	20.18	513	22.00	559
9x9	11.05	281	12.61	320	16.19	22.37	568	24.31	617
10.62	12.18	309	13.92	354	19.69	24.13	613	29.70	754
12.62	14.24	362	16.42	417	21.56	27.27	693	32.45	824
15	16.68	424	19.42	493	24	29.51	750	35.95	913
18	19.68	500	22.68	576					
20	27.27	693	28.98	736					
22.25	29.51	750	32.23	819					
24.5	32.51	826	35.45	900					

UNIT SIZE (fan diameter)	A**		B**		C		G		H (min.)	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
003 (9 x 4, 9 x 7 FC)*	38.00	965	26.00	660	32.00	813	9.93	252	8.18	208
004 (9 x 9 FC)*	40.00	1016	30.00	762	32.00	813	9.93	252	8.18	208
006 (9 x 7, 9 x 9 FC)*	52.00	1321	30.00	762	32.00	813	9.93	252	8.18	208
006 (10.62 FC)*	52.00	1321	30.00	762	36.00	914	12.74	324	9.43	240
008 (12.62 FC)	58.00	1473	34.00	864	40.00	1016	14.61	371	10.83	275
008 (13.22 AF)	58.00	1473	34.00	864	40.00	1016	13.88	353	11.25	286
010 (15 FC)	64.00	1626	36.00	914	42.00	1067	15.55	395	12.50	318
010 (13.22 AF)	64.00	1626	36.00	914	40.00	1016	13.88	353	11.25	286
012 (18 FC)	66.00	1676	42.00	1067	46.00	1168	16.80	427	14.62	371
012 (16.19 AF)	66.00	1676	42.00	1067	46.00	1168	14.28	363	13.33	339
014 (18 FC)	74.00	1880	42.00	1067	46.00	1168	16.80	427	14.62	371
014 (16.19 AF)	74.00	1880	42.00	1067	46.00	1168	14.28	363	13.33	339
017 (18 FC)	80.00	2032	46.00	1168	46.00	1168	16.80	427	14.62	371
017 (19.69 AF)	80.00	2032	46.00	1168	50.00	1270	16.09	409	15.78	401
021 (20 FC)	82.00	2083	52.00	1321	52.00	1321	15.82	402	16.00	406
021 (21.56 AF)	82.00	2083	52.00	1321	52.00	1321	16.06	408	17.09	434
025 (24.50 FC)*	86.00	2184	60.00	1524	60.00	1524	18.63	473	19.15	486
025 (24 AF)	86.00	2184	60.00	1524	58.00	1473	17.55	446	18.80	478
030 (24.50 FC)	98.00	2489	60.00	1524	60.00	1524	18.63	473	19.15	486
030 (24 AF)	98.00	2489	60.00	1524	58.00	1473	17.55	446	18.80	478
035 (24.50 FC)	102.00	2591	66.00	1676	60.00	1524	18.63	473	19.15	486
035 (24 AF)	102.00	2591	66.00	1676	58.00	1473	17.55	446	18.80	478

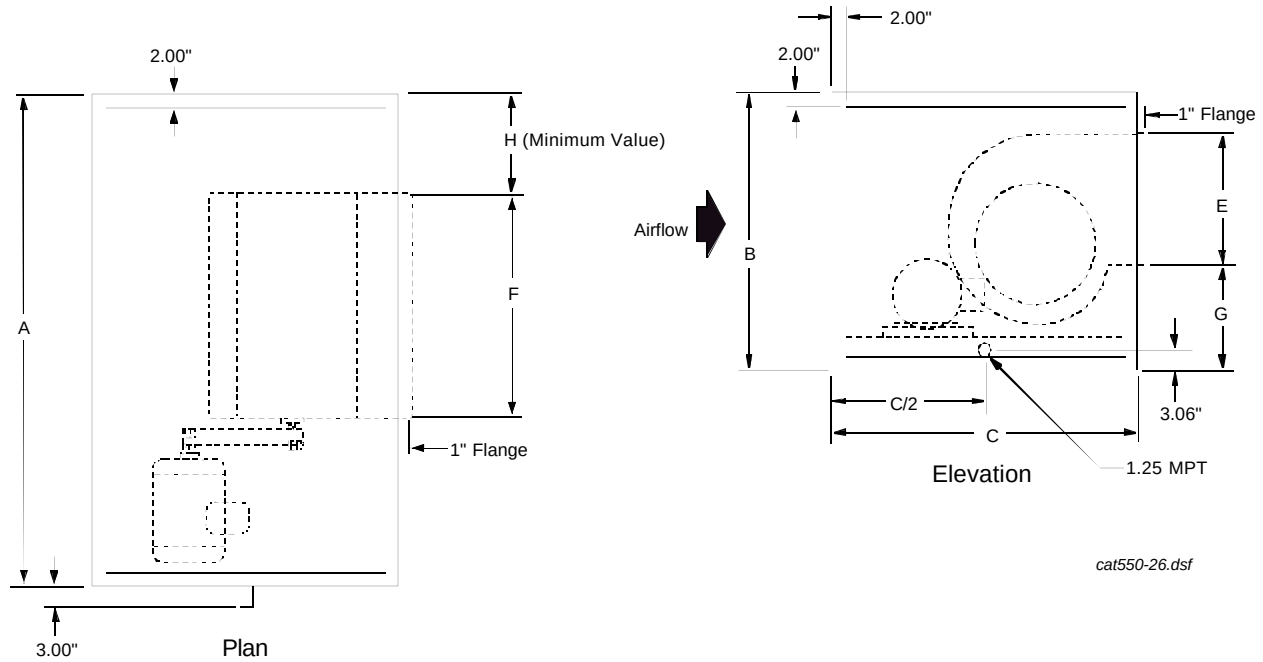
* Not available with vanes.

**These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height.

A base rail is mandatory on unit sizes 025 and above and will add a minimum of 6" (152 mm) to the unit height.

An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Top horizontal housed fan discharge



Note:

The H dimension shown is the minimum value.
This dimension will vary with the motor size used.

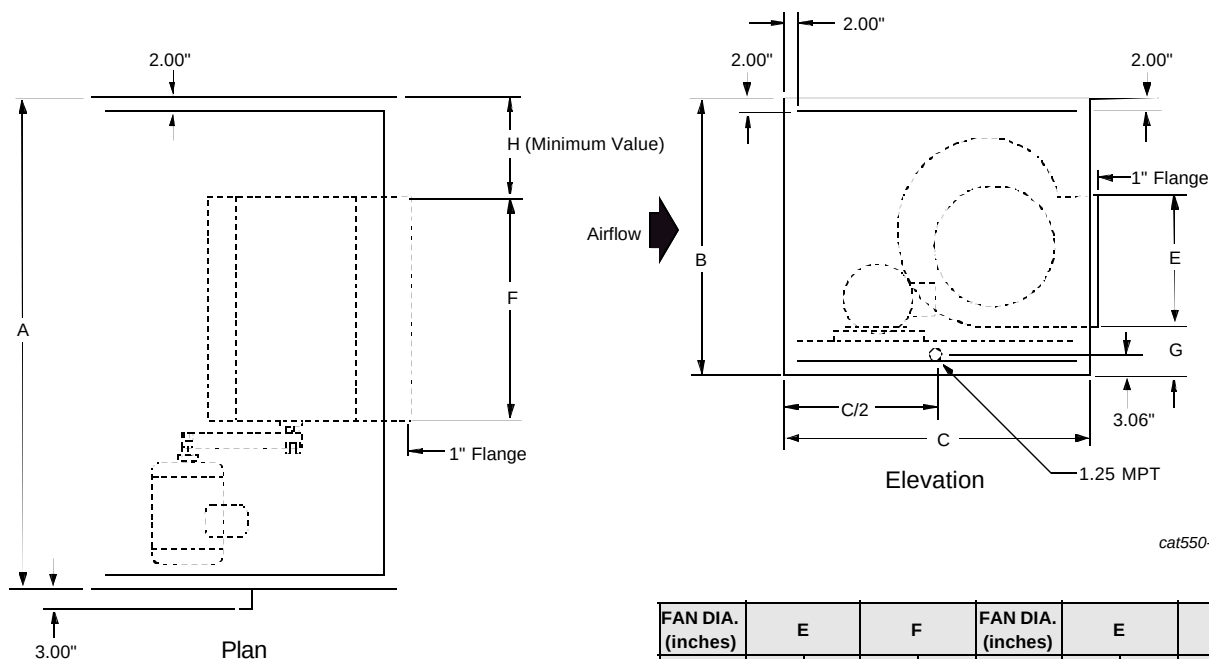
FAN DIA. (inches)	E		F		FAN DIA. (inches)	E		F	
FC	in.	mm	in.	mm	Airfoil	in.	mm	in.	mm
20	25.55	649	25.55	649	20	22.05	560	29.49	749
22.38	28.05	712	28.05	712	22.25	24.43	621	32.18	817
25	32.05	814	32.05	814	24.5	26.74	679	35.68	906
27.62	35.05	890	35.05	890	27	29.43	748	39.11	993
30.25	37.55	954	37.55	954	30	32.61	828	43.24	1098
33	43.74	1111	40.55	1030	33	35.74	908	47.99	1219
36	43.74	1111	43.55	1106	36.5	39.49	1003	52.55	1335
40.25	43.36	1101	58.30	1481	40.25	43.36	1101	58.30	1481

UNIT SIZE (fan diameter)	A**		B**		C		G		H (min.)	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
040 (27.62 FC)	116.00	2946	68.00	1727	74.00	1880	26.28	668	40.47	1028
040 (27 AF)	116.00	2946	68.00	1727	70.00	1778	33.10	841	38.45	977
050 (36 FC)	120.00	3048	80.00	2032	82.00	2083	25.90	658	36.22	971
050 (33 AF)	120.00	3048	80.00	2032	80.00	2032	38.15	969	30.00	863
065 (40.25 FC)*	136.00	3454	92.00	2337	92.00	2337	43.65	1109	34.85	987
065 (40.25 AF)	136.00	3454	92.00	2337	92.00	2337	43.65	1109	34.85	987
080 (40.25 FC)*	136.00	3454	98.00	2489	92.00	2337	43.65	1109	34.85	987
080 (40.25 AF)	136.00	3454	98.00	2489	92.00	2337	43.65	1109	34.85	987
085 (40.25 FC)*	136.00	3454	104.00	2641	92.00	2337	43.65	1109	34.85	987
085 (40.25 AF)	136.00	3454	104.00	2641	92.00	2337	43.65	1109	34.85	987
090 (40.25 FC)*	136.00	3454	110.00	2794	92.00	2337	43.65	1109	34.85	987
090 (40.25 AF)	136.00	3454	110.00	2794	92.00	2337	43.65	1109	34.85	987

* Not available with vanes.

**These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height.
A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Bottom horizontal housed fan discharge



Note:

The H dimension shown is the minimum value.
This dimension will vary with the motor size used.

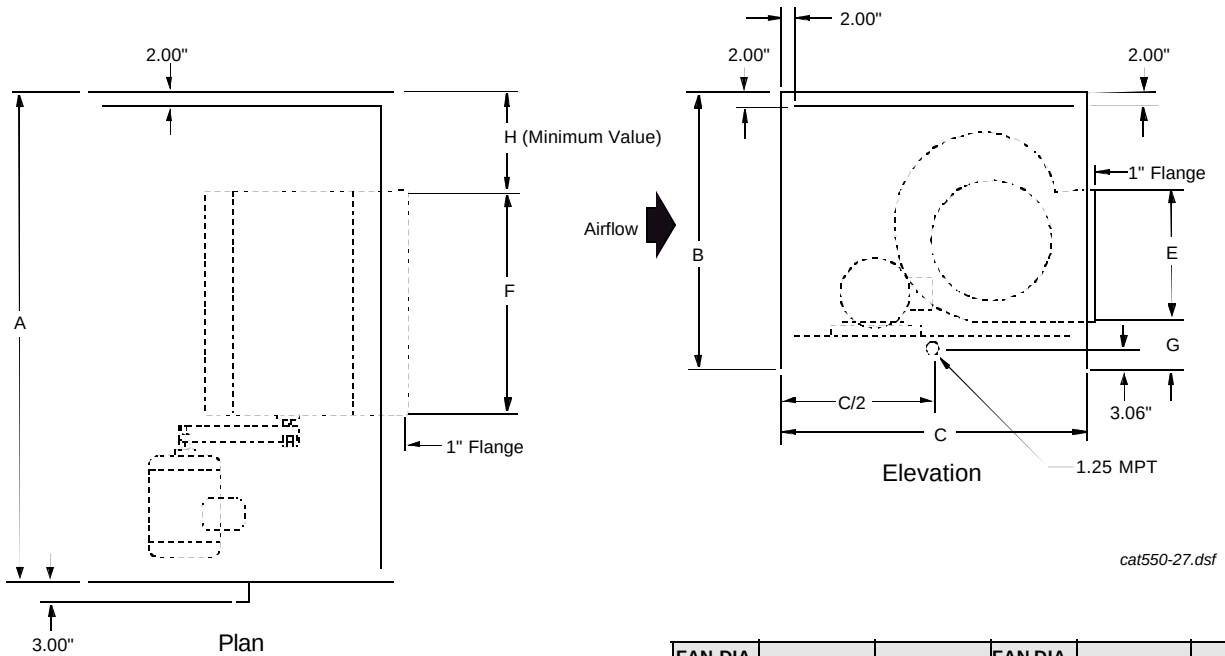
FAN DIA. (inches)	E		F		FAN DIA. (inches)	E		F	
FC	in.	mm	in.	mm	Airfoil	in.	mm	in.	mm
9x4	11.05	281	7.61	193	13.22	16.68	424	19.94	506
9x7	11.05	281	9.99	254	14.56	20.18	513	22.00	559
9x9	11.05	281	12.61	320	16.19	22.37	568	24.31	617
10.62	12.18	309	13.92	354	19.69	24.13	613	29.70	754
12.62	14.24	362	16.42	417	21.56	27.27	693	32.45	824
15	16.68	424	19.42	493	24	29.51	750	35.95	913
18	19.68	500	22.68	576					
20	27.27	693	28.98	736					
22.25	29.51	750	32.23	819					
24.5	32.51	826	35.45	900					

UNIT SIZE (fan diameter)	A**		B**		C		G		H (min.)	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
003 (9 x 4, 9 x 7 FC)*	38.00	965	26.00	660	32.00	813	6.43	163	8.18	208
004 (9 x 9 FC)*	40.00	1016	30.00	762	32.00	813	6.43	163	8.18	208
006 (9 x 7, 9 x 9 FC)*	52.00	1321	30.00	762	32.00	813	6.43	163	8.18	208
006 (10.62 FC)*	52.00	1321	30.00	762	36.00	914	8.50	216	9.43	240
008 (12.62 FC)	58.00	1473	34.00	864	40.00	1016	8.50	216	10.83	275
008 (13.22 AF)	58.00	1473	34.00	864	40.00	1016	8.50	216	11.25	286
010 (15 FC)	64.00	1626	36.00	914	42.00	1067	8.50	216	12.50	318
010 (13.22 AF)	64.00	1626	36.00	914	40.00	1016	8.50	216	11.25	286
012 (18 FC)	66.00	1676	42.00	1067	46.00	1168	8.50	216	14.62	371
012 (16.19 AF)	66.00	1676	42.00	1067	46.00	1168	8.50	216	13.33	339
014 (18 FC)	74.00	1880	42.00	1067	46.00	1168	8.50	216	14.62	371
014 (16.19 AF)	74.00	1880	42.00	1067	46.00	1168	8.50	216	13.33	339
017 (18 FC)	80.00	2032	46.00	1168	46.00	1168	8.50	216	14.62	371
017 (19.69 AF)	80.00	2032	46.00	1168	50.00	1270	8.50	216	15.78	401
021 (20 FC)	82.00	2083	52.00	1321	52.00	1321	8.50	216	16.00	406
021 (21.56 AF)	82.00	2083	52.00	1321	52.00	1321	8.50	216	17.09	434
025 (24.50 FC)*	86.00	2184	60.00	1524	60.00	1524	8.50	216	19.15	486
025 (24 AF)	86.00	2184	60.00	1524	58.00	1473	8.50	216	18.80	478
030 (24.50 FC)	98.00	2489	60.00	1524	60.00	1524	8.50	216	19.15	486
030 (24 AF)	98.00	2489	60.00	1524	58.00	1473	8.50	216	18.80	478
035 (24.50 FC)	102.00	2591	66.00	1676	60.00	1524	8.50	216	19.15	486
035 (24 AF)	102.00	2591	66.00	1676	58.00	1473	8.50	216	18.80	478

* Not available with vanes.

**These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height.
A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Bottom horizontal housed fan discharge



cat550-27.dsf

Note:

The H dimension shown is the minimum value.
This dimension will vary with the motor size used.

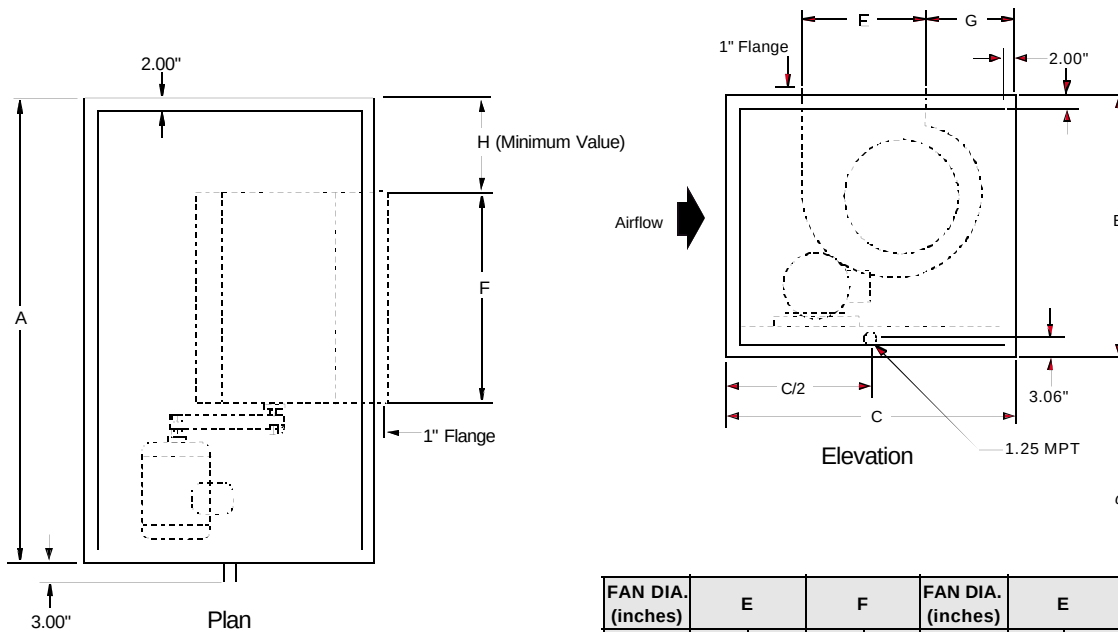
FAN DIA. (inches)	E		F		FAN DIA. (inches)	E		F	
FC	in.	mm	in.	mm	Airfoil	in.	mm	in.	mm
20	25.55	649	25.55	649	20	22.05	560	29.49	749
22.38	28.05	712	28.05	712	22.25	24.43	621	32.18	817
25	32.05	814	32.05	814	24.5	26.74	679	35.68	906
27.62	35.05	890	35.05	890	27	29.43	748	39.11	993
30.25	37.55	954	37.55	954	30	32.61	828	43.24	1098
33	43.74	1111	40.55	1030	33	35.74	908	47.99	1219
36	43.74	1111	43.55	1106	36.5	39.49	1003	52.55	1335
40.25	43.36	1101	58.30	1481	40.25	43.36	1101	58.30	1481

UNIT SIZE (fan diameter)	A**		B**		C		G		H (min.)	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
040 (27.62 FC)	116.00	2946	68.00	1727	74.00	1880	13.03	331	40.47	1028
040 (27 AF)	116.00	2946	68.00	1727	70.00	1778	13.47	342	38.45	977
050 (36 FC)	120.00	3048	80.00	2032	82.00	2083	13.09	333	36.22	971
050 (33 AF)	120.00	3048	80.00	2032	80.00	2032	14.21	361	30.00	863
065 (40.25 FC)*	136.00	3454	92.00	2337	92.00	2337	14.09	358	34.85	987
065 (40.25 AF)	136.00	3454	92.00	2337	92.00	2337	14.09	358	34.85	987
080 (40.25 FC)*	136.00	3454	98.00	2489	92.00	2337	14.09	358	34.85	987
080 (40.25 AF)	136.00	3454	98.00	2489	92.00	2337	14.09	358	34.85	987
085 (40.25 FC)*	136.00	3454	104.00	2641	92.00	2337	14.09	358	34.85	987
085 (40.25 AF)	136.00	3454	104.00	2641	92.00	2337	14.09	358	34.85	987
090 (40.25 FC)*	136.00	3454	110.00	2794	92.00	2337	14.09	358	34.85	987
090 (40.25 AF)	136.00	3454	110.00	2794	92.00	2337	14.09	358	34.85	987

* Not available with vanes.

**These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height.
A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Upblast housed fan discharge



cat550-28.dsf

Note:

The H dimension shown is the minimum value.
This dimension will vary with the motor size used.

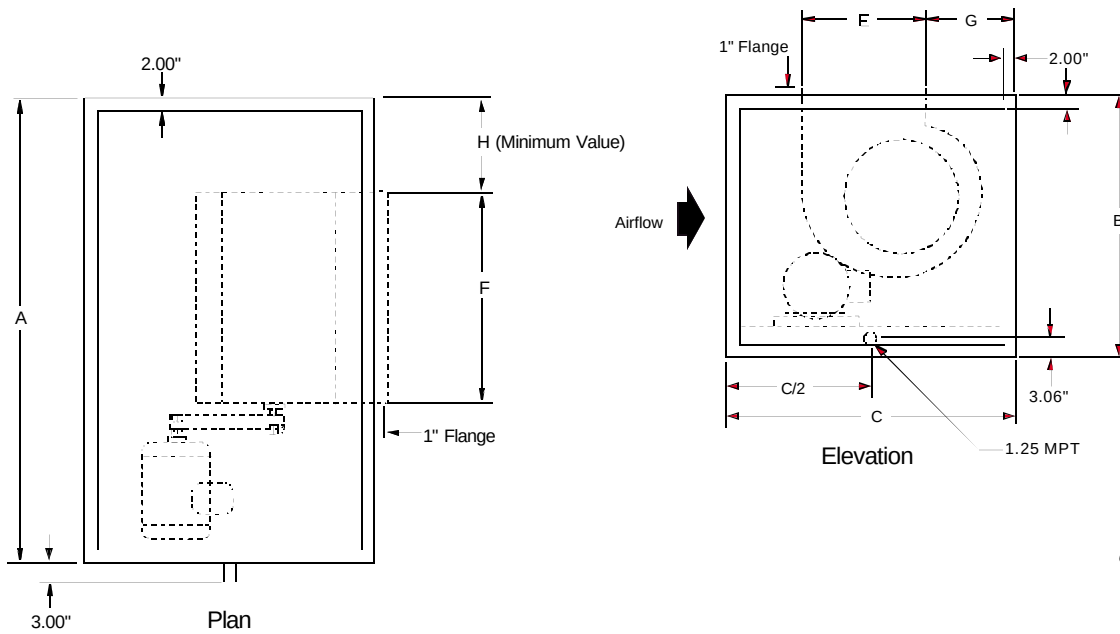
FAN DIA. (inches)	E		F		FAN DIA. (inches)	E		F	
FC	in.	mm	in.	mm	Airfoil	in.	mm	in.	mm
9x4	11.05	281	7.61	193	13.22	16.68	424	19.94	506
9x7	11.05	281	9.99	254	14.56	20.18	513	22.00	559
9x9	11.05	281	12.61	320	16.19	22.37	568	24.31	617
10.62	12.18	309	13.92	354	19.69	24.13	613	29.70	754
12.62	14.24	362	16.42	417	21.56	27.27	693	32.45	824
15	16.68	424	19.42	493	24	29.51	750	35.95	913
18	19.68	500	22.68	576					
20	27.27	693	28.98	736					
22.25	29.51	750	32.23	819					
24.5	32.51	826	35.45	900					

UNIT SIZE (fan diameter)	A**		B**		C		G		H (min.)	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
003 (9 x 4, 9 x 7 FC)*	38.00	965	26.00	660	32.00	813	8.49	216	8.18	208
004 (9 x 7, 9 x 9 FC)*	40.00	1016	30.00	762	32.00	813	8.49	216	8.18	208
006 (10.62 FC)*	52.00	1321	30.00	762	36.00	914	12.03	306	9.43	240
008 (15 FC)	58.00	1473	34.00	864	42.00	1067	13.76	350	12.50	318
008 (13.22 AF)	58.00	1473	34.00	864	40.00	1016	12.30	312	11.25	286
010 (18 FC)	64.00	1626	36.00	914	46.00	1168	15.14	385	14.62	371
010 (16.19 AF)	64.00	1626	36.00	914	46.00	1168	12.56	319	13.33	339
012 (18 FC)	66.00	1676	42.00	1067	46.00	1168	15.17	385	14.62	371
012 (16.19 AF)	66.00	1676	42.00	1067	46.00	1168	12.56	319	13.33	339
014 (20 FC)	74.00	1880	42.00	1067	52.00	1321	14.07	357	16.00	406
014 (19.69 AF)	74.00	1880	42.00	1067	50.00	1270	14.54	369	15.78	401
017 (20 FC)	80.00	2032	46.00	1168	52.00	1321	14.07	357	16.00	406
017 (21.56 AF)	80.00	2032	46.00	1168	52.00	1321	14.16	360	17.09	434
021 (20 FC)	82.00	2083	52.00	1321	52.00	1321	14.07	357	16.00	406
021 (21.56 AF)	82.00	2083	52.00	1321	52.00	1321	14.16	360	17.09	434
025 (24.50 FC)*	86.00	2184	60.00	1524	60.00	1524	16.34	415	19.15	486
025 (24 AF)	86.00	2184	60.00	1524	58.00	1473	15.48	393	18.80	478
030 (24.50 FC)	98.00	2489	60.00	1524	60.00	1524	16.34	415	19.15	486
030 (24 AF)	98.00	2489	60.00	1524	58.00	1473	15.48	393	18.80	478
035 (24.50 FC)	102.00	2591	66.00	1676	60.00	1524	16.34	415	19.15	486
035 (24 AF)	102.00	2591	66.00	1676	58.00	1473	15.48	393	18.80	478

* Not available with vanes.

**These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height. A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Upblast housed fan discharge



cat550-28.dsf

Note:

The H dimension shown is the minimum value.
This dimension will vary with the motor size used.

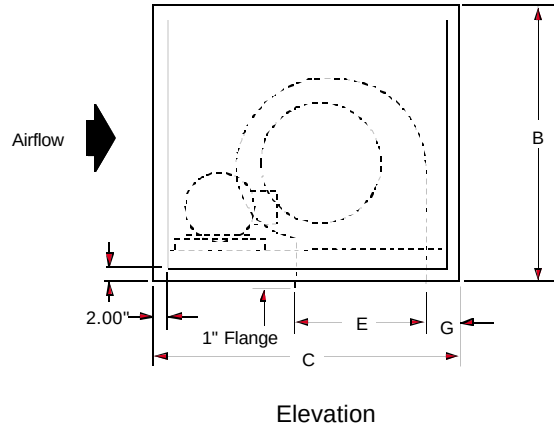
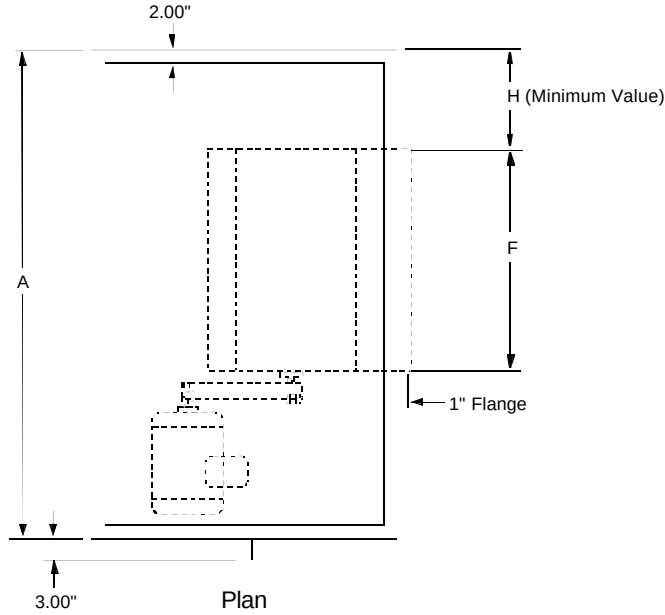
FAN DIA. (inches)	E		F		FAN DIA. (inches)	E		F	
FC	in.	mm	in.	mm	Airfoil	in.	mm	in.	mm
20	25.55	649	25.55	649	20	22.05	560	29.49	749
22.38	28.05	712	28.05	712	22.25	24.43	621	32.18	817
25	32.05	814	32.05	814	24.5	26.74	679	35.68	906
27.62	35.05	890	35.05	890	27	29.43	748	39.11	993
30.25	37.55	954	37.55	954	30	32.61	828	43.24	1098
33	43.74	1111	40.55	1030	33	35.74	908	47.99	1219
36	43.74	1111	43.55	1106	36.5	39.49	1003	52.55	1335
40.25	43.36	1101	58.30	1481	40.25	43.36	1101	58.30	1481

UNIT SIZE (fan diameter)	A**		B**		C		G		H (min.)	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
040 (27.62 FC)	116.00	2946	68.00	1727	74.00	1880	31.35	796	40.47	1028
040 (27 AF)	116.00	2946	68.00	1727	70.00	1778	36.98	939	38.45	977
050 (36 FC)	120.00	3048	80.00	2032	82.00	2083	28.41	722	36.22	917
050 (33 AF)	120.00	3048	80.00	2032	80.00	2032	38.66	982	30.00	762
065 (40.25 FC)*	136.00	3454	92.00	2337	92.00	2337	43.04	1093	34.85	887
065 (40.25 AF)	136.00	3454	92.00	2337	92.00	2337	43.04	1093	34.85	887
080 (40.25 FC)*	136.00	3454	98.00	2489	92.00	2337	43.04	1093	34.85	887
080 (40.25 AF)	136.00	3454	98.00	2489	92.00	2337	43.04	1093	34.85	887
085 (40.25 FC)*	136.00	3454	104.00	2641	92.00	2337	43.04	1093	34.85	887
085 (40.25 AF)	136.00	3454	104.00	2641	92.00	2337	43.04	1093	34.85	887
090 (40.25 FC)*	136.00	3454	110.00	2794	92.00	2337	43.04	1093	34.85	887
090 (40.25 AF)	136.00	3454	110.00	2794	92.00	2337	43.04	1093	34.85	887

* Not available with vanes.

**These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height.
A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Downblast housed fan discharge



cat550-29.dsf

FAN DIA. (inches)	E		F		FAN DIA. (inches)	E		F	
FC	in.	mm	in.	mm	Airfoil	in.	mm	in.	mm
9x4	11.05	281	7.61	193	13.22	16.68	424	19.94	506
9x7	11.05	281	9.99	254	14.56	20.18	513	22.00	559
9x9	11.05	281	12.61	320	16.19	22.37	568	24.31	617
10.62	12.18	309	13.92	354	19.69	24.13	613	29.70	754
12.62	14.24	362	16.42	417	21.56	27.27	693	32.45	824
15	16.68	424	19.42	493	24	29.51	750	35.95	913
18	19.68	500	22.68	576					
20	27.27	693	28.98	736					
22.25	29.51	750	32.23	819					
24.5	32.51	826	35.45	900					

Note:

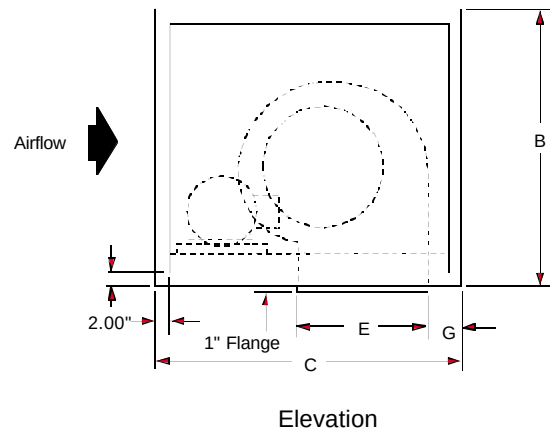
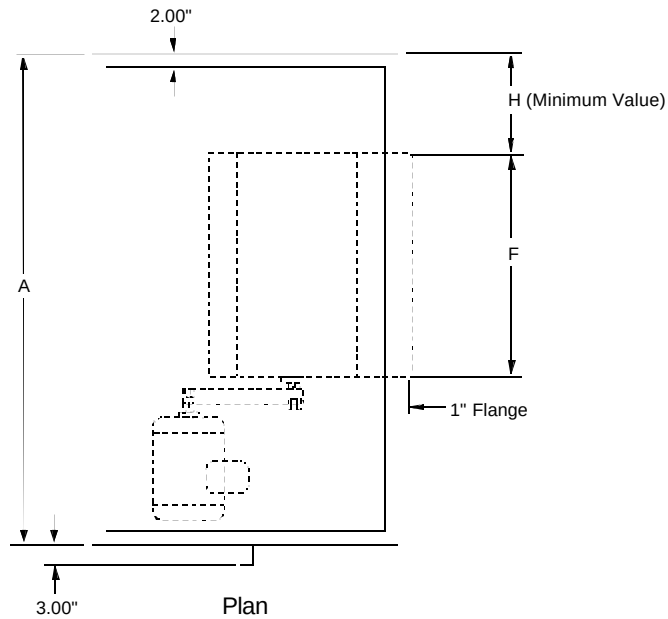
The H dimension shown is the minimum value.
This dimension will vary with the motor size used.

UNIT SIZE (fan diameter)	A**		B**		C		G		H (min.)	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
003 (9 x 4, 9 x 7 FC)*	38.00	965	26.00	660	36.00	914	6.43	163	8.18	208
004 (9 x 7, 9 x 9 FC)*	40.00	1016	30.00	762	36.00	914	6.43	163	8.18	208
006 (10.62 FC)*	52.00	1321	30.00	762	40.00	1016	8.50	216	9.43	240
008 (15 FC)	58.00	1473	34.00	864	46.00	1168	8.50	216	12.50	318
008 (13.22 AF)	58.00	1473	34.00	864	44.00	1118	8.50	216	11.25	286
010 (15FC)	64.00	1626	36.00	914	46.00	1168	8.50	216	12.50	318
010 (14.56 AF)	64.00	1626	36.00	914	48.00	1219	8.50	216	12.19	310
012 (18 FC)	66.00	1676	42.00	1067	50.00	1270	8.50	216	14.62	371
012 (16.19 AF)	66.00	1676	42.00	1067	52.00	1321	8.50	216	13.33	339
014 (18 FC)	74.00	1880	42.00	1067	50.00	1270	8.50	216	14.62	371
014 (19.69 AF)	74.00	1880	42.00	1067	56.00	1422	8.50	216	15.78	401
017 (20 FC)	80.00	2032	46.00	1168	56.00	1422	8.50	216	16.00	406
017 (21.56 AF)	80.00	2032	46.00	1168	58.00	1473	8.50	216	17.09	434
021 (20 FC)	82.00	2083	52.00	1321	56.00	1422	8.50	216	16.00	406
021 (21.56 AF)	82.00	2083	52.00	1321	58.00	1473	8.50	216	17.09	434
025 (24.50 FC)*	86.00	2184	60.00	1524	62.00	1575	8.50	216	19.15	486
025 (24 AF)	86.00	2184	60.00	1524	62.00	1575	8.50	216	18.80	478
030 (24.50 FC)	98.00	2489	60.00	1524	62.00	1575	8.50	216	19.15	486
030 (24 AF)	98.00	2489	60.00	1524	62.00	1575	8.50	216	18.80	478
035 (24.50 FC)	102.00	2591	66.00	1676	62.00	1575	8.50	216	19.15	486
035 (24 AF)	102.00	2591	66.00	1676	62.00	1575	8.50	216	18.80	478

* Not available with vanes.

**These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height.
A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Downblast housed fan discharge



cat550-29.dsf

FAN DIA. (inches)	E		F		FAN DIA. (inches)	E		F	
FC	in.	mm	in.	mm	Airfoil	in.	mm	in.	mm
20	25.55	649	25.55	649	20	22.05	560	29.49	749
22.38	28.05	712	28.05	712	22.25	24.43	621	32.18	817
25	32.05	814	32.05	814	24.5	26.74	679	35.68	906
27.62	35.05	890	35.05	890	27	29.43	748	39.11	993
30.25	37.55	954	37.55	954	30	32.61	828	43.24	1098
33	43.74	1111	40.55	1030	33	35.74	908	47.99	1219
36	43.74	1111	43.55	1106	36.5	39.49	1003	52.55	1335
40.25	43.36	1101	58.30	1481	40.25	43.36	1101	58.30	1481

Note:

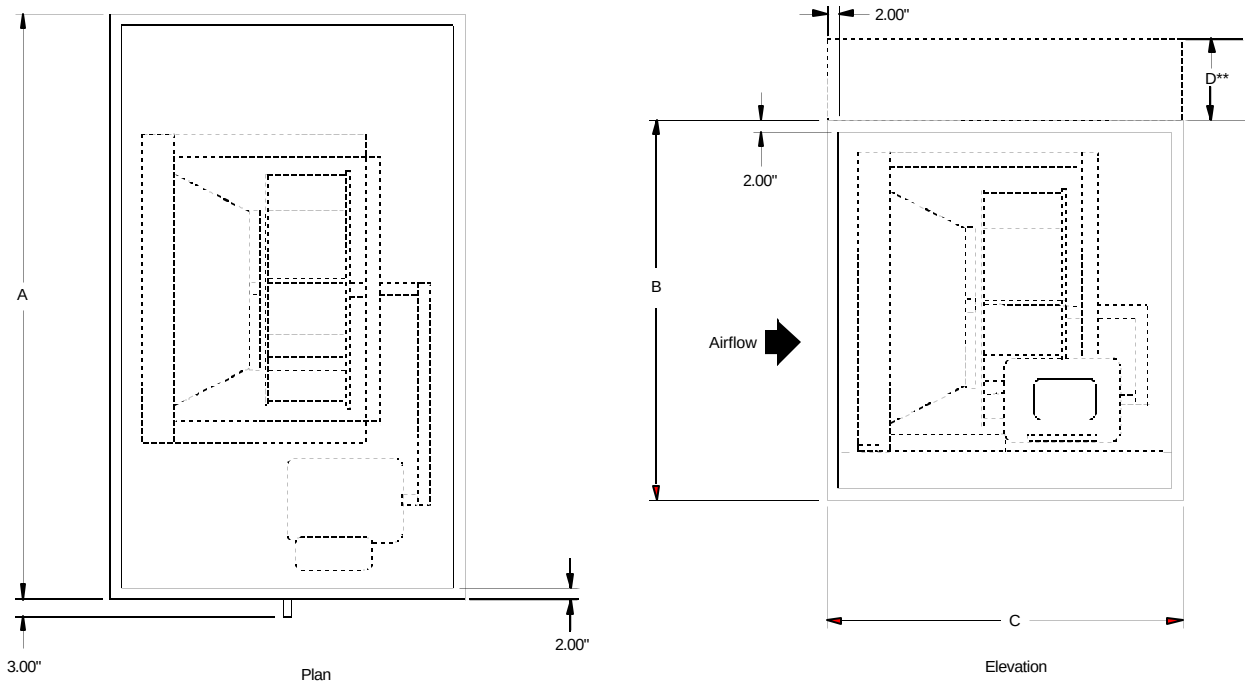
The H dimension shown is the minimum value.
This dimension will vary with the motor size used.

UNIT SIZE (fan diameter)	A**		B**		C		G		H (min.)	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
040 (27.62 FC)	116.00	2946	68.00	1727	74.00	1880	13.03	331	40.47	1028
040 (27 AF)	116.00	2921	68.00	1727	76.00	1930	13.47	342	38.45	977
050 (36 FC)	120.00	3048	80.00	2032	80.00	2032	13.09	333	36.22	917
050 (33 AF)	120.00	3048	80.00	2032	86.00	2184	14.21	361	30.00	763
065 (40.25 FC)*	136.00	3454	92.00	2337	96.00	2438	14.09	358	34.85	887
065 (40.25 AF)	136.00	3454	92.00	2337	96.00	2438	14.09	358	34.85	887
080 (40.25 FC)*	136.00	3454	98.00	2489	96.00	2438	12.6	320	34.85	887
080 (40.25 AF)	136.00	3454	98.00	2489	96.00	2438	12.6	320	34.85	887
085 (40.25 FC)*	136.00	3454	104.00	2641	96.00	2438	12.6	320	34.85	887
085 (40.25 AF)	136.00	3454	104.00	2641	96.00	2438	12.6	320	34.85	887
090 (40.25 FC)*	136.00	3454	110.00	2794	96.00	2438	12.6	320	34.85	887
090 (40.25 AF)	136.00	3454	110.00	2794	96.00	2438	12.6	320	34.85	887

* Not available with vanes.

**These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height.
A base rail is mandatory on unit sizes 025 and above and will add a minimum of 6" (152 mm) to the unit height.
An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Belt drive plenum fan discharge



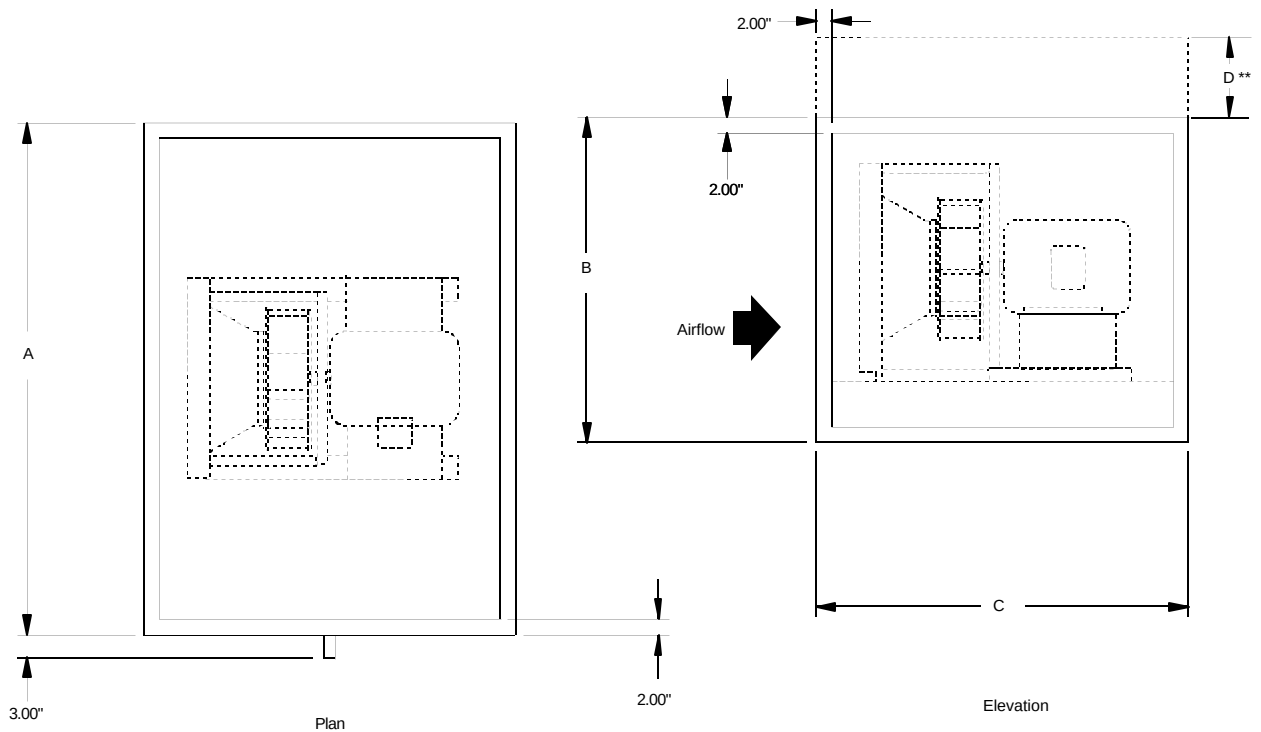
UNIT SIZE	A*		B*		C		D**	
	in.	mm	in.	mm	in.	mm	in.	mm
010 (13.50 AF)	64.00	1626	36.00	914	32.00	813	16.00	406
010 (15.00 AF)	64.00	1626	36.00	914	34.00	864	16.00	406
012 (18.25 AF)	66.00	1676	42.00	1067	40.00	1016	18.00	457
012 (20.00 AF)	66.00	1676	42.00	1067	42.00	1067	18.00	457
014 (18.25 AF)	74.00	1880	42.00	1067	40.00	1016	18.00	457
014 (20.00 AF)	74.00	1880	42.00	1067	42.00	1067	18.00	457
017 (24.50 AF)	80.00	2032	46.00	1168	46.00	1168	20.00	508
017 (27.00 AF)	80.00	2032	46.00	1168	48.00	1219	20.00	508
021 (27.00 AF)	82.00	2083	52.00	1321	48.00	1219	22.00	559
021 (30.00 AF)	82.00	2083	52.00	1321	52.00	1321	22.00	559
025 (30.00 AF)	86.00	2184	60.00	1524	52.00	1321	24.00	610
025 (33.00 AF)	86.00	2184	60.00	1524	56.00	1422	24.00	610
030 (33.00 AF)	98.00	2489	60.00	1524	56.00	1422	24.00	610
030 (36.50 AF)	98.00	2489	60.00	1524	58.00	1473	24.00	610
035 (36.50 AF)	102.00	2591	66.00	1676	58.00	1473	26.00	660
035 (40.25 AF)	102.00	2591	66.00	1676	62.00	1575	26.00	660
040 (36.50 AF)	116.00	2946	68.00	1727	58.00	1473	26.00	660
040 (40.25 AF)	116.00	2946	68.00	1727	62.00	1575	26.00	660
050 (44.50 AF)	120.00	3048	80.00	2032	66.00	1676	30.00	762
050 (49.00 AF)	120.00	3048	80.00	2032	70.00	1778	30.00	762
065 (54.25 AF)	136.00	3454	92.00	2337	76.00	1930	36.00	914
065 (60.00 AF)	136.00	3454	92.00	2337	82.00	2083	36.00	914
080 (54.25 AF)	136.00	3454	98.00	2489	76.00	1930	38.00	965
080 (60.00 AF)	136.00	3454	98.00	2489	82.00	2083	38.00	965
085 (54.25 AF)	136.00	3454	104.00	2642	76.00	1930	40.00	1016
085 (60.00 AF)	136.00	3454	104.00	2642	82.00	2083	40.00	1016
090 (54.25 AF)	136.00	3454	110.00	2794	76.00	1930	42.00	1067
090 (60.00 AF)	136.00	3454	110.00	2794	82.00	2083	42.00	1067

cat550-33.dsf

*These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height. A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

**This value represents the height for an optional top plenum. Additional discharge openings include axial, top and single opening, opposite drive side, bottom single opening, split and removable panels. Use Vision SelectTOOLS™ Software for more detailed dimensional information.

Direct drive plenum fan discharge



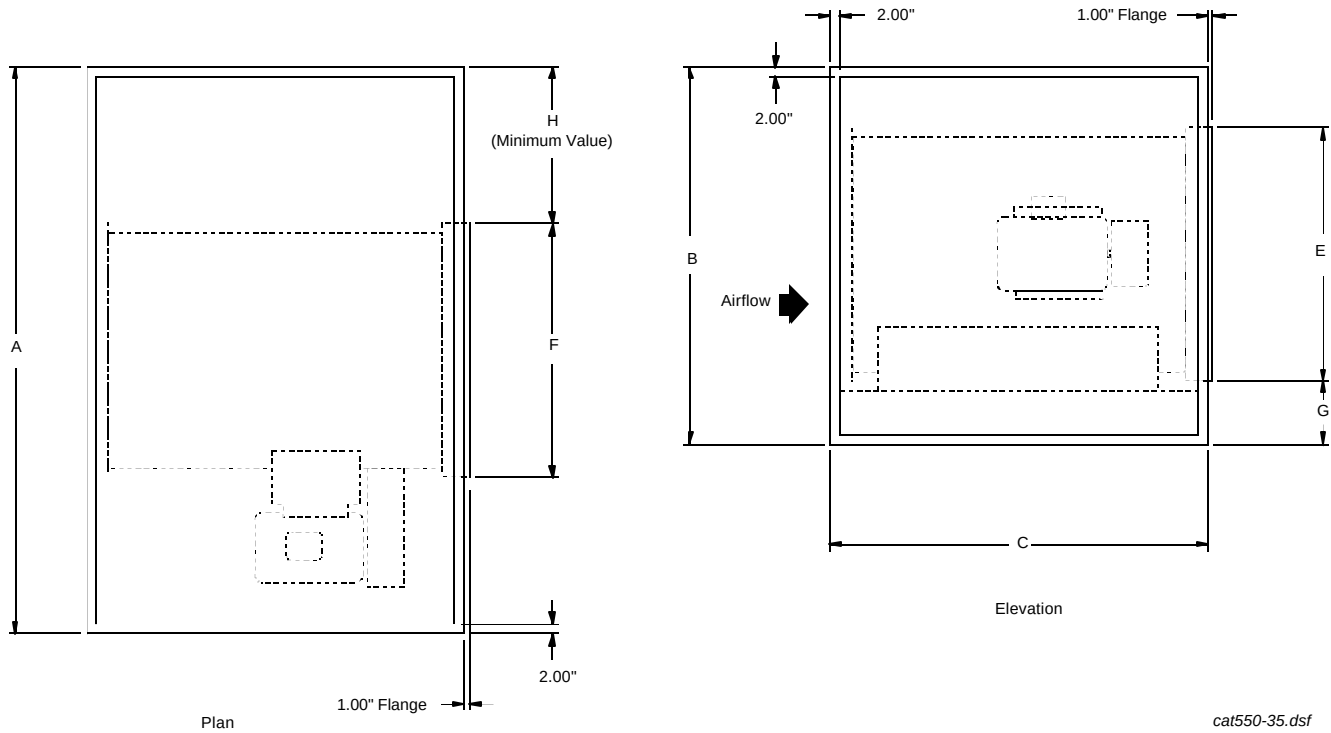
cat550-34.dsf

UNIT SIZE	A*		B*		C		D**	
	in.	mm	in.	mm	in.	mm	in.	mm
010 (13.50 AF)	36.00	914	64.00	1626	44.00	1118	16.00	406
010 (15.00 AF)	36.00	914	64.00	1626	46.00	1168	16.00	406
012 (16.50 AF)	42.00	1067	66.00	1676	52.00	1321	18.00	457
012 (18.25 AF)	42.00	1067	66.00	1676	54.00	1372	18.00	457
014 (16.50 AF)	42.00	1067	74.00	1880	52.00	1321	18.00	457
014 (18.25 AF)	42.00	1067	74.00	1880	54.00	1372	18.00	457
017 (24.50 AF)	46.00	1168	80.00	2032	60.00	1524	20.00	508
017 (27.00 AF)	46.00	1168	80.00	2032	66.00	1372	20.00	508
021 (27.00 AF)	52.00	1321	82.00	2083	66.00	1372	22.00	559
021 (30.00 AF)	52.00	1321	82.00	2083	68.00	1727	22.00	559
025 (33.00 AF)	60.00	1524	86.00	2184	74.00	1880	24.00	610
025 (36.50 AF)	60.00	1524	86.00	2184	78.00	1981	24.00	610
030 (33.00 AF)	60.00	1524	98.00	2489	74.00	1880	24.00	610
030 (36.50 AF)	60.00	1524	98.00	2489	78.00	1981	24.00	610
035 (36.50 AF)	66.00	1676	102.00	2591	78.00	1981	26.00	660
035 (40.25 AF)	66.00	1676	102.00	2591	84.00	2134	26.00	660
040 (36.50 AF)	68.00	1727	116.00	2946	80.00	2032	26.00	660
040 (40.25 AF)	68.00	1727	116.00	2946	86.00	2184	26.00	660
050 (40.25 AF)	80.00	2032	120.00	3048	86.00	2184	30.00	762
065 (40.25 AF)	92.00	2337	136.00	3454	86.00	2184	36.00	914
080 (40.25 AF)	98.00	2489	136.00	3454	86.00	2184	38.00	965
085 (40.25 AF)	104.00	2642	136.00	3454	86.00	2184	40.00	1016
090 (40.25 AF)	110.00	2794	136.00	3454	86.00	2184	42.00	1067

*These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height. A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

**This value represents the height for an optional top plenum. Additional discharge openings include axial, top and single opening, opposite drive side, bottom single opening, split and removable panels. Use Vision SelectTOOLS™ Software for more detailed dimensional information.

Inline fan discharge

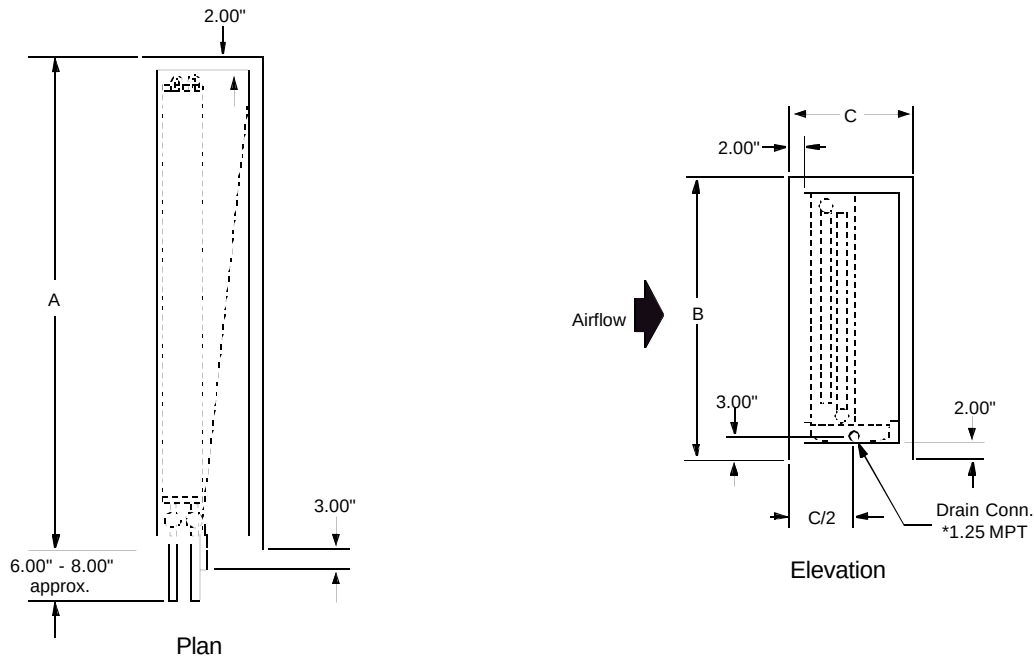


cat550-35.dsf

UNIT SIZE	A*		B*		C		E		F		G		H (min)	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
012 (150 Inline)	66.00	1676	42.00	1067	42.00	1067	27.80	706	25.80	655	8.10	206	20.10	511
012 (165 Inline)	66.00	1676	42.00	1067	44.00	1118	29.80	757	27.80	706	8.10	206	18.10	460
014 (150 Inline)	74.00	1880	42.00	1067	42.00	1067	27.80	706	25.80	655	8.10	206	24.10	612
014 (165 Inline)	74.00	1880	42.00	1067	44.00	1118	29.80	757	27.80	706	8.10	206	22.10	561
017 (165 Inline)	80.00	2032	46.00	1168	44.00	1118	29.80	757	27.80	706	8.10	206	26.10	663
017 (182 Inline)	80.00	2032	46.00	1168	48.00	1219	31.80	808	31.80	808	8.10	206	24.10	612
021 (200 Inline)	82.00	2083	52.00	1321	50.00	1270	33.80	859	35.80	909	8.10	206	14.10	358
021 (222 Inline)	82.00	2083	52.00	1321	54.00	1372	35.80	909	35.80	909	10.10	257	16.10	409
025 (245 Inline)	86.00	2184	60.00	1524	58.00	1473	41.80	1062	39.80	1011	8.10	206	16.10	409
025 (270 Inline)	86.00	2184	60.00	1524	64.00	1626	43.80	1113	43.80	1113	12.10	307	14.10	358
030 (245 Inline)	98.00	2489	60.00	1524	58.00	1473	41.80	1062	39.80	1011	8.10	206	28.10	714
030 (270 Inline)	98.00	2489	60.00	1524	64.00	1626	43.80	1113	43.80	1113	12.10	307	20.10	511
035 (270 Inline)	102.00	2591	66.00	1676	64.00	1626	43.80	1113	43.80	1113	12.10	307	28.10	714
035 (300 Inline)	102.00	2591	66.00	1676	70.00	1778	47.80	1214	47.80	1214	12.10	307	20.10	511
040 (270 Inline)	116.00	2946	68.00	1727	66.00	1676	43.80	1113	43.80	1113	12.10	307	36.10	917
040 (300 Inline)	116.00	2946	68.00	1727	70.00	1778	49.80	1265	47.80	1214	12.10	307	34.10	866
050 (365 Inline)	120.00	3048	80.00	2032	82.00	2083	57.80	1468	57.80	1468	12.10	307	30.10	765
050 (402 Inline)	120.00	3048	80.00	2032	90.00	2286	63.80	1621	63.80	1621	12.10	307	22.10	561
065 (402 Inline)	136.00	3454	92.00	2337	90.00	2286	63.80	1621	63.80	1621	12.10	307	36.10	917
065 (445 Inline)	136.00	3454	92.00	2337	96.00	2438	69.80	1773	69.80	1773	12.10	307	32.10	815
080 (402 Inline)	136.00	3454	98.00	2489	90.00	2286	63.80	1621	63.80	1621	12.10	307	36.10	917
080 (445 Inline)	136.00	3454	98.00	2489	96.00	2438	69.80	1773	69.80	1773	12.10	307	32.10	815
085 (402 Inline)	136.00	3454	104.00	2642	90.00	2286	63.80	1621	63.80	1621	12.10	307	36.10	917
085 (445 Inline)	136.00	3454	104.00	2642	96.00	2438	69.80	1773	69.80	1773	12.10	307	32.10	815
090 (402 Inline)	136.00	3454	110.00	2794	90.00	2286	63.80	1621	63.80	1621	12.10	307	36.10	917
090 (445 Inline)	136.00	3454	110.00	2794	96.00	2438	69.80	1773	69.80	1773	12.10	307	32.10	815

*These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height.
A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Horizontal blow-through, draw-through, and vertical single cooling section



cat550-36.dsf

Note: The single cooling coil section is available in numerous depths. This provides the option of varying the amount of space downstream from the coil for access to inspect and clean the coil and drain pan. When using a single bank of coils, the 18" deep section will hold up to a 4-row water or DX coil.

Also available are staggered cooling coils. The staggered arrangement maximizes the cooling coil face area by providing 2 banks of coils. These 2 coils are not stacked, but are staggered in the direction of airflow. This configuration does require more section depth in the direction of airflow. Staggered coils are only available on unit sizes 025 and larger. The

section depth required will vary between 30" - 54". This will depend on the unit size and number of rows in the coil.

When a coil section is used in a vertical application, the section must be deep enough to support the fan above it, and keep the face velocity through the top opening below 1000 ft./mn. The coil section should be a minimum of 36" deep in direction of airflow. As the fan and airflow get larger, it may be necessary to add a plenum downstream from the coil section. *Drain connections for units with width (A) greater than 108" are 1.50 MPT.

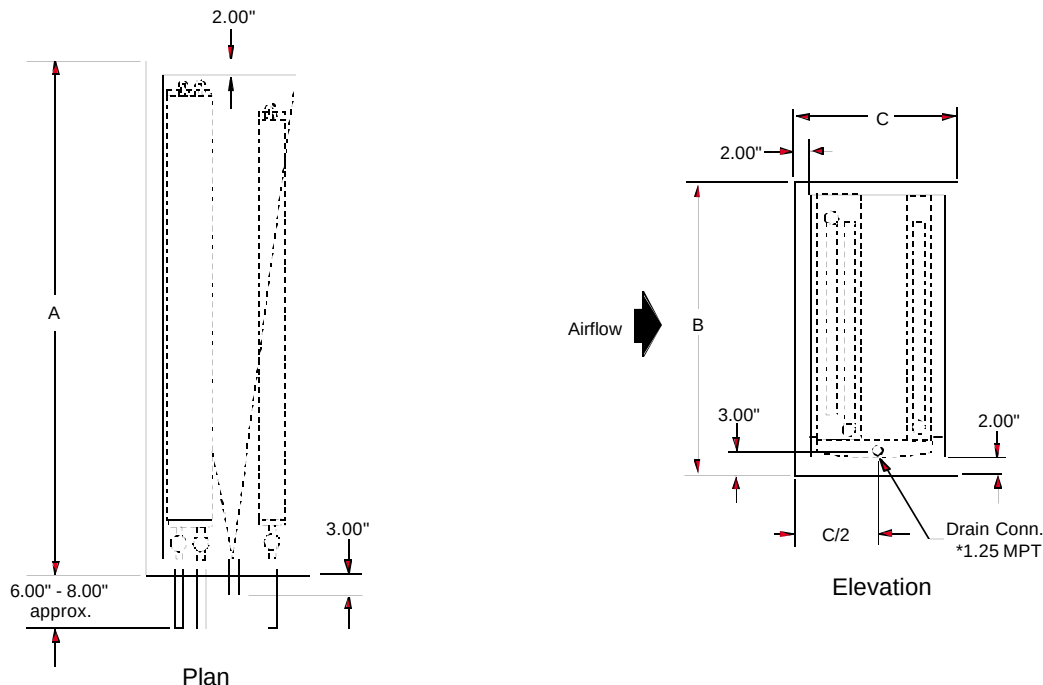
UNIT SIZE	Staggered A* (large, medium coils)		A* (extended coils)		B*		C (available all sizes)	
	in.	mm	in.	mm	in.	mm	in.	mm
003	38.00	965	44.00	1118	26.00	660	18.00	457
004	40.00	1016	46.00	1168	30.00	762	24.00	610
006	52.00	1321	58.00	1473	30.00	762	30.00	762
008	58.00	1473	64.00	1626	34.00	864	36.00	914
010	64.00	1626	70.00	1778	36.00	914	42.00	1067
012	66.00	1676	72.00	1829	42.00	1067	48.00	1219
014	74.00	1880	80.00	2032	42.00	1067	54.00	1372
017	80.00	2032	86.00	2184	46.00	1168	—	—
021	82.00	2083	88.00	2235	52.00	1321	—	—
025	86.00	2184	N/A	N/A	60.00	1524	—	—
030	98.00	2489	N/A	N/A	60.00	1524	—	—
035	102.00	2591	N/A	N/A	66.00	1676	—	—
040	116.00	2946	N/A	N/A	68.00	1727	—	—
050	120.00	3048	N/A	N/A	80.00	2032	—	—
065	136.00	3454	N/A	N/A	92.00	2337	—	—
080	136.00	3454	N/A	N/A	98.00	2489	—	—
085	136.00	3454	N/A	N/A	104.00	2642	—	—
090	136.00	3454	N/A	N/A	110.00	2794	—	—

*These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height.

A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

**18.0", 24.0", 30.0", 36.0", 42.0", 48.0", and 54.0" cooling coil sections available for all unit sizes.

Horizontal blow-through, draw-through, and vertical combination cooling and reheat coil section



cat550-37.dsf

Note: The combination cooling and reheat coil section is available in numerous depths. This provides the option of varying the amount of space between the cooling coil and reheat coil for access to inspect and clean the coil and drain pan. All reheat and most cooling coils are arranged in a single bank, which may be 1, 2 or 3 coils high. When using a single bank of cooling coils, the 24" deep section will hold up to a 4-row water or DX coil, and a 1 or 2 row heating coil. A 36" deep section will hold a 12 row water coil and a 1 or 2 row reheat coil.

Also available are staggered cooling coils. The staggered arrangement maximizes the cooling coil face area by providing

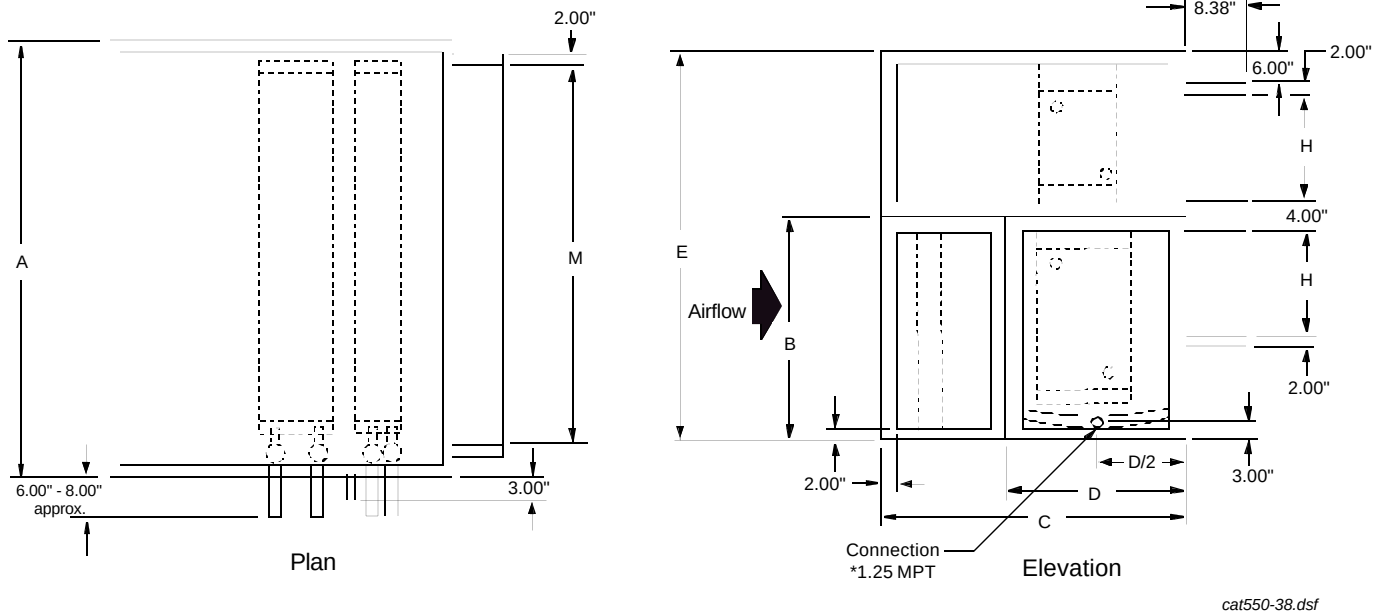
2 banks of coils. These 2 coils are not stacked, but are staggered in the direction of airflow. This configuration does require more section depth in the direction of airflow. Staggered coils are only available on unit sizes 025 and larger. The section depth required will vary between 30" - 54". This will depend on the unit size and number of rows in the coil.

When a combination coil section is used in a vertical application, a plenum must be provided on the leaving air side of the coil. The plenum is used for discharging air into the bottom of the fan section. *Drain connections for units with width (A) greater than 108" are 1.50 MPT.

UNIT SIZE	Staggered A* (large, medium coils)		A* (extended coils)		B*		C (available all sizes)	
	in.	mm	in.	mm	in.	mm	in.	mm
003	38.00	965	44.00	1118	26.00	660	24.00	610
004	40.00	1016	46.00	1168	30.00	762	30.00	762
006	52.00	1321	58.00	1473	30.00	762	36.00	914
008	58.00	1473	64.00	1626	34.00	864	42.00	1067
010	64.00	1626	70.00	1778	36.00	914	48.00	1219
012	66.00	1676	72.00	1829	42.00	1067	54.00	1372
014	74.00	1880	80.00	2032	42.00	1067	—	—
017	80.00	2032	86.00	2184	46.00	1168	—	—
021	82.00	2083	88.00	2235	52.00	1321	—	—
025	86.00	2184	N/A	N/A	60.00	1524	—	—
030	98.00	2489	N/A	N/A	60.00	1524	—	—
035	102.00	2591	N/A	N/A	66.00	1676	—	—
040	116.00	2946	N/A	N/A	68.00	1727	—	—
050	120.00	3048	N/A	N/A	80.00	2032	—	—
065	136.00	3454	N/A	N/A	92.00	2337	—	—
080	136.00	3454	N/A	N/A	98.00	2489	—	—
085	136.00	3454	N/A	N/A	104.00	2642	—	—
090	136.00	3454	N/A	N/A	110.00	2794	—	—

*These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm - 305 mm) to the unit height.
A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.
**18.0", 24.0", 30.0", 36.0", 42.0", 48.0", and 54.0" cooling coil sections available for all unit sizes.

Horizontal discharge 2-deck multizone coil section

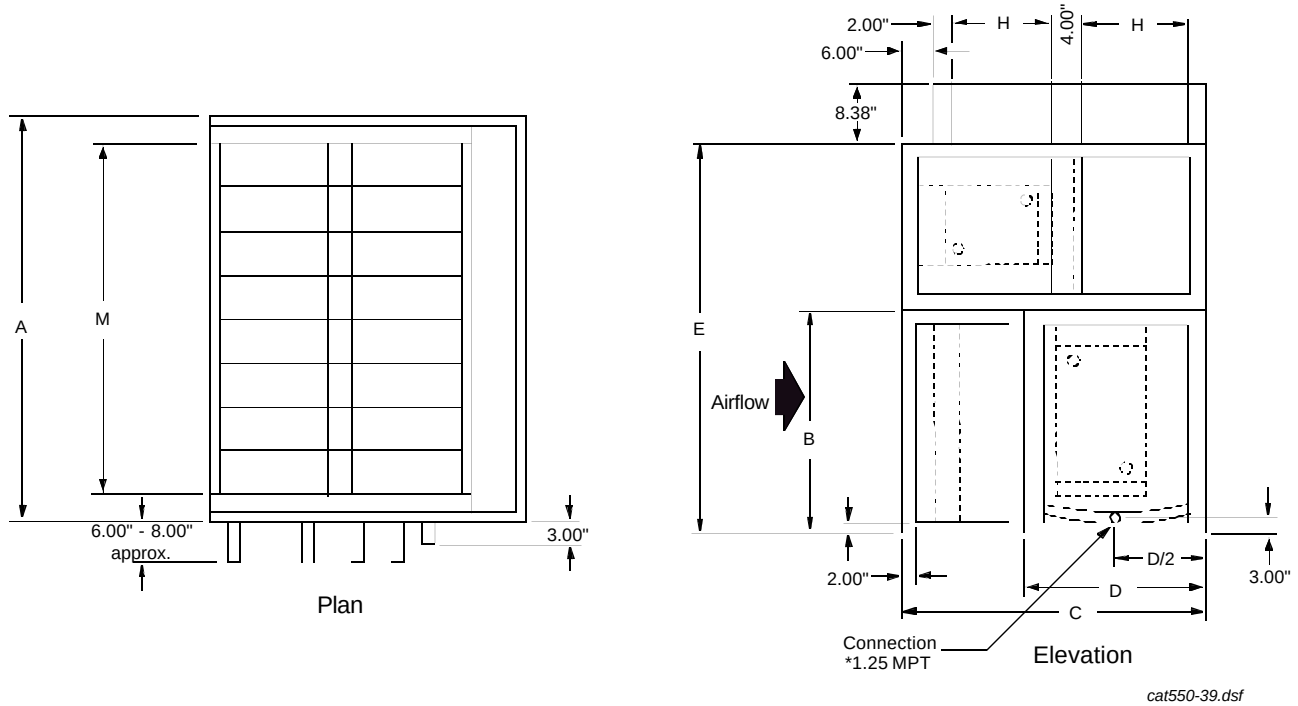


***Note:** Drain connection for units with width (A) greater than 108\" are 1.50 MPT.

UNIT SIZE	Max # of Zones	A (1)		B		C (2)		D (2)		E (1)		H		M	
		in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
006	5	52.00	1321	30.00	762	40.00	1016	24.00	610	52.00	1321	12.00	305	40.60	1031
008	6	58.00	1473	34.00	864	42.00	1067	24.00	610	58.00	1473	14.00	356	48.60	1234
010	7	64.00	1626	36.00	914	42.00	1067	24.00	610	60.00	1524	14.00	356	56.60	1438
012	7	66.00	1676	42.00	1067	52.00	1321	30.00	762	70.00	1778	18.00	457	56.60	1438
014	8	74.00	1880	42.00	1067	52.00	1321	30.00	762	70.00	1778	18.00	457	64.60	1641
017	9	80.00	2032	46.00	1168	54.00	1372	30.00	762	76.00	1930	20.00	508	72.60	1844
021	9	82.00	2083	52.00	1321	56.00	1422	30.00	762	84.00	2134	22.00	559	72.60	1844
025	9	86.00	2184	60.00	1524	60.00	1524	30.00	762	96.00	2438	26.00	660	72.60	1844
030	11	98.00	2489	60.00	1524	60.00	1524	30.00	762	96.00	2438	26.00	660	88.60	2250
035	11	102.00	2591	66.00	1676	70.00	1778	36.00	914	106.00	2692	30.00	762	88.60	2250
040	13	116.00	2946	68.00	1727	70.00	1778	36.00	914	108.00	2743	30.00	762	104.60	2657
050	14	120.00	3048	80.00	2032	76.00	1930	36.00	914	126.00	3200	36.00	914	112.60	2860
065	16	136.00	3454	92.00	2337	88.00	2235	42.00	1067	144.00	3658	42.00	1067	128.60	3266
080	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
085	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
090	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

- (1) These are the dimensions of the unit only. A base rail adds 8\" (203 mm) to the width, and between 4\"-12\" (102 mm-305 mm) to the unit height. A base rail is mandatory on unit sizes 025-065 and will add a minimum of 6\" (152 mm) to the unit height. Unit sizes greater than 080 will require a base rail that adds a minimum of 8\" (203 mm) to the unit height.
- (2) For 10-row water and DX coils, the C & D dimensions are 6\" longer. If additional access is required downstream from the cooling coil, refer to Vision SelectTOOLS™ Software for available selection options.

Vertical discharge 2-deck multizone coil section



cat550-39.ds

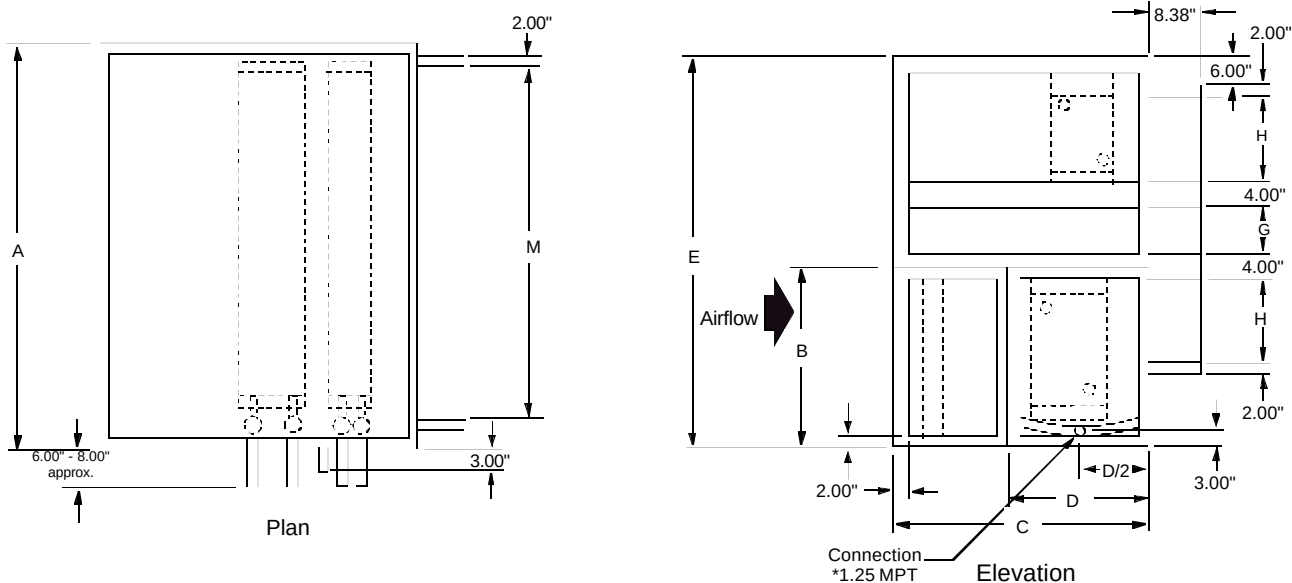
***Note:** Drain connection for units with width (A) greater than 108\" are 1.50 MPT.

UNIT SIZE	Max # of Zones	A (1)		B		C (2)		D (2)		E (1)		H		M	
		in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
006	5	52.00	1321	30.00	762	46.00	1168	30.00	762	48.00	1219	12.00	305	40.60	1031
008	6	58.00	1473	34.00	864	48.00	1219	30.00	762	54.00	1372	14.00	356	48.60	1234
010	7	64.00	1626	36.00	914	48.00	1219	30.00	762	56.00	1422	14.00	356	56.60	1438
012	7	66.00	1676	42.00	1067	58.00	1473	36.00	914	66.00	1676	18.00	457	56.60	1438
014	8	74.00	1880	42.00	1067	58.00	1473	36.00	914	66.00	1676	18.00	457	64.60	1641
017	9	80.00	2032	46.00	1168	60.00	1524	36.00	914	72.00	1829	20.00	508	72.60	1844
021	9	82.00	2083	52.00	1321	62.00	1575	36.00	914	80.00	2032	22.00	559	72.60	1844
025	9	86.00	2184	60.00	1524	72.00	1829	42.00	1067	92.00	2337	26.00	660	72.60	1844
030	11	98.00	2489	60.00	1524	72.00	1829	42.00	1067	92.00	2337	26.00	660	88.60	2250
035	11	102.00	2591	66.00	1676	82.00	2083	48.00	1219	102.00	2591	30.00	762	88.60	2250
040	13	116.00	2946	68.00	1727	82.00	2083	48.00	1219	104.00	2642	30.00	762	104.60	2657
050	14	120.00	3048	80.00	2032	94.00	2388	54.00	1372	122.00	3099	36.00	914	112.60	2860
065	16	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
080	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
085	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
090	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

(1) These are the dimensions of the unit only. A base rail adds 8\" (203 mm) to the width, and between 4\"-12\" (102 mm-305 mm) to the unit height. A base rail is mandatory on unit sizes 025-065 and will add a minimum of 6\" (152 mm) to the unit height. Unit sizes greater than 080 will require a base rail that adds a minimum of 8\" (203 mm) to the unit height.

(2) For 10-row water and DX coils, the C & D dimensions are 6\" longer. If additional access is required downstream from the cooling coil, refer to Vision SelectTOOLS™ Software for available selection options.

Horizontal discharge 3-deck multizone coil section



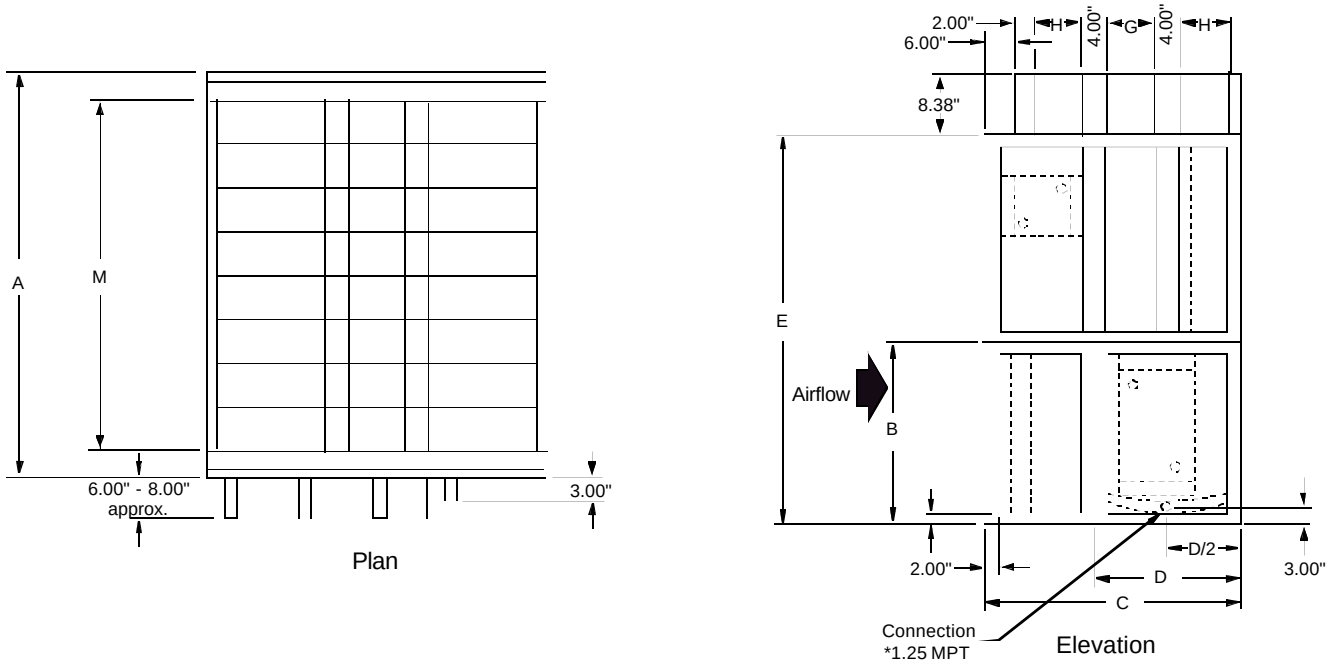
cat550-40.dsf

***Note:** Drain connection for units with width (A) greater than 108\" are 1.50 MPT.

UNIT SIZE	Max # of Zones	A (1)		B (1)(2)		C (1)(2)		D (2)		E (1)		G		H		M	
		in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
006	5	52.00	1321	30.00	762	40.00	1016	24.00	610	64.00	1626	8.00	203	12.00	305	40.60	1031
008	6	58.00	1473	34.00	864	42.00	1067	24.00	610	70.00	1778	8.00	203	14.00	356	48.60	1234
010	7	64.00	1626	36.00	914	42.00	1067	24.00	610	72.00	1829	8.00	203	14.00	356	56.60	1438
012	7	66.00	1676	42.00	1067	52.00	1321	30.00	762	84.00	2134	10.00	254	18.00	457	56.60	1438
014	8	74.00	1880	42.00	1067	52.00	1321	30.00	762	84.00	2134	10.00	254	18.00	457	64.60	1641
017	9	80.00	2032	46.00	1168	54.00	1372	30.00	762	92.00	2337	12.00	305	20.00	508	72.60	1844
021	9	82.00	2083	52.00	1321	56.00	1422	30.00	762	102.00	2591	14.00	356	22.00	559	72.60	1844
025	9	86.00	2184	60.00	1524	60.00	1524	30.00	762	116.00	2946	16.00	406	26.00	660	72.60	1844
030	11	98.00	2489	60.00	1524	60.00	1524	30.00	762	116.00	2946	16.00	406	26.00	660	88.60	2250
035	11	102.00	2591	66.00	1676	70.00	1778	36.00	914	128.00	3251	18.00	457	30.00	762	88.60	2250
040	13	116.00	2946	68.00	1727	70.00	1778	36.00	914	130.00	3302	18.00	457	30.00	762	104.60	2657
050	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
060	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
080	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
085	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
090	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

- (1) These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm-305 mm) to the unit height. A base rail is mandatory on unit sizes 025-065 and will add a minimum of 6" (152 mm) to the unit height. Unit sizes greater than 080 will require a base rail that adds a minimum of 8" (203 mm) to the unit height.
- (2) For 10-row water and DX coils, the C & D dimensions are 6" longer. If additional access is required downstream from the cooling coil, refer to Vision SelectTOOLS™ Software for available selection options.

Vertical discharge 3-deck multizone coil section



cat550-41.ds

***Note:** Drain connection for units with width (A) greater than 108\"

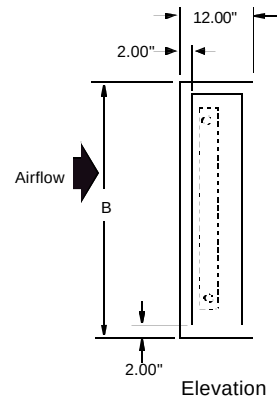
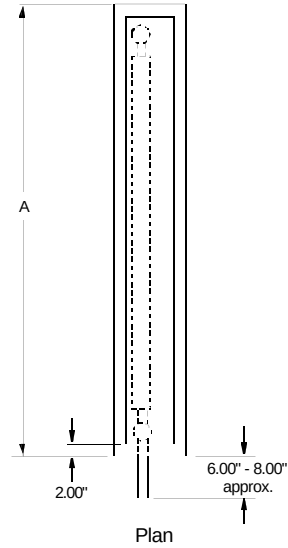
UNIT SIZE	Max # of Zones	A (1)		B		C (2)		D (2)		E (1)		G		H		M	
		in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
006	5	52.00	1321	30.00	762	50.00	1270	30.00	762	58.00	1473	8.00	203	12.00	305	40.60	1031
008	6	58.00	1473	34.00	864	54.00	1372	36.00	914	64.00	1626	8.00	203	14.00	356	48.60	1234
010	7	64.00	1626	36.00	914	54.00	1372	36.00	914	66.00	1676	8.00	203	14.00	356	56.60	1438
012	7	66.00	1676	42.00	1067	64.00	1626	36.00	914	76.00	1930	10.00	254	18.00	457	56.60	1438
014	8	74.00	1880	42.00	1067	64.00	1626	36.00	914	76.00	1930	10.00	254	18.00	457	64.60	1641
017	9	80.00	2032	46.00	1168	70.00	1778	42.00	1067	82.00	2083	12.00	305	20.00	508	72.60	1844
021	9	82.00	2083	52.00	1321	76.00	1930	42.00	1067	90.00	2286	14.00	356	22.00	559	72.60	1844
025	9	86.00	2184	60.00	1524	86.00	2184	48.00	1219	102.00	2591	16.00	406	26.00	660	72.60	1844
030	11	98.00	2489	60.00	1524	86.00	2184	48.00	1219	102.00	2591	16.00	406	26.00	660	88.60	2250
035	11	102.00	2591	66.00	1676	96.00	2438	48.00	1219	112.00	2845	18.00	457	30.00	762	88.60	2250
040	13	116.00	2946	68.00	1727	96.00	2438	48.00	1219	114.00	2896	18.00	457	30.00	762	104.60	2657
050	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
065	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
080	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
085	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
090	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

(1) These are the dimensions of the unit only. A base rail adds 8\"

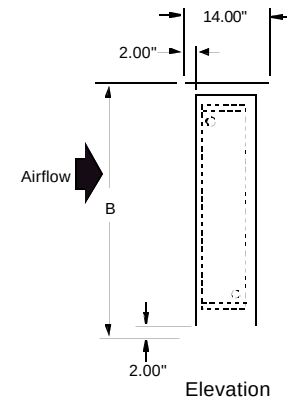
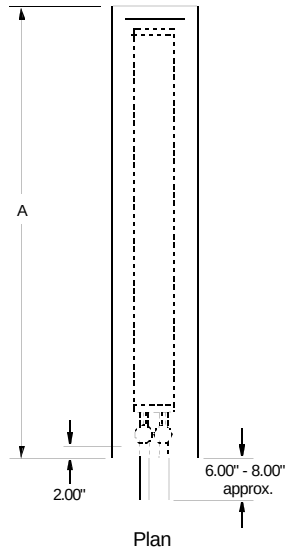
(2) For 10-row water and DX coils, the C & D dimensions are 6\"

Heating coil section

**1 and 2 row heating coil section
(uncased coils)**



**3 and 4 row heating coil section
(cased coils)**



cat550-42.dsf

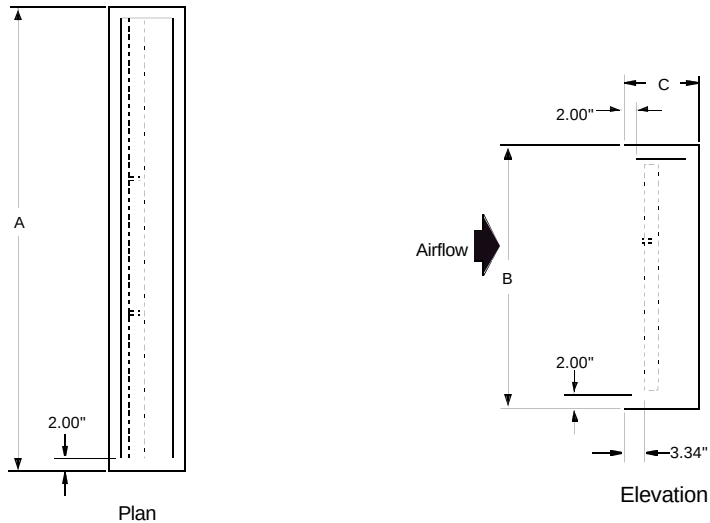
NOTE: 1.25 MPT drain connections are available as an option for heating coil sections.

These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm-305 mm) to the unit height. A base rail is mandatory on unit sizes 025-065 and will add a minimum of 6" (152 mm) to the unit height. Unit sizes greater than 080 require a base rail that adds a minimum of 8" (203 mm) to the unit height.

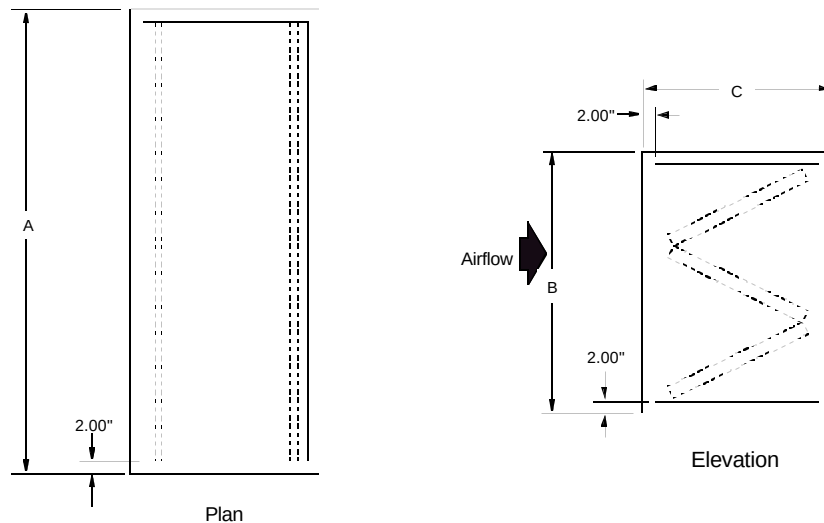
UNIT SIZE	A		B	
	in.	mm	in.	mm
003	38.00	965	26.00	660
004	40.00	1016	30.00	762
006	52.00	1321	30.00	762
008	58.00	1473	34.00	864
010	64.00	1626	36.00	914
012	66.00	1676	42.00	1067
014	74.00	1880	42.00	1067
017	80.00	2032	46.00	1168
021	82.00	2083	52.00	1321
025	86.00	2184	60.00	1524
030	98.00	2489	60.00	1524
035	102.00	2591	66.00	1676
040	116.00	2946	68.00	1727
050	120.00	3048	80.00	2032
065	136.00	3454	92.00	2337
080	136.00	3454	98.00	2489
085	136.00	3454	104.00	2642
090	136.00	3454	110.00	2794

Side load flat and angular filter section

Flat filter section



Angular filter section



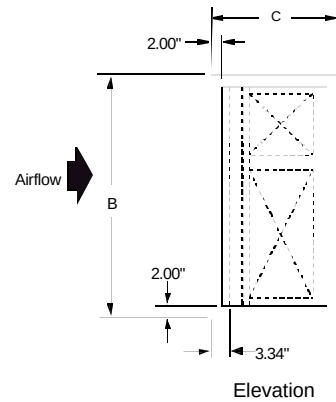
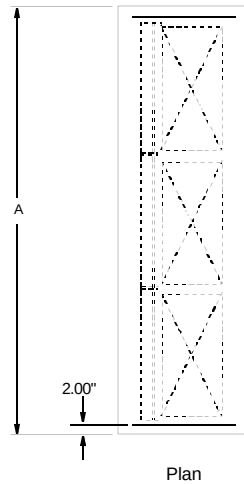
cat550-43.dsf

*These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm-305 mm) to the unit height. A base rail is mandatory on unit sizes 025-065 and will add a minimum of 6" (152 mm) to the unit height. Unit sizes greater than 080 require a base rail that adds a minimum of 8" (203 mm) to the unit height.

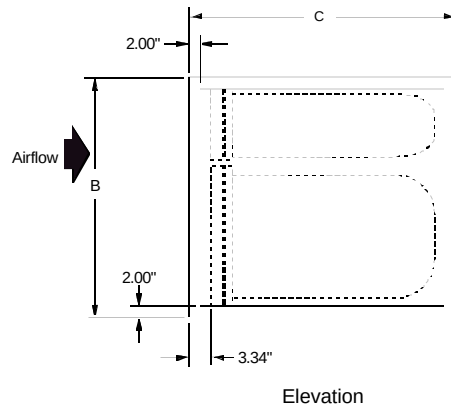
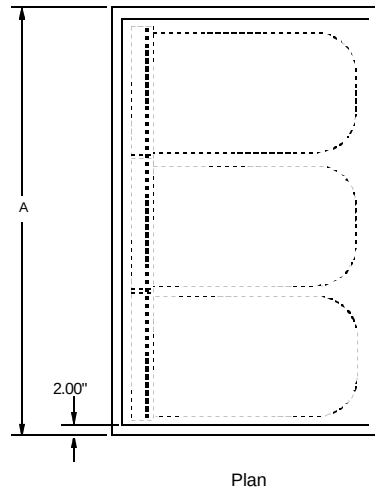
UNIT SIZE	A*		B*		C		C		C	
	in.	mm	in.	mm	2" & 4" Flat Filter		2" Angular Filter		4" Angular Filter	
					in.	mm	in.	mm	in.	mm
003	38.00	965	26.00	660	12.00	305	32.00	813	N/A	N/A
004	40.00	1016	30.00	762	12.00	305	30.00	762	N/A	N/A
006	52.00	1321	30.00	762	12.00	305	30.00	762	N/A	N/A
008	58.00	1473	34.00	864	12.00	305	30.00	762	32.00	813
010	64.00	1626	36.00	914	12.00	305	30.00	762	32.00	813
012	66.00	1676	42.00	1067	12.00	305	30.00	762	32.00	813
014	74.00	1880	42.00	1067	12.00	305	30.00	762	32.00	813
017	80.00	2032	46.00	1168	12.00	305	30.00	762	32.00	813
021	82.00	2083	52.00	1321	12.00	305	30.00	762	32.00	813
025	86.00	2184	60.00	1524	12.00	305	32.00	813	32.00	813
030	98.00	2489	60.00	1524	12.00	305	32.00	813	32.00	813
035	102.00	2591	66.00	1676	12.00	305	32.00	813	32.00	813
040	116.00	2946	68.00	1727	12.00	305	32.00	813	32.00	813
050	120.00	3048	80.00	2032	12.00	305	32.00	813	32.00	813
065	136.00	3454	92.00	2337	12.00	305	32.00	813	32.00	813
080	136.00	3454	98.00	2489	12.00	305	32.00	813	32.00	813
085	136.00	3454	104.00	2642	12.00	305	32.00	813	32.00	813
090	136.00	3454	110.00	2794	12.00	305	32.00	813	32.00	813

Side load cartridge and bag filter section

Cartridge filter section



Bag filter section



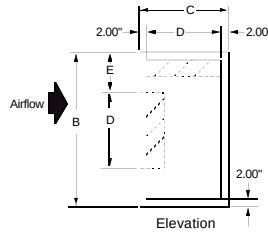
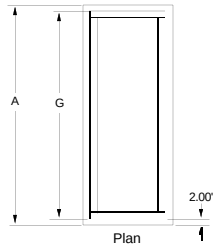
cat550-44.dsf

*These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm-305 mm) to the unit height. A base rail is mandatory on unit sizes 025-065 and will add a minimum of 6" (152 mm) to the unit height. Unit sizes greater than 080 require a base rail that adds a minimum of 8" (203 mm) to the unit height.

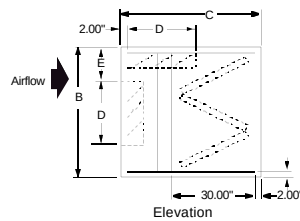
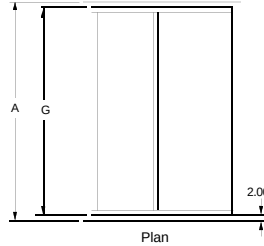
UNIT SIZE	A*		B*		FILTER TYPE	C – ALL SIZES	
	in.	mm	in.	mm		in.	mm
003	38.00	965	26.00	660	4" w/2" Pre	14.00	356
004	40.00	1016	30.00	762	4" w/4" Pre	16.00	406
006	52.00	1321	30.00	762	12" w/2" Pre	22.00	559
008	58.00	1473	34.00	864	12" w/4" Pre	24.00	610
010	64.00	1626	36.00	914	Bag	in.	mm
012	66.00	1676	42.00	1067	12" w/2" Pre	18.00	457
014	74.00	1880	42.00	1067	12" w/4" Pre	20.00	508
017	80.00	2032	46.00	1168	15" w/2" Pre	22.00	559
021	82.00	2083	52.00	1321	15" w/4" Pre	24.00	610
025	86.00	2184	60.00	1524	19" w/2 Pre	26.00	660
030	98.00	2489	60.00	1524	19" w/4" Pre	28.00	711
035	102.00	2591	66.00	1676	22" w/2" Pre	28.00	711
040	116.00	2946	68.00	1727	22" w/4" Pre	30.00	762
050	120.00	3048	80.00	2032	30" w/2" Pre	36.00	914
060	136.00	3454	92.00	2337	30" w/4" Pre	38.00	965
080	136.00	3454	98.00	2489	36" w/2" Pre	42.00	1067
085	136.00	3454	104.00	2642	36" w/4" Pre	44.00	1118
090	136.00	3454	110.00	2794			

Mixing box and combination filter mixing box

Mixing box



Mixing box with angular filter



cat550-45.dsf

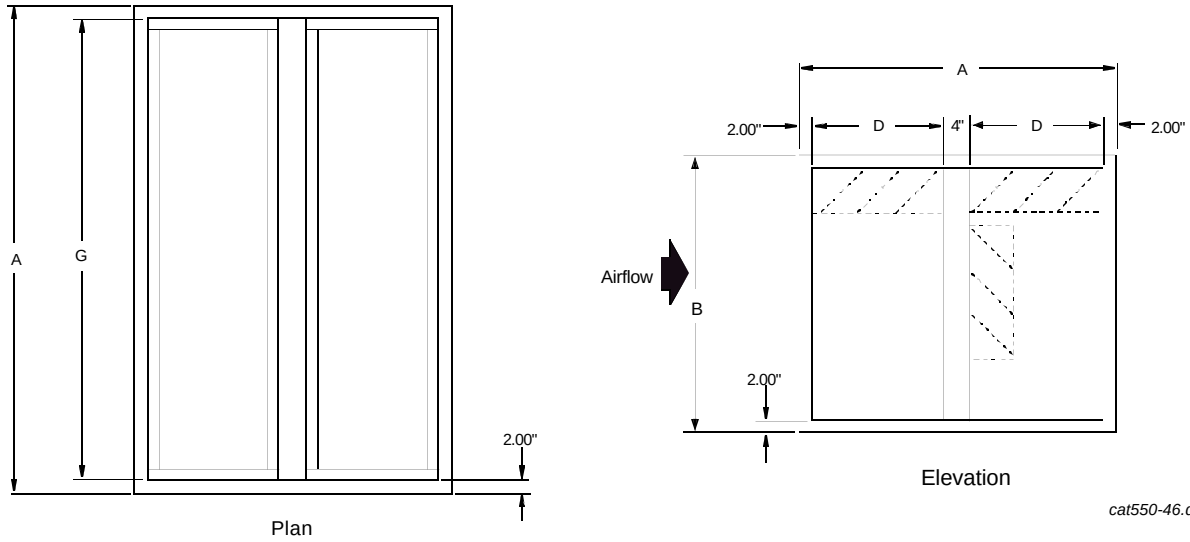
Note: Dampers can be located in the top, bottom, and back of the section. 1.25 MPT drain connections are available on mixing box sections without filters.

UNIT SIZE	A*		B*		C Mixing Box Only		C w/2" Flat Filter		C w/4" Flat Filter		C w/2" Ang Filter		C w/4" Ang Filter	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
003	38.00	965	26.00	660	20.00	508	24.00	610	26.00	660	42.00	1067	N/A	N/A
004	40.00	1016	30.00	762	20.00	508	24.00	610	26.00	660	42.00	1067	N/A	N/A
006	52.00	1321	30.00	762	20.00	508	24.00	610	26.00	660	42.00	1067	N/A	N/A
008	58.00	1473	34.00	864	20.00	508	24.00	610	26.00	660	42.00	1067	46.00	1168
010	64.00	1626	36.00	914	22.00	559	26.00	660	28.00	711	44.00	1118	48.00	1219
012	66.00	1676	42.00	1067	24.00	610	28.00	711	30.00	762	46.00	1168	50.00	1270
014	74.00	1880	42.00	1067	24.00	610	28.00	711	30.00	762	46.00	1168	50.00	1270
017	80.00	2032	46.00	1168	26.00	660	30.00	762	32.00	813	48.00	1219	52.00	1321
021	82.00	2083	52.00	1321	30.00	762	34.00	864	36.00	914	52.00	1321	56.00	1422
025	86.00	2184	60.00	1524	32.00	813	36.00	914	38.00	965	54.00	1372	58.00	1473
030	98.00	2489	60.00	1524	32.00	813	36.00	914	38.00	965	54.00	1372	58.00	1473
035	102.00	2591	66.00	1676	36.00	914	40.00	1016	42.00	1067	58.00	1473	62.00	1575
040	116.00	2946	68.00	1727	36.00	914	40.00	1016	42.00	1067	58.00	1473	62.00	1575
050	120.00	3048	80.00	2032	40.00	1016	44.00	1118	46.00	1168	62.00	1575	66.00	1676
065	136.00	3454	92.00	2337	46.00	1168	50.00	1270	52.00	1321	68.00	1727	72.00	1829
080	136.00	3454	98.00	2489	50.00	1270	54.00	1372	56.00	1422	72.00	1829	76.00	1930
085	136.00	3454	104.00	2642	54.00	1372	58.00	1473	60.00	1524	76.00	1930	80.00	2032
090	136.00	3454	110.00	2794	56.00	1422	60.00	1524	62.00	1575	78.00	1981	82.00	2083

UNIT SIZE	D		E		G		Damper Width		Damper Length	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
003	16.00	406	8.00	203	34.00	864	12.00	305	24.00	610
004	16.00	406	12.00	305	36.00	914	12.00	305	26.00	660
006	16.00	406	12.00	305	48.00	1219	12.00	305	38.00	965
008	16.00	406	10.00	254	54.00	1372	12.00	305	44.00	1118
010	18.00	457	10.00	254	60.00	1524	14.00	356	50.00	1270
012	20.00	508	12.00	305	62.00	1575	16.00	406	52.00	1321
014	20.00	508	12.00	305	70.00	1778	16.00	406	60.00	1524
017	22.00	559	12.00	305	76.00	1930	18.00	457	66.00	1676
021	26.00	660	14.00	356	78.00	1981	22.00	559	68.00	1727
025	28.00	711	16.00	406	82.00	2083	24.00	610	72.00	1829
030	28.00	711	16.00	406	94.00	2388	24.00	610	84.00	2134
035	32.00	813	18.00	457	98.00	2489	28.00	711	88.00	2235
040	32.00	813	18.00	457	112.00	2845	28.00	711	102.00	2591
050	36.00	914	22.00	559	116.00	2946	32.00	813	106.00	2692
065	42.00	1067	26.00	660	132.00	3353	38.00	965	122.00	3099
080	46.00	3454	26.00	660	132.00	3353	42.00	1067	122.00	3099
085	50.00	3454	27.00	686	132.00	3353	46.00	1168	122.00	3099
090	52.00	3454	29.00	737	132.00	3353	48.00	1219	122.00	3099

*These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm-305 mm) to the unit height. A base rail is mandatory on unit sizes 025-065 and will add a minimum of 6" (152 mm) to the unit height. Unit sizes greater than 080 require a base rail that adds a minimum of 8" (203 mm) to the unit height.

Economizer section

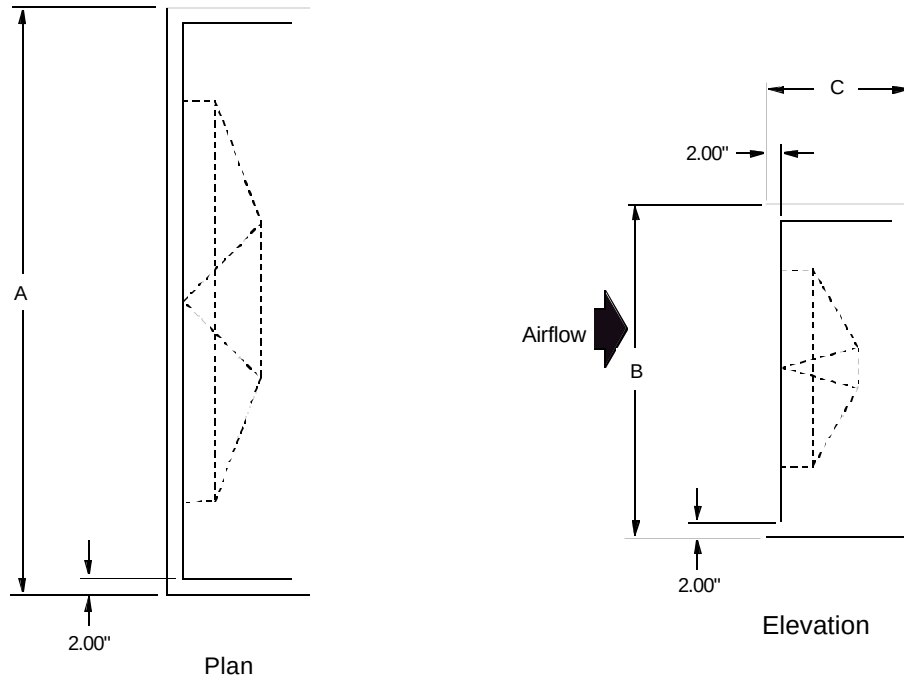


Note: Dampers can be located in the top, bottom, and back of the section. 1.25 MPT drain connections are optional for economizer sections.

UNIT SIZE	A*		B*		D		G		Damper Width		Damper Length	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
003	38.00	965	26.00	660	16.00	406	34.00	864	12.00	305	24.00	610
004	40.00	1016	30.00	762	16.00	406	36.00	914	12.00	305	26.00	660
006	52.00	1321	30.00	762	16.00	406	48.00	1219	12.00	305	38.00	965
008	58.00	1473	34.00	864	16.00	406	54.00	1372	12.00	305	44.00	1118
010	64.00	1626	36.00	914	18.00	457	60.00	1524	14.00	356	50.00	1270
012	66.00	1676	42.00	1067	20.00	508	62.00	1575	16.00	406	52.00	1321
014	74.00	1880	42.00	1067	20.00	508	70.00	1778	16.00	406	60.00	1524
017	80.00	2032	46.00	1168	22.00	559	76.00	1930	18.00	457	66.00	1676
021	82.00	2083	52.00	1321	26.00	660	78.00	1981	22.00	559	68.00	1727
025	86.00	2184	60.00	1524	28.00	711	82.00	2083	24.00	610	72.00	1829
030	98.00	2489	60.00	1524	28.00	711	94.00	2388	24.00	610	84.00	2134
035	102.00	2591	66.00	1676	32.00	813	98.00	2489	28.00	711	88.00	2235
040	116.00	2946	68.00	1727	32.00	813	112.00	2845	28.00	711	102.00	2591
050	120.00	3048	80.00	2032	36.00	914	116.00	2946	32.00	813	106.00	2692
065	136.00	3454	92.00	2337	42.00	1067	132.00	3353	38.00	965	122.00	3099
080	136.00	3454	98.00	3489	46.00	1168	132.00	3353	42.00	1067	122.00	3099
085	136.00	3454	104.00	2642	50.00	1270	132.00	3353	46.00	1168	122.00	3099
090	136.00	3454	110.00	2794	52.00	1321	132.00	3353	48.00	1219	122.00	3099
UNIT SIZE	C Econ. Only		C w/2" Flat Filter		C w/4" Flat Filter		C w/2" Ang Filter		C w/4" Ang Filter			
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm		
003	40.00	1016	44.00	1118	46.00	1168	62.00	1575	N/A	N/A		
004	40.00	1016	44.00	1118	46.00	1168	62.00	1575	N/A	N/A		
006	40.00	1016	44.00	1118	46.00	1168	62.00	1575	N/A	N/A		
008	40.00	1016	44.00	1118	46.00	1168	62.00	1575	66.00	1676		
010	44.00	1118	48.00	1219	50.00	1270	66.00	1676	70.00	1778		
012	48.00	1219	52.00	1321	54.00	1372	70.00	1778	74.00	1880		
014	48.00	1219	52.00	1321	54.00	1372	70.00	1778	74.00	1880		
017	52.00	1321	56.00	1422	58.00	1473	74.00	1880	78.00	1981		
021	60.00	1524	64.00	1626	66.00	1676	82.00	2083	86.00	2184		
025	64.00	1626	68.00	1727	70.00	1778	86.00	2184	90.00	2286		
030	64.00	1626	68.00	1727	70.00	1778	86.00	2184	90.00	2286		
035	72.00	1829	76.00	1930	78.00	1981	94.00	2388	98.00	2489		
040	72.00	1829	76.00	1930	78.00	1981	94.00	2388	98.00	2489		
050	80.00	2032	84.00	2134	86.00	2184	102.00	2591	106.00	2692		
065	92.00	2337	96.00	2438	98.00	2489	114.00	2896	118.00	2997		
080	100.00	2540	108.00	2743	112.00	2845	144.00	3658	152.00	3861		
085	108.00	2743	116.00	2946	120.00	3048	152.00	3861	160.00	4064		
090	112.00	2845	120.00	3048	124.00	3150	156.00	3962	164.00	4166		

*These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4"-12" (102 mm-305 mm) to the unit height. A base rail is mandatory on unit sizes 025-065 and will add a minimum of 6" (152 mm) to the unit height. Unit sizes greater than 080 require a base rail that adds a minimum of 8" (203 mm) to the unit height.

Blenders



cat550-47.dsf

Note:

The depth of an air mixer section does vary depending on components adjacent to the mixing device. The catalog dimensions shown are for a 2 blender system except sizes 040 - 090 which are 3 blender systems, using a Blender Products Series IV Blender. For applications other than this, the depth dimension will vary. Use Vision SelectTOOLS™ Software for more detailed dimensional information.

Blender Product Series IV

UNIT SIZE	A*		B*		C	
	in.	mm	in	mm	in	mm
003	38.00	965	26.00	660	N/A	N/A
004	40.00	1016	30.00	762	12.00	305
006	52.00	1321	30.00	762	14.00	356
008	58.00	1473	34.00	864	16.00	406
010	64.00	1626	36.00	914	18.00	457
012	66.00	1676	42.00	1067	20.00	508
014	74.00	1880	42.00	1067	22.00	559
017	80.00	2032	46.00	1168	24.00	610
021	82.00	2083	52.00	1321	28.00	711
025	86.00	2184	60.00	1524	32.00	813
030	98.00	2489	60.00	1524	34.00	864
035	102.00	2591	66.00	1676	36.00	914
040	116.00	2946	68.00	1727	38.00	965
050	120.00	3048	80.00	2032	46.00	1168
060	136.00	3454	92.00	2337	52.00	1321
080	136.00	3454	98.00	3489	54.00	1372
085	136.00	3454	104.00	2642	56.00	1422
090	136.00	3454	110.00	2794	58.00	1473

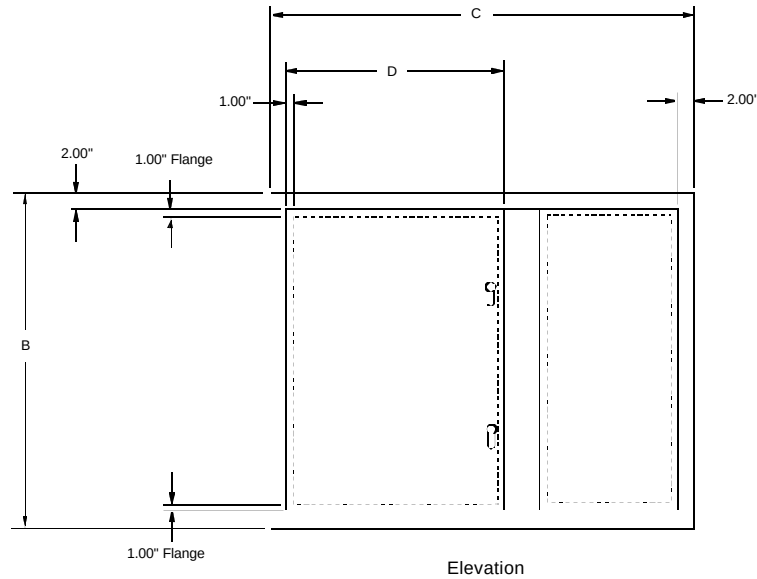
Kees Blender

UNIT SIZE	A**		B*		C	
	in.	mm	in	mm	in	mm
003	38.00	965	26.00	660	N/A	N/A
004	40.00	1016	30.00	762	N/A	N/A
006	52.00	1321	30.00	762	22.00	559
008	58.00	1473	34.00	864	24.00	610
010	64.00	1626	36.00	914	26.00	660
012	66.00	1676	42.00	1067	28.00	711
014	74.00	1880	42.00	1067	30.00	762
017	80.00	2032	46.00	1168	32.00	813
021	82.00	2083	52.00	1321	36.00	914
025	86.00	2184	60.00	1524	38.00	965
030	98.00	2489	60.00	1524	40.00	1016
035	102.00	2591	66.00	1676	42.00	1067
040	116.00	2946	68.00	1727	38.00	965
050	120.00	3048	80.00	2032	42.00	1067
060	136.00	3454	92.00	2337	46.00	1168
080	136.00	3454	98.00	3489	56.00	1422
085	136.00	3454	104.00	2642	58.00	1473
090	136.00	3454	110.00	2794	58.00	1473

*These are the dimensions of the unit only. A base rail adds 8" (203 mm) to the width, and between 4" -12" (102 mm - 305 mm) to the unit height. A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on all sizes 080 and larger.

**Kees Blender dimensions shown are using a 2-blender system except sizes 040-090 which are 3-blender systems. All values represent Kees blenders located after filters.

Access doors



cat550-48.dsf

Note: Access doors are an option for most sections. Fan sections are always provided with a door on one side of the unit. The size of the access door varies with the size of the section. Access doors are never wider than 30". On most components the doors may be selected as inward opening.

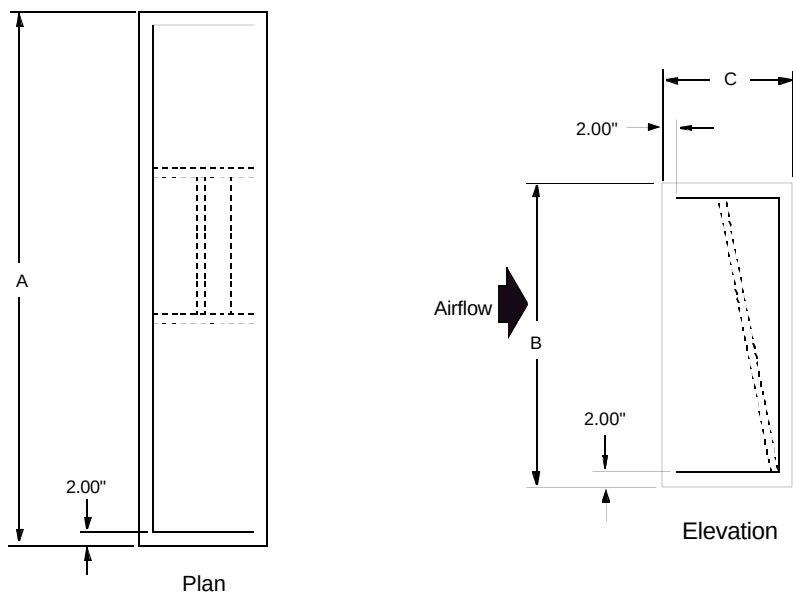
Once the door is opened outward, the opening will be 2" less than the door height and width. This is because there is a 1" flange around the opening. Once the door is opened

inward, the opening will be 3" less than the door height and width. There is no flange around the opening. Door height is always 4" shorter than the section height except if the section has a drain connection. If the section has a drain pan, then the door will be 6" shorter than the section height. The maximum door height is 50", except filter section doors, which are equal to section height less 4".

UNIT SIZE	A*		B*		C		D	
	(available for all size units)							
	in.	mm	in.	mm	in.	mm	in.	mm
003	38.00	965	26.00	660	12.00	305	8.00	203
004	40.00	1016	30.00	762	14.00	356	10.00	254
006	52.00	1321	30.00	762	16.00	406	12.00	305
008	58.00	1473	34.00	864	18.00	457	14.00	356
010	64.00	1626	36.00	914	20.00	508	16.00	406
012	66.00	1676	42.00	1067	22.00	559	18.00	457
014	74.00	1880	42.00	1067	24.00	610	20.00	508
017	80.00	2032	46.00	1168	26.00	660	22.00	559
021	82.00	2083	52.00	1321	28.00	711	24.00	610
025	86.00	2184	60.00	1524	30.00	762	26.00	660
030	98.00	2489	60.00	1524	32.00	813	28.00	711
035	102.00	2591	66.00	1676	34.00	864	30.00	762
040	116.00	2946	68.00	1727	36.00	914	28.00	711
050	120.00	3048	80.00	2032	38.00	965	30.00	762
065	136.00	3454	92.00	2337	40.00	1016	30.00	762
080	136.00	3454	98.00	3489	42.00	1067	30.00	762
085	136.00	3454	104.00	2642	44.00	1118	30.00	762
090	136.00	3454	110.00	2794	46.00	1168	30.00	762
					48.00	1219	30.00	762
					50.00	1270	30.00	762
					52.00	1321	30.00	762
					54.00	1372	30.00	762

* These are dimensions of the unit only. A base rail adds a minimum of 8" (203 mm) to the width, and between 4"-12" to the unit height. A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Diffuser

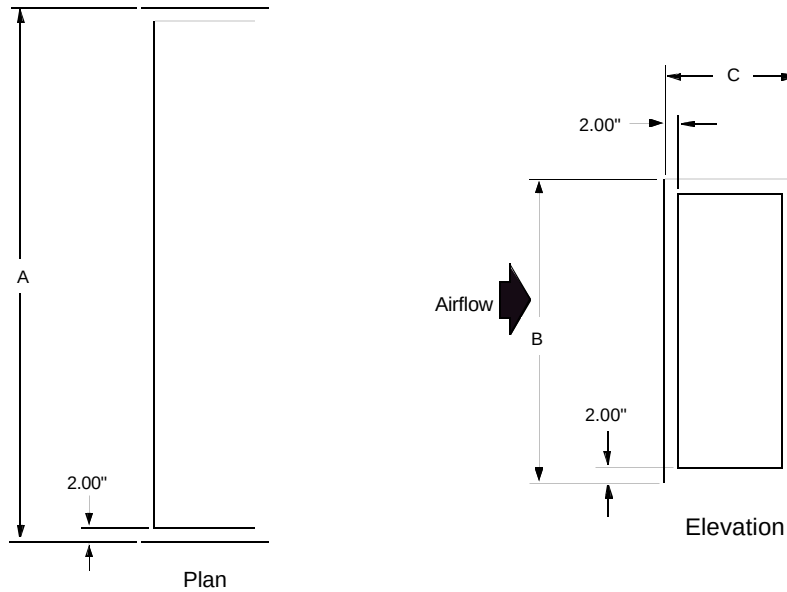


cat550-49.dsf

UNIT SIZE	A*		B*		C			
	in.	mm	in.	mm	Housed Fans		Inline Fans	
					in.	mm	in.	mm
003	38.00	965	26.00	660	10.00	254	N/A	N/A
004	40.00	1016	30.00	762	10.00	254	N/A	N/A
006	52.00	1321	30.00	762	10.00	254	N/A	N/A
008	58.00	1473	34.00	864	12.00	305	N/A	N/A
010	64.00	1626	36.00	914	12.00	305	18.00	457
012	66.00	1676	42.00	1067	16.00	406	18.00	457
014	74.00	1880	42.00	1067	16.00	406	18.00	457
017	80.00	2032	46.00	1168	16.00	406	22.00	559
021	82.00	2083	52.00	1321	16.00	406	26.00	660
025	86.00	2184	60.00	1524	24.00	610	26.00	660
030	98.00	2489	60.00	1524	24.00	610	26.00	660
035	102.00	2591	66.00	1676	24.00	610	28.00	711
040	116.00	2946	68.00	1727	30.00	762	30.00	762
050	120.00	3048	80.00	2032	30.00	762	36.00	914
065	136.00	3454	92.00	2337	30.00	762	38.00	965
080	136.00	3454	98.00	3489	30.00	762	38.00	965
085	136.00	3454	104.00	2642	30.00	762	38.00	965
090	136.00	3454	110.00	2794	30.00	762	38.00	965

* These are dimensions of the unit only. A base rail adds a minimum of 8" (203 mm) to the width, and between 4"-12" to the unit height. A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Access section



cat550-50.dsf

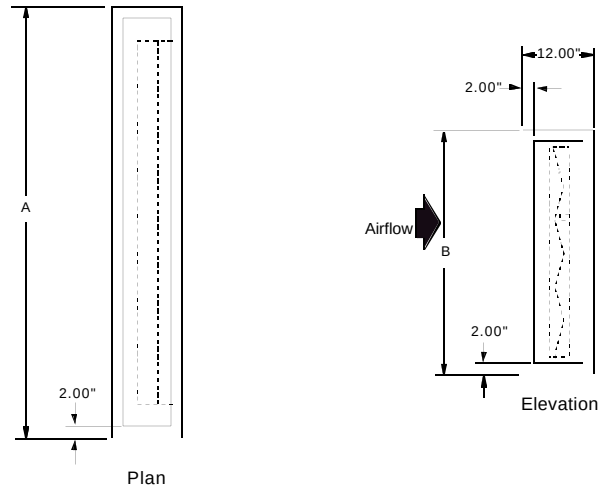
NOTE: 1.25 MPT drain connection available as an option for access sections.

UNIT SIZE	A*		B*		C (available all sizes)	
	in.	mm	in.	mm	in.	mm
003	38.00	965	26.00	660	16.00	406
004	40.00	1016	30.00	762	24.00	610
006	52.00	1321	30.00	762	30.00	762
008	58.00	1473	34.00	864	36.00	914
010	64.00	1626	36.00	914	42.00	1067
012	66.00	1676	42.00	1067	48.00	1219
014	74.00	1880	42.00	1067	54.00	1372
017	80.00	2032	46.00	1168	—	—
021	82.00	2083	52.00	1321	—	—
025	86.00	2184	60.00	1524	—	—
030	98.00	2489	60.00	1524	—	—
035	102.00	2591	66.00	1676	—	—
040	116.00	2946	68.00	1727	—	—
050	120.00	3048	80.00	2032	—	—
065	136.00	3454	92.00	2337	—	—
080	136.00	3454	98.00	2489	—	—
085	136.00	3454	104.00	2642	—	—
090	136.00	3454	110.00	2794	—	—

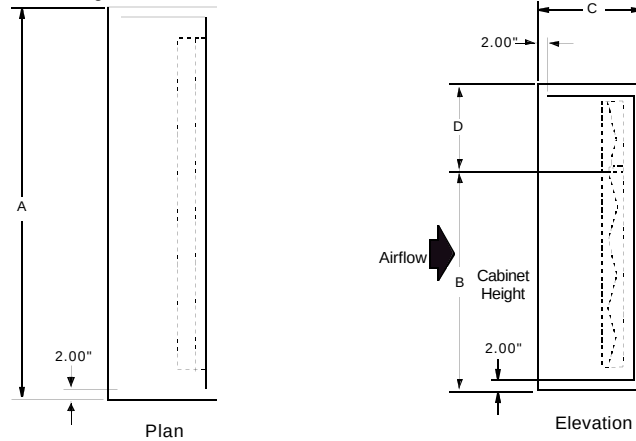
* These are dimensions of the unit only. A base rail adds a minimum of 8" (203 mm) to the width, and between 4"-12" to the unit height. A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger. Access doors can be either inward or outward opening.

Internal and external face and bypass sections

Internal face and bypass with small and medium face area coil



External face and bypass with large face area coil

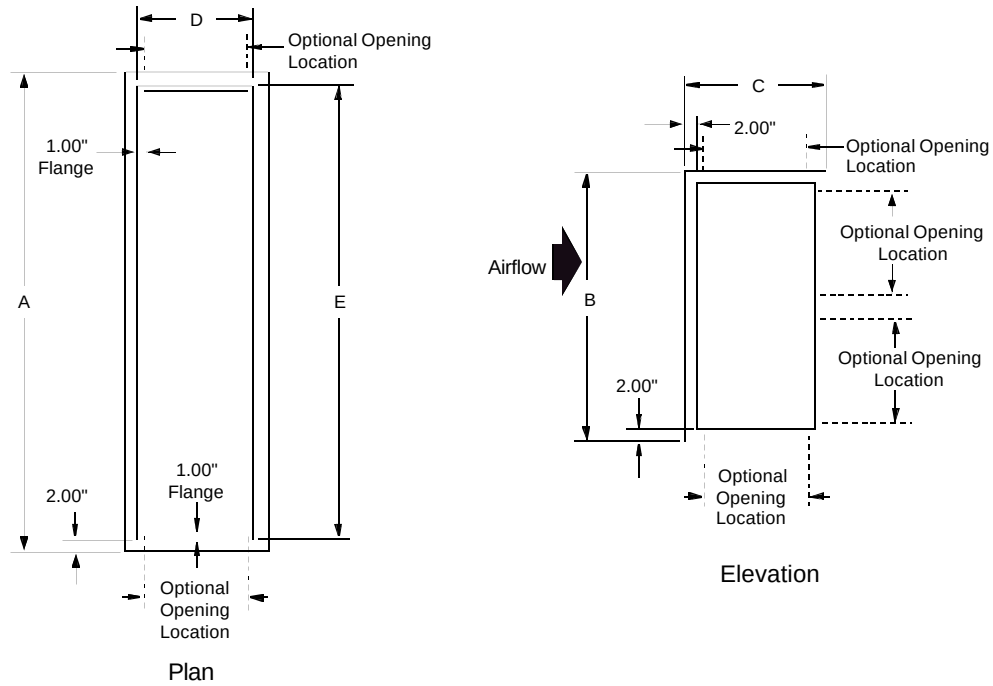


cat550-51.dsf

UNIT SIZE	A*		B*		C - Internal		C - External		D	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
003	38.00	965	26.00	660	12.00	305	18.00	457	10.00	254
004	40.00	1016	30.00	762	12.00	305	18.00	457	10.00	254
006	52.00	1321	30.00	762	12.00	305	18.00	457	10.00	254
008	58.00	1473	34.00	864	12.00	305	20.00	508	10.00	254
010	64.00	1626	36.00	914	12.00	305	22.00	559	12.00	305
012	66.00	1676	42.00	1067	12.00	305	24.00	610	16.00	406
014	74.00	1880	42.00	1067	12.00	305	24.00	610	16.00	406
017	80.00	2032	46.00	1168	12.00	305	26.00	660	16.00	406
021	82.00	2083	52.00	1321	12.00	305	30.00	762	22.00	559
025	86.00	2184	60.00	1524	12.00	305	32.00	813	24.00	610
030	98.00	2489	60.00	1524	12.00	305	32.00	813	24.00	610
035	102.00	2591	66.00	1676	12.00	305	34.00	864	26.00	660
040**	116.00	2946	68.00	1727	12.00	305	38.00	965	26.00	660
050**	120.00	3048	80.00	2032	12.00	305	44.00	1118	32.00	813
065**	136.00	3454	92.00	2337	12.00	305	50.00	1270	36.00	914
080**	136.00	3454	98.00	3489	12.00	305	54.00	1372	38.00	965
085**	136.00	3454	104.00	2642	12.00	305	56.00	1422	42.00	1067
090**	136.00	3454	110.00	2794	12.00	305	58.00	1473	1473	1118

* These are dimensions of the unit only. A base rail adds a minimum of 8" (203 mm) to the width, and between 4"-12" to the unit height.
A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.
**Right angle face and bypass dampers are mandatory on unit sizes 040 and larger.

Discharge or return plenum



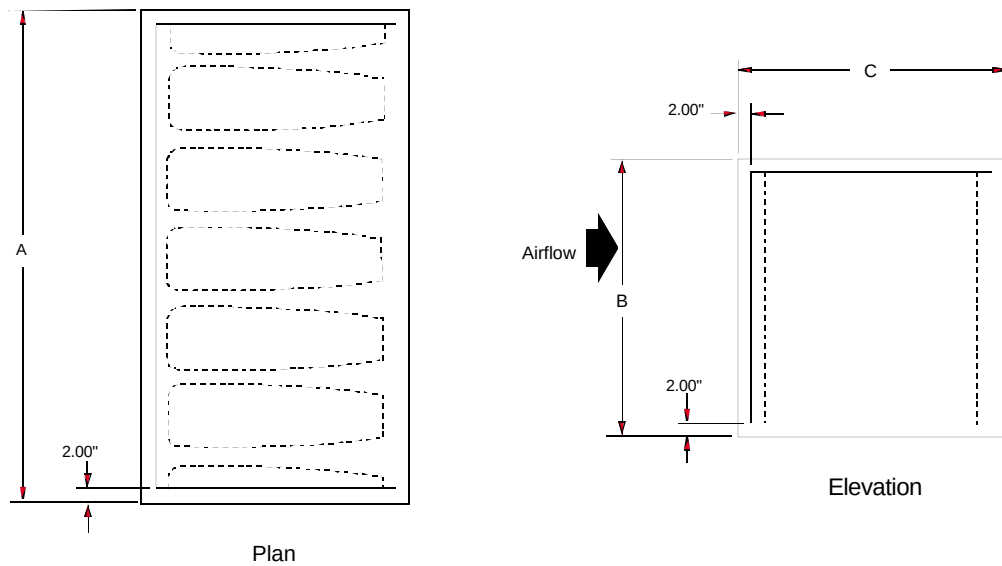
cat550-52.dsf

Note: Return and discharge plenums are available with openings in the top, bottom, upper end, lower end, drive side or opposite drive side. The opening size does not vary with the location. All openings have a 1" flange recessed in opening. 1.25 MPT drain connection available as an option for plenum sections.

UNIT SIZE	A*		B*		C		D		E	
	in.	mm	in.	mm	in.	mm	in.	mm	in.	mm
003	38.00	965	26.00	660	14.00	356	10.00	254	34.00	864
004	40.00	1016	30.00	762	16.00	406	12.00	305	36.00	914
006	52.00	1321	30.00	762	16.00	406	12.00	305	48.00	1219
008	58.00	1473	34.00	864	18.00	457	14.00	356	54.00	1372
010	64.00	1626	36.00	914	20.00	508	16.00	406	60.00	1524
012	66.00	1676	42.00	1067	22.00	559	18.00	457	62.00	1575
014	74.00	1880	42.00	1067	22.00	559	18.00	457	70.00	1778
017	80.00	2032	46.00	1168	24.00	610	20.00	508	76.00	1930
021	82.00	2083	52.00	1321	28.00	711	24.00	610	78.00	1981
025	86.00	2184	60.00	1524	30.00	762	26.00	660	82.00	2083
030	98.00	2489	60.00	1524	30.00	762	26.00	660	94.00	2388
035	102.00	2591	66.00	1676	32.00	813	28.00	711	98.00	2489
040	116.00	2946	68.00	1727	32.00	813	28.00	711	112.00	2845
050	120.00	3048	80.00	2032	38.00	965	34.00	864	116.00	2946
065	136.00	3454	92.00	2337	42.00	1067	38.00	965	132.00	3353
080	136.00	3454	98.00	3489	42.00	1067	44.00	1118	132.00	3353
085	136.00	3454	104.00	2642	44.00	1118	48.00	1219	132.00	3353
090	136.00	3454	110.00	2794	46.00	1168	50.00	1270	132.00	3353

* These are dimensions of the unit only. A base rail adds a minimum of 8" (203 mm) to the width, and between 4"-12" to the unit height. A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Sound attenuator section

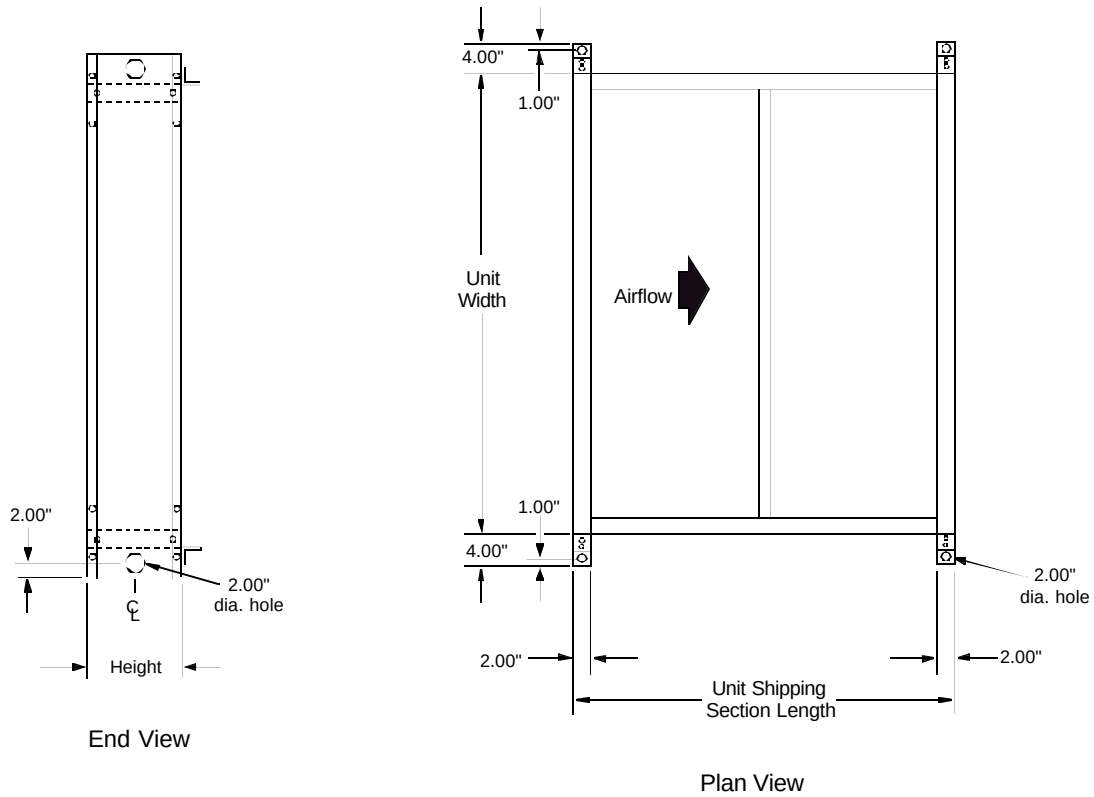


cat550-53.dsf

UNIT SIZE	A*		B*		C (available all sizes)	
	in.	mm	in.	mm	in.	mm
003	38.00	965	26.00	660	in.	mm
004	40.00	1016	30.00	762	40.00	1016
006	52.00	1321	30.00	762	52.00	1321
008	58.00	1473	34.00	864	64.00	1626
010	64.00	1626	36.00	914	—	—
012	66.00	1676	42.00	1067	—	—
014	74.00	1880	42.00	1067	—	—
017	80.00	2032	46.00	1168	—	—
021	82.00	2083	52.00	1321	—	—
025	86.00	2184	60.00	1524	—	—
030	98.00	2489	60.00	1524	—	—
035	102.00	2591	66.00	1676	—	—
040	116.00	2946	68.00	1727	—	—
050	120.00	3048	80.00	2032	—	—
065	136.00	3454	92.00	2337	—	—
080	136.00	3454	98.00	3489	42.00	1067
085	136.00	3454	104.00	2642	44.00	1118
090	136.00	3454	110.00	2794	46.00	1168

* These are dimensions of the unit only. A base rail adds a minimum of 8" (203 mm) to the width, and between 4"-12" to the unit height. A 6" (152 mm) base rail is mandatory on unit sizes 025-065. An 8" (203 mm) base rail is mandatory on unit sizes 080 and larger.

Base rail



Note: The base rail is available in heights of 4", 6", 8", 10" and 12".

cat550-54.dsf

Engineering guide specification

PART 1: GENERAL

1.01 SECTION INCLUDES

- A. Air handling units.

1.02 REFERENCES

- A. AFBMA 9 - Load Ratings and Fatigue Life for Ball Bearings.
- B. AMCA 99 - Standards Handbook.
- C. AMCA 210 - Laboratory Methods of Testing Fans for Rating Purposes.
- D. AMCA 300 - Test Code for Sound Rating Air Moving Devices.
- E. AMCA 500 - Test Methods for Louver, Dampers, and Shutters.
- F. ARI 410 - Forced-Circulation Air-Cooling and Air-Heating Coils.
- G. ARI 430 - Central-Station Air-Handling Units.
- H. ARI 435 - Application of Central-Station Air-Handling Units.
- I. ASTM B117 – Standard Practice for Operating Salt Spray Apparatus
- J. NEMA MG1 - Motors and Generators.
- K. NFPA 70 - National Electrical Code.
- L. SMACNA - HVAC Duct Construction Standards - Metal and Flexible.
- M. UL 723 – Test for Surface Burning Characteristics of Building Materials
- N. UL 900 - Test Performance of Air Filter Units.
- O. UL 1995 – Standard for Heating and Cooling Equipment
- P. UL 94 – Test for Flammability of Plastic Materials for Parts in Devices and Appliances

1.03 SUBMITTALS

- A. Shop Drawings: Indicate assembly, unit dimensions, weight loading, required clearances, construction details, field connection details, and electrical characteristics and connection requirements. Computer generated fan curves for each air handling unit shall be submitted with specific design operating point noted. A computer generated psychrometric chart shall be submitted for each cooling coil with design points and final operating point clearly noted. Sound data for discharge, radiated and return positions shall be submitted by octave band for each unit. Calculations for required baserail heights shall be included.
- B. Product Data:
 - 1. Provide literature that indicates dimensions, weights, capacities, ratings, fan performance, gauges and finishes of materials, and electrical characteristics and connection requirements.
 - 2. Provide data of filter media, filter performance data, filter assembly, and filter frames.
 - 3. Manufacturer's Installation Instructions.

1.04 OPERATION AND MAINTENANCE DATA

- A. Maintenance Data: Include instructions for lubrication, filter replacement, motor and drive replacement, spare parts lists, and wiring diagrams.

1.05 QUALIFICATIONS

- A. Manufacturer: Company specializing in manufacturing the Products specified in this section with minimum five years documented experience, who issues complete catalog data on total product.

1.06 DELIVERY, STORAGE, AND HANDLING

- A. Deliver, store, protect and handle products to site.
- B. Accept products on site on factory-installed shipping skids. Inspect for damage.
- C. Store in clean dry place and protect from weather and construction traffic. Handle carefully to avoid damage to components, enclosures, and finish.

1.07 ENVIRONMENTAL REQUIREMENTS

- A. Do not operate units for any purpose, temporary or permanent, until ductwork is clean, filters are in place, bearings lubricated, and fan has been test run under observation.

1.08 EXTRA MATERIALS

- A. Provide one set for each unit of fan belts and filters.

PART 2: PRODUCTS

2.01 MANUFACTURERS

A. The following manufacturers are approved for use. No substitutions will be permitted.

1. McQuay
2. Miller-Picking
3. Temtrol
4. Scott-Springfield
5. Racan

2.02 GENERAL DESCRIPTION

A. Configuration: Fabricate as detailed on prints, including (as detailed on drawings):

B. Performance: Conform to ARI 430. See schedules on prints.

C. Acoustics: Sound power levels (dB) for the unit shall not exceed the following specified levels. The manufacturer shall provide the necessary sound treatment to meet these levels if required.

Octave Band at Center Frequency (Hz)								
	63	125	250	500	1000	2000	4000	8000
Radiated								
Discharge								
Return								

2.03 CASING

A. Fabricate unit casing of 16 gauge channel posts and removable panels assembled with mechanical fasteners and a galvanized steel or painted galvanized steel. Assemble sections with high compression gasketing between each frame member and unit panel or door to prevent thermal bridging from interior to exterior of unit.

1. Outside casing shall consist of [G90 galvanized steel (16-gauge, 1.5 mm)(18 gauge, 1.2mm)] [G60 steel phosphatized and coated with baked on MistGrey enamel paint which meets or exceeds a 600 hour salt spray in accordance with ASTM117 standard for salt spray resistance (16-gauge, 1.5mm)(18 gauge, 1.2mm)].

2. Internal lining [fan section] [coil section] [all sections] shall be constructed of 20 gauge, 0.94 mm [G90 galvanized steel (solid) (perforated)] [(304 solid stainless steel)].

3. Floor plate [access sections] [all sections] shall be constructed of [G90 galvanized steel (20 gauge, 0.94 mm) (14 gauge, 1.8mm)] [304 solid stainless steel (20 gauge, 0.94 mm) (14 gauge, 1.8mm)] [Aluminum tread plate .125 inch thick, 3.2mm].

4. Entire unit shall have a [4] [6] [8] [10] [12] inch full perimeter base rail for structural rigidity and condensate trapping. The following calculation shall determine the required height of the baserail to allow for adequate drainage. Use the largest pressure to determine base rail height.

[(Negative)(Positive) static pressure (in)] (2) + 4" = required baserail height

Should the unit baserail not be factory supplied at this height, the contractor is required to supply a concrete housekeeping pad to make up the difference.

B. Module to module assembly shall be accomplished with an overlapping, full perimeter internal splice joint that is sealed with bulb type gasketing on both mating modules to minimize on-site labor along with meeting indoor air quality standards.

C. Unit casing leak rate shall not exceed 0.5 cfm per square foot of cabinet area at 5" static pressure in order to meet efficiency and indoor air quality standards.

D. Insulation: [1] [2] inch ([25] [50] mm) thick, [3/4] [1-1/2] [3] lb. per cu ft ([12] [24] [48] kg per cu m) density, neoprene coated, glass fiber insulation (with adhesive and pins on units without liners).

E. Access Doors: Made of galvanized steel, flush mounted to cabinetry, with minimum of two six inch long stainless steel piano-type hinges, latch and full size (4.5" minimum) handle assembly (Opt.) (provide inspection window for fan section). Door shall swing outward for unit sections under negative pressure (inward for unit sections under positive pressure). Doors limited from swinging inward (such as side access filter sections) on positive pressure sections, shall have a secondary latch to relieve pressure and prevent injury upon access.

F. Construct drain pans from (microbial resistant coated galvanized steel)(304 solid stainless steel) with cross break and double sloping pitch to drain connection. Provide drain pans under cooling coil section (fan section). Drain connection centerline shall be a minimum of 3" above the base rail to aid in proper condensate trapping. Drain connections that protrude from the base rail are not acceptable.

2.04 SUPPLY / RETURN FANS

- A. Provide (DWDI forward-curved) (DWDI airfoil) (belt-drive airfoil plenum) (direct-drive airfoil plenum) (inline) (DWDI forward curved twin) supply (return) fan(s). Fan assemblies including fan, motor and sheaves shall be dynamically balanced by the manufacturer on all three planes and at all bearing supports. Manufacturer must ensure maximum fan RPM is below the first critical speed.
- B. Bearings shall be self-aligning, grease lubricated, ball or roller bearings with extended copper lubrication lines to access side of unit. Grease fittings shall be attached to the fan base assembly near access door. If not supplied at the factory, contractor shall mount copper lube lines in the field.
- C. Fan and motor shall be mounted internally on a steel base. Factory mount motor on slide base that can be slid out the side of unit if removal is required. Provide access to motor, drive, and bearings through hinged access door. Fan and motor assembly shall be mounted on (rubber in shear)(2" deflection spring) vibration type isolators inside cabinetry. (Seismic snubbers shall be provided).

2.05 BEARINGS AND DRIVES

- A. Bearings: Basic load rating computed in accordance with AFBMA - ANSI Standards, [L-50 life at 200,000 hours – all DWDI fans] [L-50 life at 500,000 hours – DWDI fans on unit sizes 003 - 035], [L-50 life at 400,000 hours all belt-drive airfoil plenum fans and DWDI fans on unit sizes greater than 035] [L-50 life 1,000,000 hours – DWDI fans on unit sizes 003 – 035] [L-50 life at 600,000 hours – all inline fans], heavy duty pillow block type, self-aligning, grease-lubricated ball bearings.
- B. Shafts shall be solid, hot rolled steel, ground and polished, keyed to shaft, and protectively coated with lubricating oil. Hollow shafts are not acceptable.
- C. V-Belt drives shall be cast iron or steel sheaves, dynamically balanced, bored to fit shafts and keyed. [Fixed sheaves, matched belts, and drive rated based on motor horsepower] [Variable and adjustable pitch sheaves selected so required RPM is obtained with sheaves set at mid-position and rated based on motor horsepower. Contractor to furnish fixed sheaves at final RPM as determined by balancing contractor]. Minimum of 2 belts shall be provided on all fans with 10 HP motors and above. Standard drive service factor [1.1 (1/4 HP – 7.5 HP)] [1.3 (10HP and larger)] times fan brake horsepower.

2.06 ELECTRICAL

- A. Wiring Termination: Provide terminal lugs to match branch circuit conductor quantities, sizes, and materials indicated. Enclosed terminal lugs in terminal box sized to NFPA 70.
- B. (Opt.) Provide [marine light] [marine light and GFI receptacle] in (fan) (each) section mounted and wired to a junction box and on-off switch mounted on the outside of the cabinet.
- C. Motors: provide (ODP)(TEFC)(2 Speed) type with (standard) (EPACT) (premium) efficiency. Electrical characteristics shall be as shown in schedule.
- D. (Opt.) Provide factory mounted starters and disconnects.
- E. (Opt.) Provide factory mounted variable speed drives.

2.07 COOLING AND HEATING COIL SECTIONS

- A. Provide access to coils from (both sides) (connection side) (opposite side) of unit for service and cleaning. Enclose coil headers and return bends fully within unit casing. Unit shall be provided with coil connections that extend a minimum of 5" beyond unit casing for ease of installation. Drain and vent connections shall be provided exterior to unit casing. Coil connections must be factory sealed with grommets on interior and exterior and gasket sleeve between outer wall and liner where each pipe extends through the unit casing to minimize air leakage and condensation inside panel assembly. If not factory packaged, Contractor must supply all coil connection grommets and sleeves. Coils shall be removable through side and/or top panels of unit without the need to remove and disassemble the entire section from the unit.
- B. Certify air coil capacities, pressure drops, and selection procedures in accordance with ARI 410.
- C. Water Coils:
 - 1. Fins shall have a minimum thickness of [0.0075" (0.191 mm)] [0.0095" (0.241 mm)] of (aluminum)(copper) plate construction. Fins shall have full drawn collars to provide a continuous surface cover over the entire tube for maximum heat transfer. Tubes shall be mechanically expanded into the fins to provide a continuous primary to secondary compression bond over the entire finned length for maximum heat transfer rates. Bare copper tubes shall not be visible between fins.
 - 2. Coil tubes shall be 5/8 inch (16mm) OD seamless copper, (0.020") (0.025") (0.035") nominal tube wall thickness, expanded into fins, brazed at joints. Soldered U-bends shall be provided to minimize the effects of erosion and premature failure having a minimum tube wall thickness of (.025"). 1/2" tubes and hairpin construction are not acceptable.
 - 3. Water coils shall be provided with headers of seamless copper tubing with intruded tube holes to permit expansion and contraction without creating undue stress or strain. Coil connections shall be [carbon steel] [copper] [red brass] with connection size to be determined by manufacturer based upon the most efficient coil circuiting. Vent connections provided at the highest point to assure proper venting. Drain connections shall be provided at the lowest point.
 - 4. Coil casings shall be a formed channel frame of (galvanized steel) (304 solid stainless steel).

D. Refrigerant Coils:

1. Coils designed for use with Refrigerant [R-22] [R-134a] [other]. Fins shall have a minimum thickness of [0.0075" (0.191 mm)] [0.0095" (0.241 mm)] of (aluminum) (copper) with full drawn collars to provide a continuous surface cover over the entire tube for maximum heat transfer. Tubes shall be mechanically expanded into the fins to provide a continuous primary-to-secondary compression bond over the entire finned length for maximum heat transfer rates. Bare copper tube shall not be visible between fins.
2. Refrigerant coils shall be provided with round seamless 5/8" (16 mm) O.D. copper tubes on 1-1/2" (38 mm) centers, staggered in the direction of airflow. All joints shall be brazed.
3. Sweat type copper suction connections located at the bottom of the suction headers for gravity oil drainage. Coils shall be uniformly circuited in a counterflow manner to [row] [face] [row and face] control capacity reduction. Pressure type liquid distributors used. Coils shall be tested with 315 pounds air pressure under warm water, and suitable for 250 psig working pressure. Coils shall be ARI certified and Underwriters Laboratories, Inc. listed.

E. Steam Coils:

1. Fins shall have a minimum thickness of [0.0075" (0.191 mm)] [0.0095" (0.241 mm)] of (aluminum) (copper) with full drawn collars to provide a continuous surface cover over the entire tube for maximum heat transfer. Tubes shall be mechanically expanded into the fins to provide a continuous primary-to-secondary compression bond over the entire finned length for maximum heat transfer rates. Bare copper tubes shall not be visible between fins.
2. Steam coils shall be provided with round seamless [5/8" (16 mm)] [1" (25.4 mm)] O.D. [copper] tubes. Tubes on two-row coils are staggered in the direction of airflow. All joints shall be brazed.
3. Steam coil headers shall be made of nonferrous materials using seamless copper tubing with intruded tube holes to permit expansion and contraction without creating undue stress or strain. Both the supply and return headers shall be completely encased by the coil casing. Coil shall be pitched in the unit to assure positive condensate drainage. Orificed baffle plates shall be installed in the supply connection to ensure proper diffusion of entering steam. Coil connections shall be [carbon steel] [red brass] with connection size to be determined by manufacturer based upon the most efficient coil circuiting.
4. Steam coils shall be tested with 315 pounds air pressure under warm water and suitable for 150 psig working pressures.

2.08 FILTERS

- A. [Filter section with filter racks and guides with [hinged and latching access doors on either, or both sides] for side loading and removal of filters] [Filter section with front loading frames and clips].
- B. Filter media shall be UL 900 listed, Class I or Class II.
- C. [Flat] [Angle] arrangement with (2", 50mm) (4", 100mm) deep [pleated] [disposable] panel filters.
- D. [Bag] [Cartridge] type arrangement with holding frames suitable for 2" (50 mm), 4" (100 mm) prefilter and final filter media and blank-off sheets, extended surface [bag] [cartridge] media filters with [60-65] [80-85] [90-95] percent dust spot efficiency. Bag filter media [12" (305 mm)] [15" (381 mm)] [19" (483 mm)] [22" (559 mm)] [30" (762 mm)] [36" (914 mm)] deep. Cartridge filter media is [4" (50 mm)] [12" (305 mm)] deep. (Provide microbial resistant Intersept® coating on all filters). Designed for (side) (front) loading of filters.
- E. (Opt.) [Manufacturer shall supply method to read pressure drop across the filter bank for scheduling filter replacement. Design shall be equal to a Dwyer Minihelic II and be recessed into the cabinet to minimize chances for damage during shipment and installation].

2.09 ADDITIONAL SECTIONS

- A. Mixing box section shall be provided with or without factory mounted low leak airfoil blade outside and return air dampers of galvanized steel in a galvanized frame. Dampers shall be hollow core airfoil blades, fully gasketed and have continuous vinyl seals between damper blades. Dampers shall have stainless steel jamb seals along end of dampers. Linkage and ABS plastic end caps shall be provided when return and outside air dampers sized for full airflow. Return and outside air dampers of different sizes must be driven separately.
- B. Economizer section shall be provided with or without factory mounted low leak hollow core airfoil blade outside air and return air dampers, and exhaust damper. Dampers shall be constructed of galvanized steel in a galvanized frame. Outside and return dampers fully gasketed. Continuous vinyl seals between damper blades. Stainless steel end seals along end of dampers. Linkage and ABS plastic end caps provided when return and outside air dampers are the same size. Return and outside air dampers of different sizes must be driven separately.
- C. Access section shall provide access between components shall be a minimum of [16"] [24"] [30"] [36"] [42"] [48"] [54"] deep ([406mm] [610mm] [762mm] [914mm] [1067mm] [1219mm] [1372mm]). (Opt.) Access doors of galvanized steel for flush mounting, with gasket, latch and full size (minimum of 4.5") handle assembly. Provide (microbial resistant coated galvanized)(stainless) steel drip pan in access section. Heavy-duty floor thickness to be (18 gauge) (16 gauge) (14 gauge) (.125 inch tread plate) to accommodate walk-in weight of service technician.
- D. Diffuser section shall be mounted on leaving side of supply fan section providing uniform air distribution across downstream components. Perforated diffuser plate shall be secured over blast area of fan to disperse air flow.
- E. Blender / air mixer section to provide proper air mixing and distribution of the outside and return airstreams. Proper spacing provided in the direction of airflow as recommended by the blender manufacturer (Blender Products or Kees).

- F. Sound attenuator section shall be provided by the air handling unit manufacturer as an integral section of the unit to attenuate fan noise at the source. Provide acoustic performance and air pressure drop as scheduled. Silencer acoustic insulation shall be covered with perforated sheet metal. [Provide (mylar)(tedlar)(fiberglass cloth) liner between perforated sheet metal and acoustic insulation].
- G. Manual section shall be provided by the air handling unit manufacturer as an integral section of the unit for field installation of special components. Section length to be determined by component supplier.
- H. (Inlet) (Discharge) plenum section shall be provided as the (first) (last) section in the direction of airflow. The plenum shall provide single or multiple (top) (end) (bottom) openings.
- I. Face and bypass section shall be provided to modulate airflow through and around heat transfer coils. Dampers shall be an integral part of the unit assembly. [Internal face and bypass shall be contained in the standard cabinet height] [External face and bypass arrangement shall be provided with factory provided bypass duct of same cabinet construction as the remainder of the unit. Field fabricated bypass ducts are not acceptable]. Blank-off and division sheets, internal linkage, access [panel] [doors] installed by unit manufacturer. Face and bypass dampers shall be of low leak design, opposed blade, with vinyl bulb edging and stainless steel edge seals, galvanized steel frame and axles in self-lubricating nylon bearings.

2.10 DAMPERS

- A. Damper Leakage: Leakage rate shall be less than two tenths of one percent leakage at 2 inches static pressure differential. Leakage rate tested in accordance with AMCA Standard 500.
- B. Multi-Zone Dampers: Zone dampers for use with [2 deck] or [3 deck], vertical or horizontal system. Factory mounted in casing of galvanized steel blades [with vinyl bulb edging]. Low leak dampers with bronze end and side seals, locked in position to common shaft with rotation angles of 90°, requiring one actuator per zone. Damper rods rotate in nylon bushings.

PART 3: EXECUTION

3.01 INSTALLATION

- A. Install in accordance with manufacturer's instructions.



13600 Industrial Park Boulevard, Minneapolis, MN 55441 USA • (800) 432-1342 • www.mcquay.com
